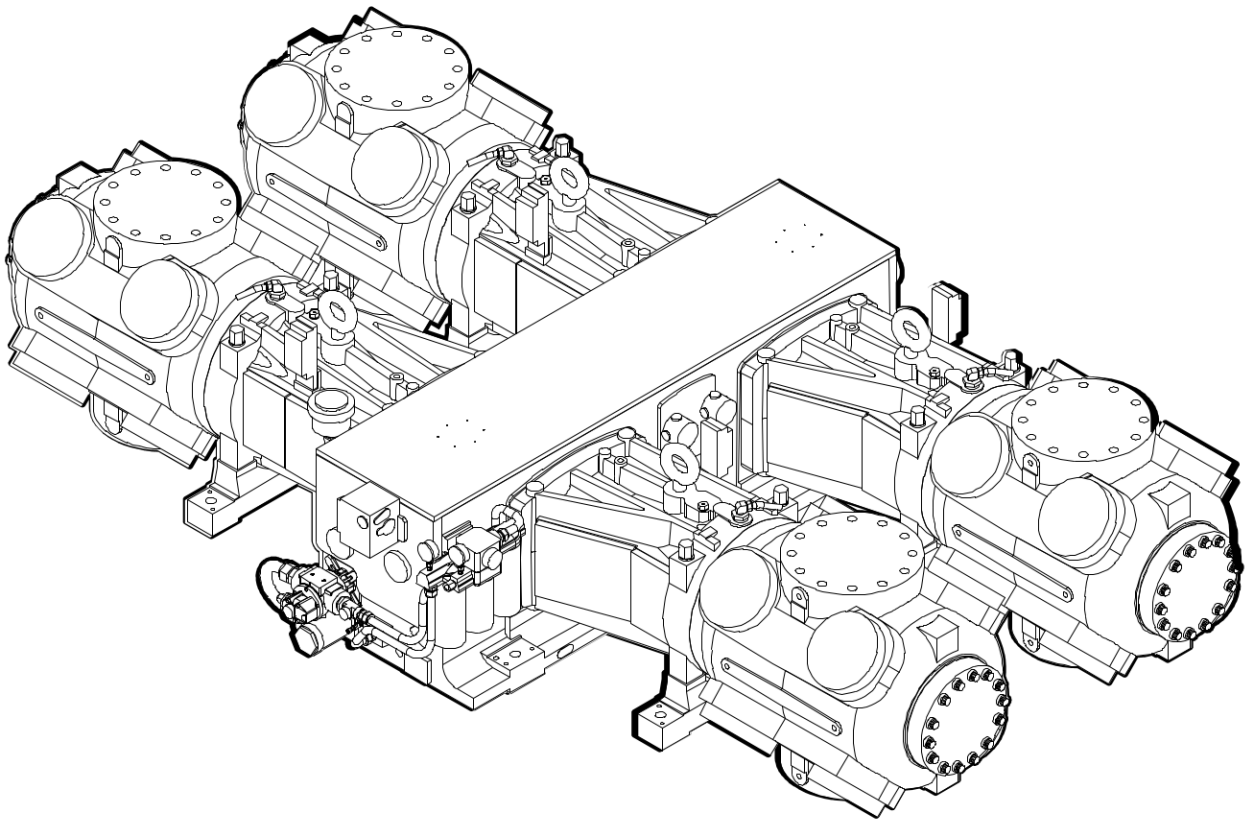
MAINTENANCE & REPAIR MANUAL

WORLD STANDARD COMPRESSORS

ARIEL CORPORATION

35 Blackjack Rd.
Mount Vernon, OH 43050
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www.arielcorp.com



KBE

Heavy Duty Balanced Opposed Reciprocating Compressors

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General Safety for Reciprocating Compressors

WARNING:

To decrease the risk of injury or death, only trained and approved personnel can do maintenance work. Approved personnel must read and know the applicable Ariel Technical Manual and Packager-supplied information.

Before you start maintenance:

- Always use personal protection equipment (PPE) and obey your company policies for safety and operation.
- Do not tighten or loosen fasteners or fittings while the unit is in operation or is pressurized.
- Make sure the driver and compressor cannot turn during maintenance.
- Obey your company and packager instructions to isolate and vent the system.
- Make sure there is no remaining pressure in the cylinders. If equipped with “Kiene” valves on the indicator ports, open them to release residual pressure after blowdown.

  WARNING: RELEASED GAS MAY BE POISONOUS AND/OR FLAMMABLE.

Before operation, purge the system. A pressurized hydrocarbon and air mixture can explode and cause injury or death.



When you see the symbol to the left on a compressor or control panel, refer to the applicable Ariel Maintenance and Repair Manual for specific information before you proceed. Manual information can change without notice and can possibly not apply to equipment built before the date on the front cover. For questions of compressor safety, operation, maintenance, or repair, speak to your packager or Ariel.

CAUTION:

Noise generated by reciprocating machinery can damage hearing. Wear hearing protection during compressor operation. Refer to Packager information for specific recommendations.

CAUTION:

Where applicable, compressor installation must comply with Zone 1 requirements. A Zone 1 environment must have correct or equivalent protection to obey electrical requirements.

CAUTION:

Hot gas temperatures (especially the cylinder discharge), 190°F (88°C) oil, and high friction areas - wear correct protection. Shut down unit and permit it to cool before maintenance.

CAUTION:

Suction or discharge valves installed in incorrect locations can cause severe personal injury and property damage.

CAUTION:

If applicable, do not remove the pin from the crosshead guide to compressor cylinder mounting face. It is there to prevent installation of misapplied cylinders and avoid personal injury.

Lifting the Compressor

CAUTION:

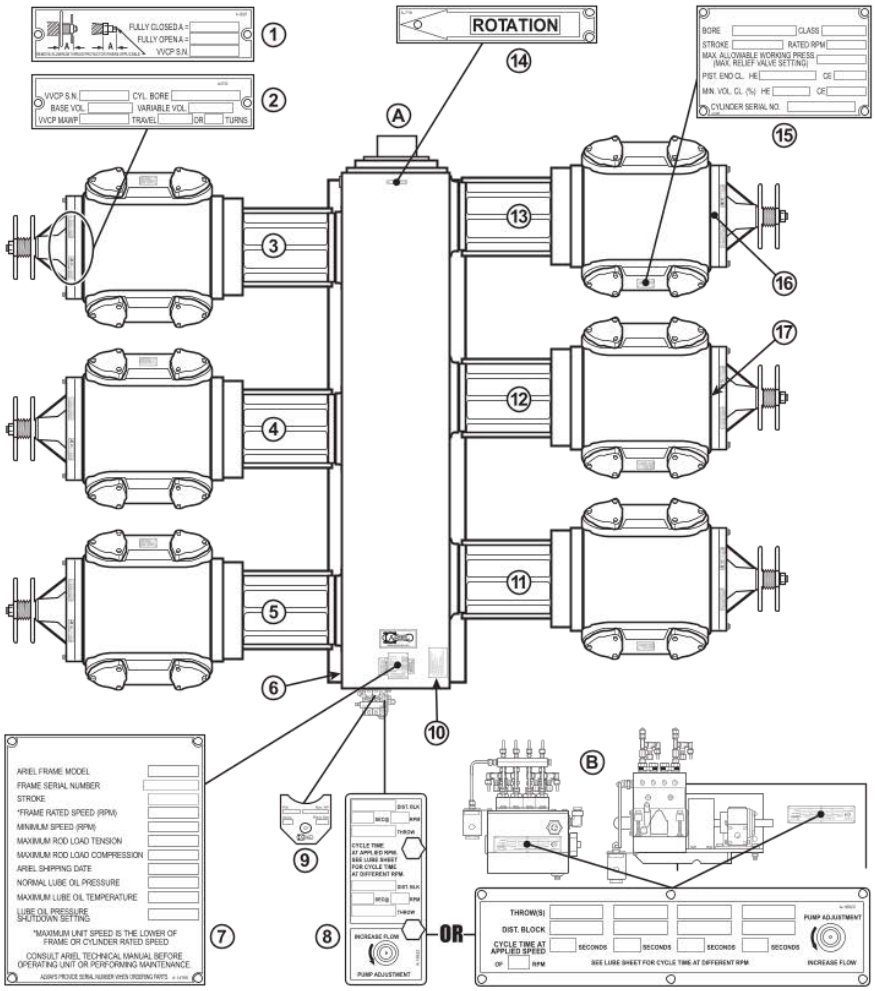
When lifting with lugs, use only Ariel-supplied lugs. Fasten lugs with correct grade cap screws that are correctly torqued. Use lifting cable spreaders. Make sure the crane can lift the weight.

- Calculate the weight before lifting. Refer to the Ariel Performance Program for frame and cylinder weights. Include extra weight like external and internal flywheels, etc.
- Each compressor ships with an attached Compressor Lifting Information packet. Find the compressor weight, center-of-gravity, and lifting hook locations through the Ariel Performance software or speak to the Ariel Response Center.
- Ariel recommends use of approved, professional riggers/movers.
- Ariel-supplied eyebolts (attached to cross- head guides), brackets (attached to cylinder nozzles) and lifting lugs (attached to frame walls) are designed to lift the Ariel compressor only. Do not attempt to lift the compressor when it is attached to the skid, driver, bottles, piping, etc. Do not use these lifting attachments on other equipment.
- Install lugs with the correct Grade 8 cap screws. Tighten them to the torque listed in the applicable Ariel Maintenance and Repair Manual or the Ariel Torque App. If lifting lugs were installed before, make sure the bolt torques are correct before lifting.
- For compressors with no lifting lugs, lift by the eyebolts at the crosshead guides. Ariel-installed eyebolts are locked in position with Loctite® thread locker.
- Remove and destroy eyebolts, brackets, and lifting lugs that have defects that could make them unserviceable. Use only Ariel-supplied brackets and lifting lugs.
- Standard lifting lug and eyebolt materials can fail with shock loads at low ambient temperatures. If necessary, replace eyebolts with forged eyebolts of materials applicable for low temperatures. Obey ASME B18.15 and ASTM F541 guidelines.
- Use spreader bars or lifting beams for a vertical pull. Chains or straps attached to eyebolts or lifting lugs must be less than 5° of vertical. For eyebolt load capacity or to pull an eyebolt in a direction other than along the threaded axis, refer to ASME B18.15.
- Make sure lifting tackle, spreader bars, and crane are correctly attached and rigged and can lift the weight. Keep compressor level when lifting and setting. Heavier cylinders and/or blank throws can have a large effect on center-of-gravity.

Throw and Data Plate Locations

When contacting Ariel with compressor questions, know throw locations and information on data plates fastened to the machine. This data helps Ariel representatives answer quickly and accurately. Contact Ariel for replacement if any data plates are missing.

Figure: Separable Guide Compressor Throw and Data Plate Locations - Typical



- **A** Compressor Top View
- **B** Auxiliary End Front Views
- **1** VVCP Dimension Plate
- **2** VVCP Data Plate
- **3** Throw #2
- **4** Throw #4
- **5** Throw #6
- **6** Mechanical Inspector Plate and Frame Serial Number Stamp
- **7** Compressor Data Plate
- **8** Force Feed Lubricator Data Plate
- **9** Force Feed Lubricator Pump Data Plate
- **10** Oil Filter Change Instruction Plate
- **11** Throw #5
- **12** Throw #3
- **13** Throw #1
- **14** Rotation Direction Plate
- **15** Cylinder Data Plate
- **16** VVCP Serial Number, MAWP, and Hydrotest Stamp
- **17** Cylinder Serial Number, MAWP, and Hydrotest Stamp

Other Ariel Resources

Ariel Website

Visit www.arielcorp.com to view and print the latest documentation, such as:

- **Customer Technical Bulletins** (CTB's) provide important information on changes, corrections, or additions to Ariel products or services. Read these bulletins before operating or servicing equipment.
- **Engineering References** (ER's) provide standard procedures and other useful information for operation, maintenance, or repair of Ariel compressors or components. Read and follow these procedures for long and trouble-free service from your Ariel compressor.
- **Maintenance and Repair Manuals** provide detailed maintenance and repair information on specific Ariel compressor models.
- **Packager Standards** provide detailed requirements and recommendations on the installation of an Ariel compressor.
- **Application Manual** provides detailed information on the use of Ariel compressors in different applications.

Ariel Technical and Service Schools

Ariel schedules several in-plant schools each year, which include classroom and hands-on training. Ariel also sends representatives to provide customized training on location. Contact Ariel for details.

Ariel Contact Information

Contact	Telephone	Fax	E-Mail
Ariel Response Center	888-397-7766 (toll-free USA & Canada) or 740-397-3602 (International)	740-397-1060	arc@arielcorp.com
Ariel Parts Sales		740-393-5054	parts@arielcorp.com
Order Entry		740-397-3856	-
Ariel World HQ	740-397-0311	740-397-3856	info@arielcorp.com
Technical Services			fieldservice@arielcorp.com

Ariel Response Center Technicians or Switchboard Operators answer telephones during Ariel business hours, Eastern Time - USA or after hours by voice mail. Contact an authorized distributor to purchase Ariel parts. Always provide Ariel equipment serial number(s) to order parts. The after-hours Telephone Emergency System works as follows:

1. Follow automated instructions to Technical Services Emergency Assistance or Parts Emergency Service. Calls are answered by voice mail.
2. Leave a message: caller name and telephone number, serial number of equipment in question (frame, cylinder, unloader), and a brief description of the emergency.
3. Your voice mail routes to an on-call representative who responds as soon as possible.

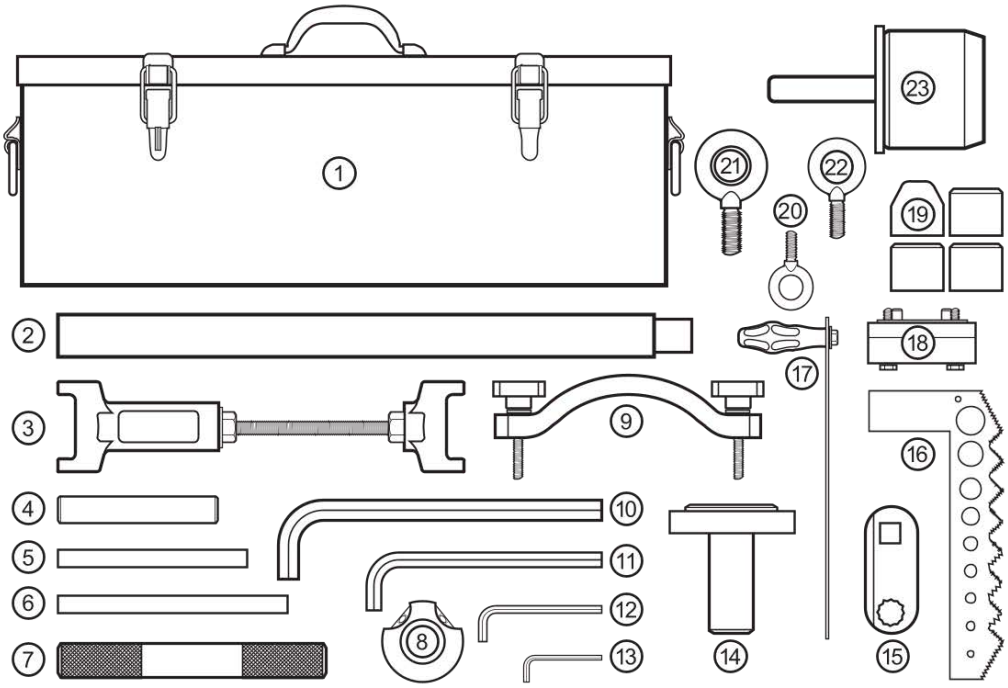
Section 1 - Tools

A variety of tools are required for safe and effective maintenance of Ariel compressors. Standard hand tools are readily purchased from hardware stores or industrial supply companies. Ariel makes certain special tools available where needed.

Ariel Optional Tools

Ariel offers an optional tool kit with every compressor. For KBE compressors, it contains the tools shown below, which are specifically designed for use on Ariel units. Clean all tools before use and verify full tool engagement with the part being removed or installed. If the Tool Kit is missing or if a single tool is missing, worn, or broken, call your distributor. Do not use worn or broken tools, or substitutes for Ariel furnished tools. See Parts List for individual tool part numbers.

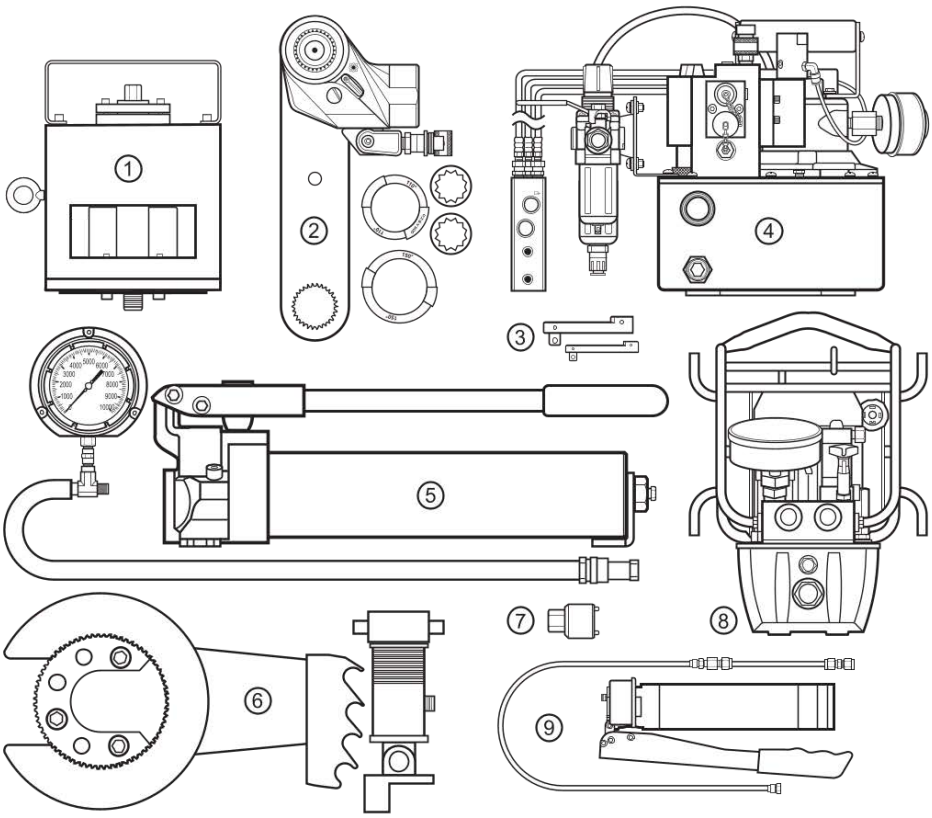
Figure: Optional Furnished Tools for KBE Compressors



1 Tool Box
2 Peg Wrench
3 Bearing Cap Installation/Removal Tool
4 5/8 x 3/4 UNF Valve Tool
5 5/16 x 1/2 UNF Valve Tool
6 1/4 x 3/8 UNF Valve Tool
7 3/4 x 1" UNC Valve Tool (for CP valves)
8 Connecting Rod Bolt 110° Turn Indicator
9 Connecting Rod Cap Removal Tool
10 3/4 Allen Wrench
11 1/2 Allen Wrench (2)
12 3/8 Allen Wrench
13 3/16 Allen Wrench (5)
14 Crosshead Installation Tool
15 Cylinder Bolt Torque Adapter
16 Ariel Bore and Thread Gauge
17 Main Bearing Removal Tool
18 Piston Nut Spanner
19 Piston Rod Entering Sleeve
20 Forged 3/8 - 16 UNC Eyebolts (6)
21 Forged 5/8 - 11 UNC Eyebolts (2)
22 Forged 1/2 - 13 UNC Eyebolts (2)
23 Crosshead Pin Alignment Tool

Ariel Separately Purchased Tools

Figure: Ariel Separately Purchased Tools for KBE Compressors



1. Hydraulic Piston Rod Tensioning Tool
2. Hydraulic Connecting Rod Torque and Turn Tool (includes one 1-inch 12-point direct drive socket, two 3/4-inch hydraulic wrench head drives, one reaction fixture, and one 15-foot hydraulic hose with tool-end quick connectors)
3. 1/2-inch & 3/4-inch offset adapters to access tough-to-reach fasteners (sold individually)
4. Pneumatic/Hydraulic Pump for connecting rod torque and turn tool kit (includes pneumatic pump, hose, coupler, and gauge)
5. Hand Pump for hydraulic crosshead balance nut torquing tool and piston rod tensioning tool (includes hand pump, hose, coupler, and gauge)
6. Hydraulic Crosshead Balance Nut Torquing Tool (includes tool and ram)
7. Force Feed Lubricator Bearing Housing Wrench
8. Electric/Hydraulic Pump for connecting rod torque and turn tool kit (includes electric pump, hose, coupler, and gauge)
9. Force Feed Lubrication Hand Purge Pump

Ariel Separately Purchased Tool Kits

1. **Ariel SAE Hand Measurement Tool Kit**
   a. 0.0005 in. needle type dial indicator
   b. 0.001 in. 1-inch travel dial indicator
   c. Magnet base for dial indicator
   d. 3/8” drive calibrated torque wrench, 40 to 200 in x lbs
   e. 3/8” drive calibrated torque wrench, 10 to 100 ft x lbs

f. 1/2” drive calibrated torque wrench, 50 to 250 ft x lbs
g. 3/4” drive calibrated torque wrench, 120 to 600 ft x lbs
h. 1” drive calibrated torque wrench, 200 to 1000 ft x lbs

2. **Ariel SAE Hand Tool Kit**
   a. 15-piece combination open/box end wrench set, 5/16” to 1-1/4”
   b. 8-piece slot and Phillips screwdriver set
   c. 3/8" square drive wrench set, including:
      - 12-piece, 12-point socket set, 5/16” to 1”
      - 7” ratchet
      - Speed handle
      - Extensions, 1-1/2”, 3”, 6”, and 12”
      - Ball type universal joint
      - 3/8” to 1/2" drive adapter
      - 1/4” hex key socket
      - 8-1/2” breaker bar
   d. 1/2" square drive wrench set, including:
      - 14-piece, 12-point socket set, 7/16” to 1-1/4” with clip rail
      - Extensions, 1-1/2”, 5”, and 10"
      - Drive adapters: 1/2” to 3/8" and 1/2” to 3/4"
      - 1-1/4” open end crow’s foot adapter
      - 10-1/4” ratchet
      - Speed handle
      - Ball type universal joint
      - 18” breaker bar
      - 1/2” and 5/8” hex key sockets
   e. 3/4" square drive wrench set, including:
      - 18-piece, 12-point socket set, 3/4” to 2”
      - Extensions, 3-1/2”, 8”, and 16”
      - Drive adapters: 3/4” to 1/2", 3/4” to 1", and 1” to 3/4"
      - 24” ratchet
      - 22” breaker bar
   f. 1" square drive wrench set, including:
      - 14-piece, 12-point socket set, 1-7/16” through 2-5/8”
      - 30” ratchet
      - Extensions: 8” and 17”
      - 22” sliding T
   g. Adjustable wrenches: 12" and 18"
   h. Dead blow semi-soft faced hammers: 3 lb. and 6 lb
   i. 3/4” x 36” pry bar
   j. 12” long feeler gauge set
   k. 12” machinist scale with 0.01 in. increments
   l. 13-piece Allen wrench set, 0.050 in. - 3/8”

Recommended Tools

Ariel compressor maintenance and repair normally requires only Ariel furnished tools and separately purchased tools and tool kits. However, Ariel also recommends purchasing the additional tools below. Contact Ariel for questions about tools for Ariel units.

1. 12-point box end torque adapter extension wrench set
2. Tape measure

3. Flashlight
4. Small mirror on a flexible extension rod
5. Small magnet on a flexible extension rod
6. Electric and/or pneumatic drill
7. Twist drill set
8. Torque multiplier

Section 2 - Instrumentation

If the compressor is equipped with Ariel Smart Compressor (ASC), see the ASC Packager Guide.

Several optional instruments can aid in the operation of an Ariel compressor. Through the data they provide, some instruments can help decrease maintenance costs and downtime, and help diagnose lubrication and other types of problems. The use of “alarm” instead of “shutdown” for any minimum instrumentation requirement may result in equipment damage.

NOTE:

Ariel recommends the use of shipboard or armored cabling as opposed to wiring in conduit to minimize nuisance shutdowns due to wire chafing and shorts. Cable in cable trays also facilitates maintenance access.

CAUTION:

DO NOT drill holes in cylinders or other pressure containing components for any purpose. Pressure containing components include cylinders, valve caps, cylinder heads, unloaders, gas passage caps, and flanges. See "Attachment of Wiring, Tubing, or Pipe Clamps to Ariel Compressor Cylinders".

CAUTION:

Arc welding on the skid and/or related equipment and piping can cause immediate failure or decreased life of solid-state electronic equipment and will void the warranty. To protect this equipment, disconnect all electrical connections and ground, and remove batteries, or fully remove the electronic equipment from the compressor before arc welding.

Attach the welding ground clamp as close as possible to where welding will occur. Use the lowest possible welder output setting. Do not cause a current flow across compressor bearing surfaces.

Table: Required Instrumentation Summary

INSTRUMENT	SUPPLIER	SETTING/REQUIREMENT
Frame Oil System (see "Frame Oil System Components")		
Oil Pressure	**Packager**	
- Start Permissive with Pre-lube Pump		30 psig (2.1 barg) for 2 minutes
- Run Permissive		45 psig (3.1 barg) within 10 seconds of start
- Alarm		50 psig (3.4 barg)
- Shutdown		45 psig (3.1 barg)
Oil Filter Differential	**Ariel**	Change at 10 psi (0.7 bar) for KBE/2/4 or 6 months of operation, whichever occurs first.
Oil Temperature	**Packager**	
- Minimum, to start-up or to load		See "Viscosity" for max. oil viscosity based on frame size.
- Minimum, operating		150°F (66°C)
- Maximum, shutdown		190°F (88°C)
Crankcase Level	**Packager**	**1/2 to 2/3 sight glass level**
- Low Level Shutdown		1/4 sight glass level
Main Bearing Temperature	**Ariel**	**Recommended, but optional**
- Alarm		+20°F (11°C) above normal not to exceed 220°F (104°C)
- Shutdown		+30°F (17°C) above normal not to exceed 230°F (110°C)
- Differential Alarm		20°F (11°C)
- Differential Shutdown		30°F (17°C)
Packing Case Vent	**Packager**	**Optional**
- Vent Flow Alarm		2 - 4 scfm (per throw)
- Vent Flow Shutdown		>4 scfm (per throw)
Scrubber Liquid Level	**Packager**	
- High Liquid Level Control		
- High Liquid Level Shutdown		
Gas Conditions	**Packager**	
- Inlet Temperature Indicator, each stage		
- Inlet Pressure Indicator, each stage		
- Discharge Pressure Indicator, each stage		
- Discharge Temperature Indicator, each cylinder		
- High Discharge Temperature Shutdown, each cylinder		10% above normal[a] (Not to exceed the maximum temperatures below)
- Lubricated Cylinders		350°F (177°C)

INSTRUMENT	SUPPLIER	SETTING/REQUIREMENT
- Relief Valve Setting above normal operating pressure (not to exceed MAWP)		Up to 150 psig (10 barg): 15 psig (1.0 barg) to 2500 psig (172 barg): 10% to 3500 psig (241 barg): 8% to 5000 psig (345 barg): 6% Over 5000 psig (345 barg): As agreed See "Relief Valves" in the Packager Standards Manual.
Cyl. Rod Load Protection	**Packager**	See Gas Rod Load Chart in the Ariel Performance Program.
High Vibration Shutdown	**Packager**	**As close to a normal level as practical. See "Vibration Protection" in the Packager Standards Manual.**
Over Speed	**Packager**	**10% over rated speed for shutdown**

a Example: Normal Discharge Temp. = 270°F; Shutdown Setting = 270 x 1.1 = 297°F.

Notes

1. Install the compressor frame low lube oil pressure shutdown to stop the unit if oil pressure downstream of the filter falls below 45 psig (3.1 barg). Compressor operation for only a few seconds without oil pressure causes major damage. Normal oil pressure is about 60 psig (4.1 barg) at full rated speed and normal operating temperature. The low oil pressure shutdown must activate after oil pressure exceeds 45 psig (3.1 barg) at start-up. Ariel provides a 1/4 inch tubing fitting to connect the low lube oil pressure shutdown and ties a tag to this connection before each compressor ships. Do not operate the compressor for prolonged periods at less than 50 psig (3.4 barg) oil pressure.
2. Automated pre-lube systems require a start permissive to sense minimum required pressure/time at oil gallery inlet.
3. Unit must shut down if the system fails to achieve 45 psig (3.1 barg) oil pressure within 10 seconds after crankshaft starts to turn.
4. Install the high lube oil inlet temperature shutdown at the filter inlet connection.
5. At a minimum, install one vibration shutdown. Mount vibration device at the crankshaft centerline with the sensitive axis parallel to the piston rod axis. Ariel provides mounting locations for vibration devices, reference applicable outline drawing.
6. Install all safety shutdowns, controls, instrumentation, ignition systems, electrical devices, and high temperature piping (gas discharge and engine exhaust) in accordance with good engineering practice and applicable codes for the area classification at the end user location. Ensure compatibility of all systems for area classification.
7. If packing vent temperature is monitored, confirm alarm and shutdown set points through vent flow rates.

Proximity Switch A-18255

CAUTION:

Any arc welding on the skid and/or associated equipment and piping can permanently damage solid-state electronic equipment. Welding can cause immediate failure or reduce electronic equipment life and void the warranty. To protect electronic equipment prior to any arc welding (including repair welding), disconnect all electrical connections including ground, and remove batteries, or completely remove the electronic equipment from the compressor.

It is good practice to attach the welding ground clamp as close as possible to the area where the welding will occur and to use the lowest practical welder output setting. Welding must not cause a current flow across any compressor bearing surface, including but not limited to crankshaft and crosshead bearing surfaces.

A proximity switch installs into a divider valve in place of a piston end plug and can be used to actuate any device. It consists of a reed switch and a magnet. When installed, the magnet rests against the divider valve piston and parallel to the reed switch. With every divider valve cycle, the piston moves the magnet, which opens and closes the reed switch contacts.

The time for the proximity switch to repeat a contact transition (ex. from open to closed) is known as the cycle time of the divider valve assembly. The proximity switch must work with a PLC or some other counter/timer device to produce a shutdown. See "Instrumentation" in the Ariel Packager Standards for interpretation of proximity switch pulse output.

Proximity Switch Installation

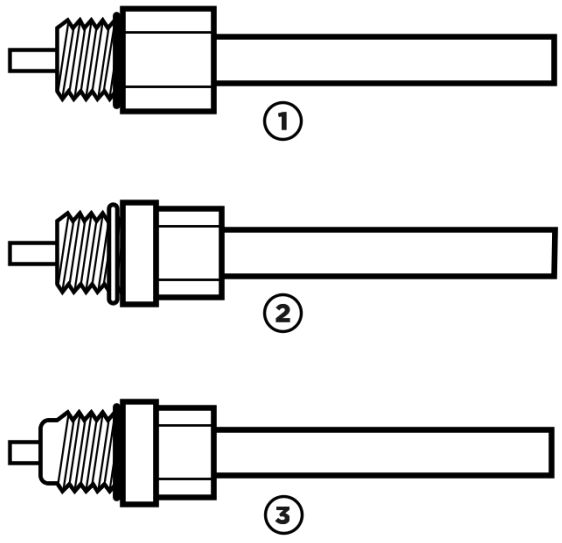
1. Loosen the Allen set screws on the proximity switch housing and remove magnet housing. Do not remove magnet, spring, or spacer from magnet housing.
2. Remove piston enclosure plug from end of desired divider valve. The proximity switch installs on any of the divider valves of the divider block. The proximity switch requires the correct magnet assembly to match the divider valve manufacturer.

   NOTE:

   Do not install a proximity switch on Lincoln divider valves with cycle indicator pins.

3. If applicable, verify O-ring or metal gasket is in place on magnet housing. Thread magnet housing into end of divider valve. Torque to 15 lbs x ft maximum.

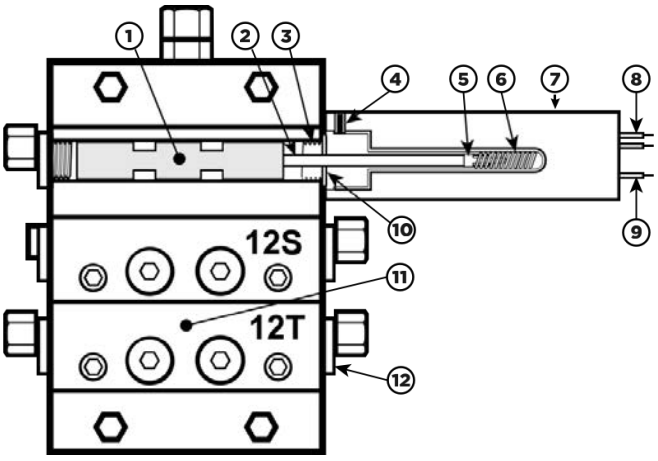
Figure: Typical Magnet Assemblies



1 Trabon Style O-ring Seal 7/16-20
2 Trabon Metal Gasket Seal 7/16-20 (1994 or earlier)
3 Lincoln O-Ring Seal 7/16-20 Extended Nose

4. Slide proximity switch all the way onto hex of magnet housing. Torque Allen set screws on hex of magnet housing to 25 lbs x inch, maximum.
5. Connect an ohm meter across the two yellow switch leads exiting the proximity switch housing.
6. The divider valve must cycle during proximity switch adjustment. To cycle it, either run the compressor or manually pump oil through the distribution block with a purge pump.
7. To adjust, slide proximity switch all the way onto hex of magnet housing. Tighten Allen set screws to 25 lbs x inch maximum. The ohm meter showing the switch changing state as the divider valve cycles indicates correct adjustment. If the switch state fails to change, slide proximity switch back on the hex of the magnet housing in 1/16" increments until it does.
8. Make all conduit and connections appropriate for area classification. Support conduit and fittings to avoid bending the magnet housing.
9. After proximity switch installation and before compressor start-up, purge all air from divider block lubrication system with a purge pump.

Figure: Typical Proximity Switch Installation

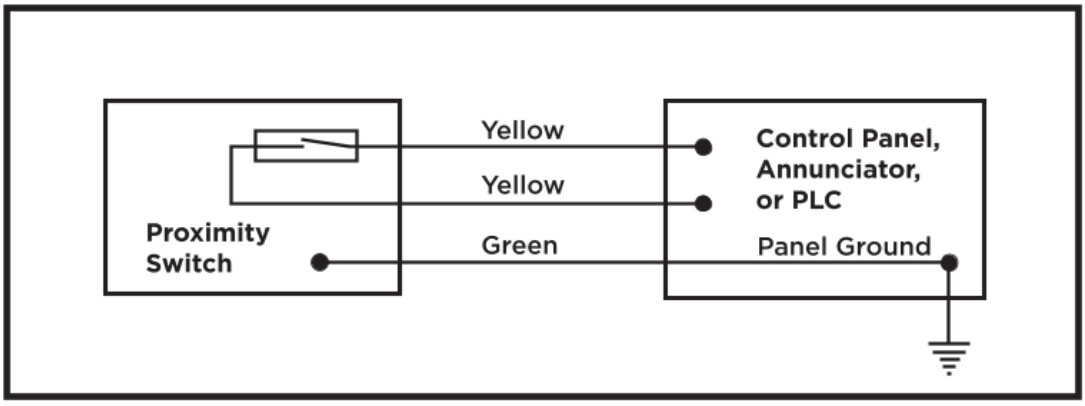


1	Divider Valve Piston	7	Switch Housing
2	Magnet	8	Wire Leads (yellow)
3	Magnet Housing	9	Ground Wire (green)
4	Allen Set Screw (2)	10	O-Ring
5	Spacer	11	Divider Valve
6	Spring	12	Piston End Plug

NOTE:

When installing multiple proximity switches, wire each to a separate alarm circuit of the control panel, annunciator, or PLC to simplify lubrication system and proximity switch troubleshooting. Always secure green ground wire to "earth ground" in the control panel. Do not ground to electrical conduit. Improper grounding can result in unreliable monitor operation.

Figure: A-18255 Proximity Switch Wiring Connections for Unit in Operation



Troubleshooting Proximity Switches

NOTE:

When installing multiple proximity switches, wire each to a separate circuit in the control panel, annunciator, or PLC to simplify lubrication system and proximity switch troubleshooting.

Problem:

- Switch does not cycle and control panel indicates lube no-flow. (also see Erratic Operation)
- ARV bypasses or rupture disk bursts and divider valve seizes after proximity switch installation.
- Erratic Operation.

Troubleshooting: Proximity Switches Lube No-Flow

Problem

Switch does not cycle and control panel indicates lube no-flow. (see also Erratic Operation)

Possible Cause

Improperly adjusted proximity switch.

Solution

1. Loosen set screws, slide proximity switch all the way onto hex of magnet housing and torque to 25 lbs x inch max. **(Do not over tighten).**
2. Either pump clean oil through lubrication system with a purge pump or run the compressor to cycle the divider valve.
3. If necessary, slide proximity switch back in 1/16“ increments until the switch opens/closes with each divider valve cycle.

Possible Cause

Broken spring or magnet in magnet housing.

Solution

1. Loosen set screws, remove proximity switch from magnet housing.
2. Remove magnet housing from divider valve.
3. Remove magnet, spring, and spacer and check for damage. Replace damaged components.
4. Re-install magnet housing on divider valve and proximity switch on magnet housing.
5. If necessary, adjust proximity switch and test switch functionality.
6. Purge air from system with purge pump.

Possible Cause

Bent magnet housing.

Solution

1. Loosen set screws, remove proximity switch from magnet housing.
2. Check for damaged or bent magnet housing.
3. Remove magnet assembly from divider valve.

4. Replace magnet housing, magnet, spring, and spacer.
5. Re-install new magnet housing on divider valve and proximity switch on magnet housing.
6. If necessary, adjust proximity switch and test switch functionality.
7. Purge air from system with purge pump.

Troubleshooting: Proximity Switches ARV Bypasses

Problem

ARV bypasses or rupture disk bursts and divider valve seizes after proximity switch installation.

Possible Cause

Wrong magnet housing installed on divider valve.

Solution

1. Loosen set screws and remove proximity switch from magnet housing.
2. Check for correct magnet housing for divider valve manufacturer.
3. Remove and replace with correct magnet housing.
4. Re-install proximity switch on new magnet housing.
5. If necessary adjust proximity switch and test switch functionality.
6. Purge air from system with purge pump.

Possible Cause

Air or debris in divider valve assembly.

Solution

1. Check system pressure to verify oil flows to divider valves.
2. If needed, install pressure gauge to monitor lubrication system operation:
   a. Loosen outlet plugs in front of valve blocks.
   b. Purge lubrication system with a purge pump until clean, clear, air-free oil flows from plugs.
   c. Loosen, but do not remove, each piston enclosure plug individually to purge air from behind piston. To ensure proper lube system operation, all tubing and components **MUST** be filled with oil and free of air before start-up.
   d. Tighten all divider valve plugs.
   e. Adjust proximity switch.

Troubleshooting: Proximity Switches Erratic Operation

Problem

Erratic Operation.

Possible Cause

Faulty wiring from proximity switch to control panel or air in system (see air in system).

Solution

1. Check electrical wiring and connections for damage or poor contact. See Figure: A-18255 Proximity Switch Wiring Connections for Unit in Operation.

Possible Cause

Faulty lube pump.

Solution

1. Check system pressure to verify oil flows to pump and divider valves. If needed, install pressure gauge to monitor lubrication system operation.
2. Check gauge to verify pump builds sufficient pressure to inject oil into cylinder. Do not remove tubing from check valve and pump oil to atmosphere to check oil flow into cylinder.
3. Replace pump.

Main Bearing Temperature Alarms and Shutdown

Main bearing thermocouples or RTD temperature sensors are optional for KBE frames.

Thermocouples - J (Iron-Constantan) or K (Chromel-Alumel)

A thermocouple is two dissimilar conductors joined together at one end to form a sensor that produces a thermoelectric voltage as an accurate function of temperature. Appropriate controls interpret the thermocouple voltage as temperature.

Resistance Temperature Devices (RTD)

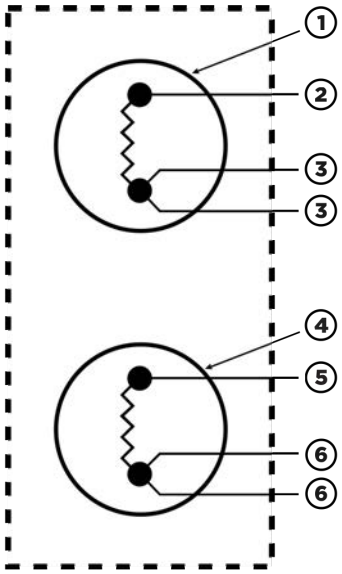
An RTD is a sensor that produces electrical resistance as an accurate function of temperature. Appropriate controls interpret the electrical resistance as temperature. A typical measurement technique sends a small constant current through the sensor and measures voltage across the sensor with a digital voltmeter to indicate resistance by a computer and wave-fitting equations.

For main or thrust bearing temperature sensing, Ariel supplies dual element, six-wire, 100 Ohm (at 0°C) RTD's. Dual element RTD's allow rewiring instead of sensor replacement if an element fails. Insulate unused wire ends from each other and conduit ground.

To simplify wiring, RTD's come with two green wires and one black for one element, and two red wires and one white for the other. See Figure: Typical Dual Element RTD Wiring Diagram.

Ariel supplied thermocouples, RTD's and the associated cables are classified as simple devices per IEC 60079-11 because they do not generate more than 1.5 V, 100 mA, or 25 mW, and have negligible inductance and capacitance.

Figure: Typical Dual Element RTD Wiring Diagram



1	Element 1	3	Red Wire	5	Black Wire
2	White Wire	4	Element 2	6	Green Wire

A Zone 1 environment may require the installation of proper, intrinsically safe or equivalent protection to fulfill electrical requirements.

Testing RTD's and Thermocouples

Ariel uses resistive temperature devices (RTD) or thermocouples (TC) to monitor main bearing and packing temperatures. An RTD has either three or six wires; a TC has two. Both types of devices are tested on the assembly floor after installation, but before the run-test.

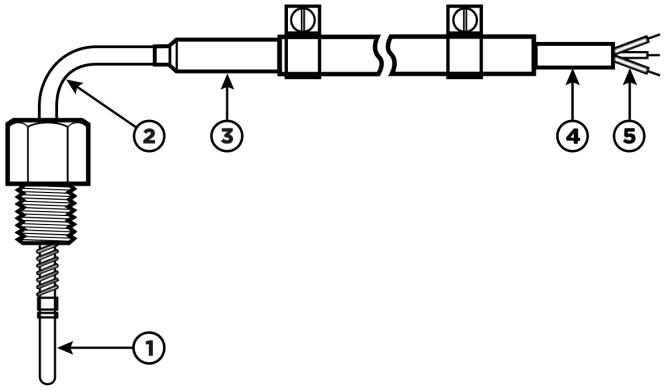
RTD Testing

Ariel uses two types of RTD: single and dual element. Single element consists of one white wire and two red wires. Dual element consists of one element of one white wire and two red wires, and another of one black wire and two green wires.

After RTD installation:

1. Set multimeter to the 100 Ohms resistance scale.
2. Touch one multimeter lead to the white element wire, and the other lead to one of the red element wires.
   - The multimeter should read between 100 Ohms (RTD at 32°F) to 112 Ohms (RTD at 90°F).
   - Test the other red wire with the white wire. Resistance should be similar for both red wires. Resistance outside the specified range indicates a damaged or broken element.
3. For dual element RTD's, perform step 2 with the black and two green wires.
4. Set the multimeter to check for continuity (⇥).
5. Isolate all element wires from the tube bend and transition piece. Check for continuity between each wire and these components.
   - The multimeter displays "1." to indicate the circuit is not continuous.
   - It displays a value between .001 and .010 and beeps to indicate a continuous circuit.
   - Only the white wire tested with either red wire and (for dual element RTD's) the black wire tested with either green wire should yield a continuous circuit.
   - A continuous circuit between any element wire and any component indicates a damaged RTD that needs replaced.

Figure: Typical RTD/Thermocouple



1	Temperature Probe	**4**	PTFE Insulated Foil Shield, FEP Jacket
2	Tube Bend	**5**	Element Wires (3 or 6 for RTD; 2 for TC)
3	Transition Piece		

Thermocouple Testing

Ariel uses two types of TC: Type J (-328 to 2192°F), which has a red and a white wire, and Type K (-328 to 2501°F), which has a red and a yellow wire. TC's generate a very low voltage that is difficult to measure with a multimeter. A digital thermometer attached to the TC measures the voltage and converts it to a temperature reading based on the thermocouple type.

After TC Installation:

1. Turn ON the digital thermometer.
2. Select the thermometer test lead with colored alligator clips that match the element wire colors of the TC being tested.

3. Plug the test lead into the top of the thermometer. Take care to align the blades to the appropriate hole.
4. Press the TYPE button until “J” or “K” appears on the screen as appropriate. If “°C” displays, press the C/F button to change the display to “°F”.
5. Attach the red alligator clip to the red element wire and the other alligator clip to the remaining element wire (either white or yellow).
   - If the connections are reversed, the thermometer displays inaccurate temperature (about 2°F less at room temperature and 135°F less at 150°F).
   - The temperature displayed should be about room temperature before TC installation. If displayed temperature is not room temperature or seems odd, verify test setup.
6. Detach alligator clips from TC.
7. Set the multimeter to check for continuity (⊣◀⊢).
8. Isolate both element wires from the tube bend and transition piece. Check for continuity between each wire and these components.
   - The multimeter displays “1.” to indicate the circuit is not continuous.
   - It displays a value between .001 and .010 and beeps to indicate a continuous circuit. Only the two TC element wires should yield a continuous circuit.
   - A continuous circuit between an element wire and any component indicates a damaged TC that needs replaced.

Section 3 - Maintenance

Ariel compressors, like all industrial equipment, require maintenance. The severity of compressor service directly influences the frequency and amount of maintenance needed. Below are recommended intervals for inspections and replacements to help determine appropriate intervals for a given compressor application. Careful documentation of inspection results is critical to establish whether recommended intervals are adequate or require more or less frequency.

Continuous Duty: The compressor operates continuously, 24 hours a day, shutting down only for short durations, usually for maintenance and unexpected shut-downs.

Short-Term Intermittent Duty: The compressor operates for short durations, starting and stopping perhaps several times per day, such as in many CNG filling station applications that require operation based on consumer demand.

Long-Term Intermittent Duty: The compressor operates continuously for extended durations, with extended shut-down periods, such as in some storage/withdrawal applications and fuel gas booster applications.

Many factors affect bearing life including the number of starts, starting conditions, and run hours. Consider site-specific conditions when adjusting maintenance intervals.

As part of your maintenance program, Ariel recommends:

- Consistent adherence to safety policies, procedures, and equipment warning labels.
- Daily operational checks.
- Routine trending and review of operational parameters.
- Routine oil analysis and trending.
- Detailed records of all maintenance.

To avoid contamination, keep all covers in place where access is not required. Use paper towels during internal maintenance.

WARNING:

To decrease the risk of injury or death, only trained and approved personnel can do maintenance work. Approved personnel must read and know the applicable Ariel Technical Manual and Packager-supplied information.

Before you start maintenance:

- Always use personal protection equipment (PPE) and obey your company policies for safety and operation.
- Do not tighten or loosen fasteners or fittings while the unit is in operation or is pressurized.
- Make sure the driver and compressor cannot turn during maintenance.
- Obey your company and packager instructions to isolate and vent the system.
- Make sure there is no remaining pressure in the cylinders. If equipped with “Kiene” valves on the indicator ports, open them to release residual pressure after blowdown.

  WARNING: RELEASED GAS MAY BE POISONOUS AND/OR FLAMMABLE.

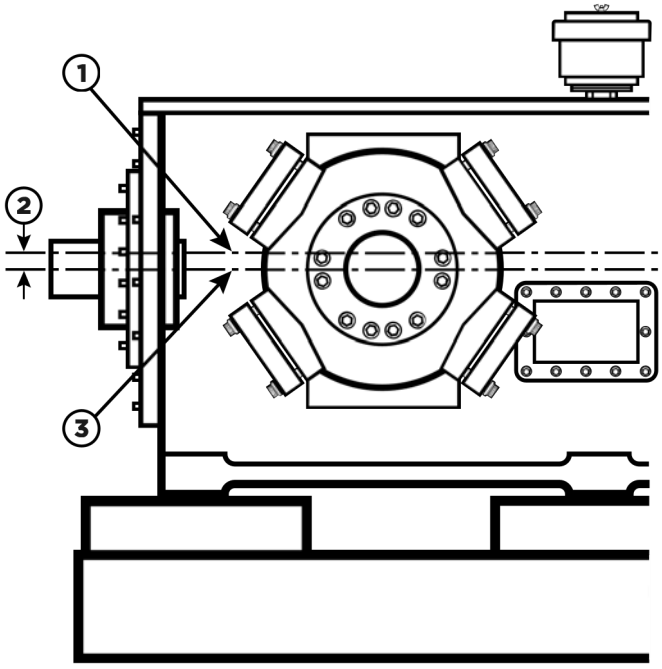
Before operation, purge the system. A pressurized hydrocarbon and air mixture can explode and cause injury or death.

Initial Maintenance

Comply with Ariel Packager Standards and the compressor Start Up Check List. Adhere to all items before and after start-up.

For cold alignment, account for the difference in thermal growth height between the compressor and driver. Table: Thermal Growth of KBE Compressors lists compressor centerline height change based on 6.5 x 10^{-6}/°F (11.7 x 10^{-6}/°C).

Figure: Thermal Growth



1 HOT Centerline
2 Thermal Growth
3 COLD Centerline

Table: Thermal Growth of KBE Compressors

Initial Frame Temp. °F (°C)	Thermal Growth Inch (mm)
0 (-18)	0.019 (0.48)
10 (-12)	0.018 (0.45)
20 (-7)	0.017 (0.42)
30 (-1)	0.015 (0.39)
40 (4)	0.014 (0.37)
50 (10)	0.013 (0.34)
60 (16)	0.012 (0.31)
70 (21)	0.011 (0.28)
80 (27)	0.010 (0.25)
90 (32)	0.009 (0.23)
100 (38)	0.008 (0.20)
110 (43)	0.007 (0.17)
120 (49)	0.006 (0.14)
130 (54)	0.004 (0.12)

NOTE:

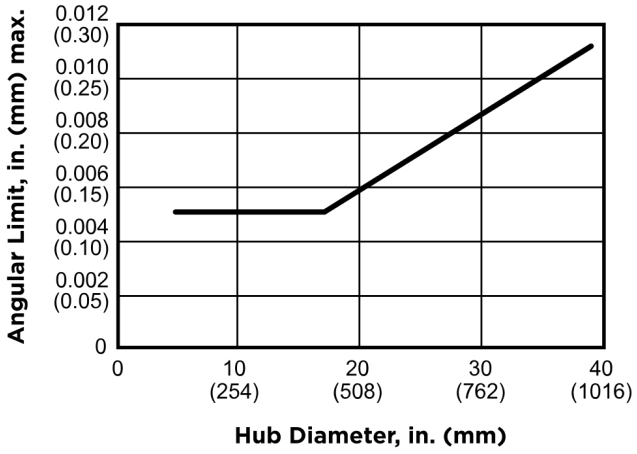
Thermal growth numbers were calculated using 170°F (77°C) as the frame operating temperature. Initial frame temperature is the temperature of the frame during cold alignment. Obtain driver thermal growth predictions from the driver manufacturer.

Center the coupling between the driver and compressor so it does not thrust or force the crankshaft against either thrust face.

To ensure parallel and concentric drive train alignment, position connected equipment so misalignment is as close to zero as possible. Limits for Total Indicator Reading (TIR):

- Maximum rim (outside diameter), (parallel) misalignment is 0.005 inch (0.13 mm).
- Face (angular) misalignment must be within 0.005 inch (0.13 mm) for hub diameters up to 17 inches (430 mm). For hubs greater than 17 inches (430 mm), use the following formula: Hub O.D. x 0.0003 = max angular misalignment. See Figure: Angular Coupling-Hub Face Alignment Limits.

Figure: Angular Coupling-Hub Face Alignment Limits



For laser alignment tools:

- Maximum parallel misalignment is 0.0025 inch (0.064 mm).
- Maximum angular misalignment is 0.0003 inch/1 inch (0.03mm/100mm).

> **NOTE:**
>
> If laser is adjusted to display TIR, use TIR limits.

After running a new, relocated, reconfigured, or overhauled compressor for 24 hours, shut down, vent the gas system, and perform the following maintenance:

1. See Appendix A for correct torques.
2. Repeat torque check after 750 hours. Recheck fasteners found loose in any of these intervals after an additional 750 hours. If loosening continues, contact your package supplier immediately.

Daily Maintenance

1. Log and trend the following:
   - Operating RPM, gas pressure and temperatures - determine if the unit is operating within design parameters and expectations.

     > **NOTE:**
     >
     > Verify high and low pressure shutdowns are set as close as practical to normal operating conditions. Set points must protect the machine from exceeding compressor limits.

   - Bearing temperatures - if the unit is equipped with main bearing temperature sensors.
   - Compressor frame oil pressure - at operating temperature (190°F (88°C) max. inlet oil temperature), it should be 50 to 60 psig (3.5 to 4.2 barg) at the filter gauges. If pressure falls below 50 psig (3.5 barg), shut down the compressor then determine and correct the cause.
   - Compressor frame inlet oil temperature.
   - Cylinder valve cap temperatures.
   - Compressor frame oil filter differential pressure - differential pressure exceeding the filter change value indicates a need for a filter change. See filter information plate on top cover or "Oil Filter" in Frame Oil System Components for procedure.
2. Check compressor frame oil level. It should be about mid-level in the sight glass and free of foam when running. If not, determine and correct the cause. **Do not overfill.** Check oil makeup tank for sufficient oil supply. For dry sump frames, check the package sump oil level. Do not add oil to the crankcase through the breather hole while the unit runs. This causes oil foaming and unnecessary no-flow shutdowns in the force feed lubrication system.
3. Check force feed lubricator box oil level. It should be full to the overflow line.
4. Log and trend piston rod packing vent/drain temperature and check distance piece and VVCP vents for leakage. If the piston rod packing vent/drain temperature is trending high, then verify leak rate before performing maintenance.
5. If applicable, check suction valve unloader actuator vents for leakage.
6. If applicable, check head end clearance pocket vents for leakage.
7. Verify all cylinder high discharge gas temperature shutdowns are set to within 10% or as close as practical above the normal operating discharge temperature. Do not exceed the maximum discharge temperature shutdown setting for the application.
8. Check lubricator block cycle time. See lubricator box data plate for correct cycle time. Changes in gas composition may require a review of cycle time. Check lube sheet for units not running at the speed on the data tag. If applicable, verify auto relief valve indicator does not show sign of over-pressurization.
9. Check for gas, oil, and coolant leaks.

> **CAUTION:**
> Do not attempt to repair leaks while the unit is operating or pressurized.

10. Check for unusual noises or vibrations.
11. See packager documentation for additional recommended checks, i.e. scrubber liquid levels, dump valve operation, cooler louver positions, etc.

Recommended Maintenance Intervals for Continuous Duty Operation

Ariel compressors, like all industrial equipment, require maintenance. The severity of compressor service directly influences the frequency and amount of maintenance needed. What follows are the recommended intervals for inspections and replacements for **continuous duty** operation. This will help you determine the appropriate intervals for a given compressor application.

> **NOTE:**
> For intermittent duty service, see "Appendix I".

> **NOTE:**
> Careful documentation of inspection results is critical to establish whether recommended intervals are adequate or require more or less frequency.

Monthly Maintenance

1. Perform all Daily maintenance.
2. Verify safety shutdown functionality.
3. Sample compressor frame oil and send it to a reputable lubricant lab for analysis. See "Oil Maintenance" for a list of what an oil analysis should provide. If analysis results indicate increasing levels of lead, tin, or copper particles in the oil, shut down unit. Remove frame top cover and crosshead guide side covers. Visually inspect for debris. Do not disassemble further without good reason. If debris indicates, replace affected parts, then change oil, oil filter, and clean the strainer with a suitable solvent.

   > **NOTE:**
   > Operate the unit to bring oil temperature to a minimum operating temperature of 150°F (66°C) to drive off any entrained water.

4. Check and log cylinder clearance devices in use and their settings.

Six-Month (4,000-Hour) Maintenance

1. Perform all Daily and Monthly maintenance.
2. Shut down unit and allow sufficient time for components to cool.
3. Inspect crankcase breather filter and clean with suitable solvent.
4. Drain and replace force feed lubricator box oil.
5. Where applicable, replace force feed lubricator inlet filter in independent supply system.

6. Change compressor oil filter. See top cover filter information plate or “Oil Filter” in Frame Oil System Components for procedure. Do not exceed 12 month installed period for filter element.

   NOTE:

   For replaceable element style filters, drain the canister completely before removing the element. Inspect elements for cleanliness and damage. Do not install dirty or damaged elements.

7. Change compressor oil. Extremely dirty environments, oil supplier recommendations, or oil analysis results may dictate a different oil change interval. Follow these steps:
   a. Drain oil from frame, associated piping, and oil cooler.
   b. Clean oil strainer with a suitable solvent. In dry sump applications, the strainer is located at the oil reservoir.
   c. Remove frame top cover and crosshead guide side covers. Visually inspect for debris. Do not disassemble further without good reason. If debris indicates, replace affected parts, then change the oil filter and clean the strainer with a suitable solvent.
   d. Refill compressor frame with fresh, clean oil.
8. Grease VVCP stem threads at grease fitting, with 2 to 3 pumps of cylinder lube compatible grease using a standard manual grease pump. Turn VVCP adjustment handle all the way in and all the way out to lubricate the threads. Measure or count turns to return the handle to its original position.
9. Check for adequate cylinder lubrication. See “Cylinder Lubrication Paper Test” for procedure.
10. Re-tighten hold down fasteners to proper torque (compressor frame feet and crosshead guide/distance piece support feet). Inspect for frame twist or bending to verify main bearing bore alignment. See Appendix H for flatness and soft foot requirements.
11. Perform a coupling alignment (see "Initial Maintenance").
12. Remove and inspect inlet screens on each stage; clean if clogged with debris or replace if damaged. Clean inlet screens regularly to avoid increased pressure that may collapse them. More debris in the screen may dictate more frequent inspections. Inlet screens may be removed if they catch little or no debris and the gas stream is clean (optional).

One-Year (8,000-Hour) Maintenance

1. Perform all Daily, Monthly, and Six-Month maintenance.
2. Open force feed lubricator box and visually inspect pump followers, cams, and gears for wear.
3. Pressure test distribution blocks. See “Divider Valve Bypass Pressure Test” for procedure.
4. Measure, log, and trend the following:

   - Main bearing, connecting rod bearing, and crankshaft jack and thrust clearances. See "Main Bearing Vertical Jack Clearance Measurement", "Connecting Rod Bearing Vertical Jack Clearance Measurement", and "Crankshaft Thrust (End) Clearance Measurement" in Section 4.

     NOTE:

     Clearance trends along with oil analysis and crankcase visual inspection can indicate the need for bearing replacement. Do not disassemble to check clearances. Disassemble only if clearance checks indicate excessive clearance. Contact Ariel for original assembly clearances.

   - Crosshead to guide clearances. See “Crosshead Installation” in Section 4.
   - Piston rod runout. See “Piston Rod Runout” in Section 4.

If any of the above items are outside limits specified in Appendix B, find the cause and correct it. It may require part replacement.

5. Remove valves and valve gaskets:

> **NOTE:**
> If using CP/CPs/XP valves, valve maintenance may be extended to the 2 year (16,000 hour) interval.

   a. Visually inspect valve pockets for damage. Verify all old valve seat gaskets are removed.
   b. Have a qualified valve repair shop disassemble, visually inspect, and refurbish the valves where needed. Provide the valve repair shop an Ariel torque chart and valve service guide.
   c. Visually inspect cylinder gas passages and clean them of all debris.

6. If applicable, remove, disassemble and visually inspect the variable volume clearance pockets (VVCP). Check for damage or wear of all sealing components and surfaces, and repair or replace if necessary.
7. If applicable, visually inspect suction valve unloader actuator stems for damage or wear. Visually inspect stem seals for damage or wear and confirm that the actuator moves freely.
8. If applicable, visually inspect pneumatic clearance pockets for damage or wear (seating surface and stems/ stem seals). Confirm that actuator moves freely.
9. Inspect cylinder bores for damage or wear. Replace the cylinder body or restore the bore if any of the following conditions exist:
   - Bore surface blemishes or gouges.
   - Bore out of round more than 0.001 inch per inch of bore diameter (0.001 mm/mm) or tapered.
   - Main bore is worn to counterbore diameter or larger.
10. Inspect piston rings and wearband:
   a. Measure and log piston ring condition, end gap, side clearance, and piston groove width.
   b. Replace rings that are damaged or outside limits listed in Appendix B.
   c. When replacing rings, re-measure and log ring side clearance to check for groove wear.
   d. Measure and log radial projection of wear band.
11. Inspect piston rods for damage and excessive wear. Replace rod if any of these conditions exist (see "Piston and Rod Removal" in Section 4):
   - Gouges or scratches on the rod.
   - Under size more than 0.005 inch (0.13 mm).
   - In the packing travel length, out of round more than 0.001 inch (0.03 mm).
   - Anywhere in the packing travel area, tapered more than 0.002 inch (0.05 mm).
12. Rebuild piston rod pressure packing cases. See "Piston Rod Packing" in Section 4 for procedure.
13. Re-install valves, retainers, and valve caps using new valve seat gaskets and valve cap O-rings/seals. See "Valve Installation" in Section 4. Use proper installation techniques and torque procedures for valve caps.
14. Check and re-calibrate all required instrumentation.
15. Clean crankcase breather filter with suitable solvent.
16. Check and, if needed, adjust drive chains. See "Chain Adjustment" in Section 4 for procedure.
17. If the compressor is equipped with crankcase over-pressure relief valves, visually inspect and exercise valves to manufacturer recommendations.
18. Check fastener torques of gas nozzle flanges, head end head or unloader, valve caps, piston rod packings, crosshead pin thru- bolts, crosshead guide to frame, frame hold down, crosshead guide to cylinder, cylinder

mounting flange to forged steel cylinder, distance piece to cylinder, distance piece to crosshead guide, and tandem cylinder to cylinder.

Two-Year (16,000-Hour) Maintenance

1. Perform all Daily, Monthly, Six-Month, and One-Year maintenance.
2. If using CP/CPs/XP valves, remove valves and valve gaskets:
   a. Visually inspect valve pockets for damage. Verify all old valve seat gaskets are removed.
   b. Have a qualified valve repair shop disassemble, visually inspect, and refurbish the valves where needed. Provide the valve repair shop an Ariel torque chart and valve service guide.
   c. Visually inspect cylinder gas passages and clean them of all debris.
3. Rebuild oil wiper cases.
4. If applicable, use new piston and stem seals to rebuild actuators on suction valve unloaders and fixed volume pneumatic pockets. Replace piston stem assemblies where stem is damaged or worn.
5. Check auxiliary end chain drive for undercut sprocket teeth and chain for excessive stretching. Replace as required.

Four-Year (32,000-Hour) Maintenance

1. Perform all Daily, Monthly, Six-Month, One-Year, and Two-Year maintenance.
2. Remove crosshead pins.
   a. Measure and log crosshead pin to crosshead pin bore and connecting rod bushing bore clearances.
   b. Check crosshead pin end caps and thru-bolt for wear.
   c. Replace if needed.
3. Check for ring groove wear in pistons.

Six-Year (48,000-Hour) Maintenance

1. Perform all Daily, Monthly, Six-Month, One-Year and Two-Year maintenance.
2. Replace lubricator distribution blocks.
3. Replace crosshead bushings and connecting rod bushings. See “Crossheads” and “Connecting Rods” in Section 4.
4. Replace connecting rod bearings. See “Connecting Rods” in Section 4.
5. Replace main bearings. See “Crankshaft” in Section 4.
6. Replace auxiliary end torsional damper, if applicable. See assembly drawing in compressor build book for procedure.
7. Replace idler sprocket assembly in the auxiliary end chain drive tensioner(s).

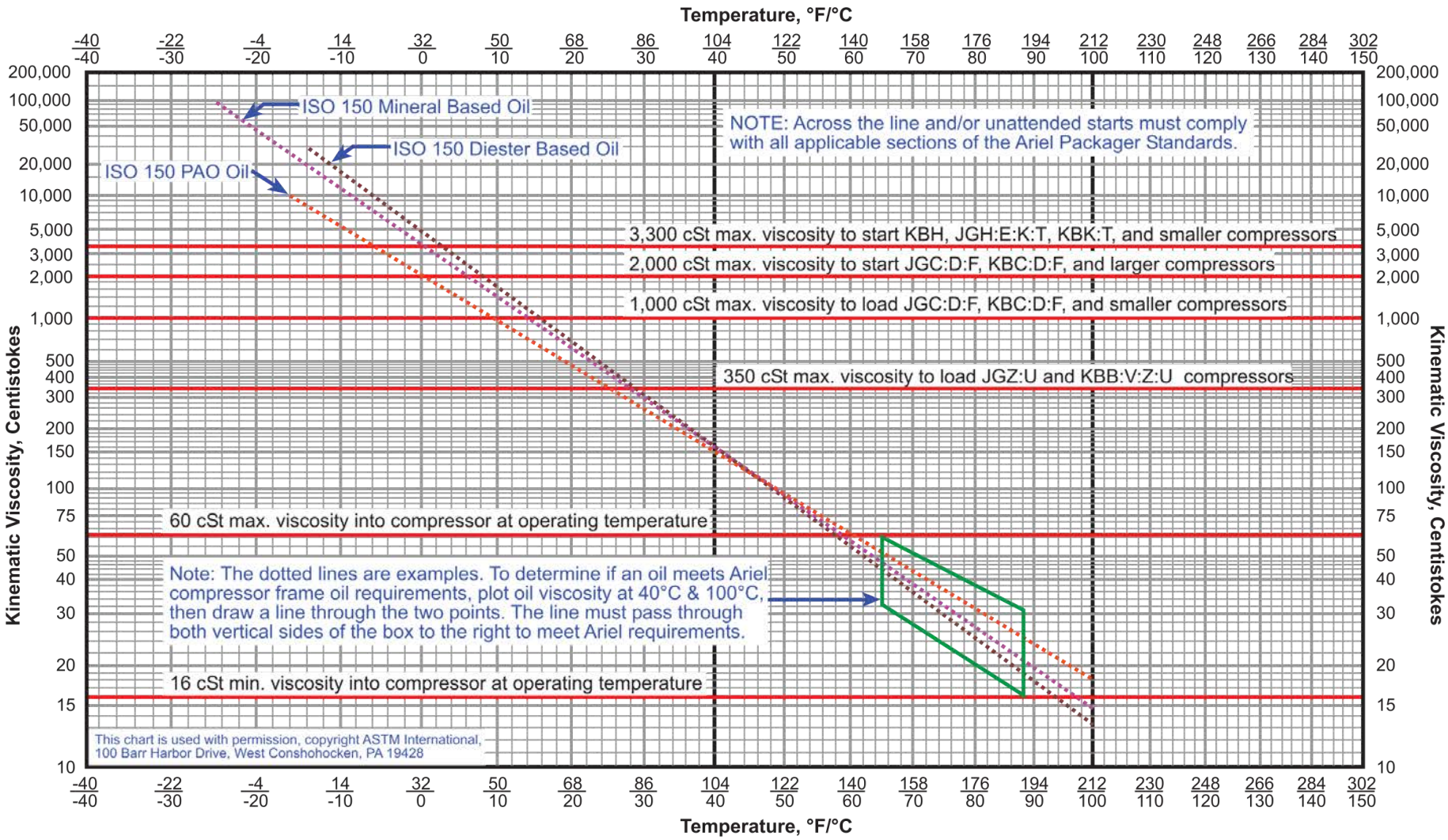
Frame Oil

There are several oil specifications, maintenance procedures, and operating conditions which affect oil performance. For optimal performance, ensure the oil meets the specifications below and is suitable for the given application, and diligently complete maintenance procedures.

Viscosity

The minimum allowable viscosity of the oil entering the frame during normal operation is 16 cSt. Typically, this is the viscosity of ISO 150 grade oil at about 190°F (88°C). The maximum viscosity at this temperature is 60 cSt. For KBE frames, the maximum viscosity to start the compressor is 3,300 cSt and the maximum viscosity to load the compressor is 1,000 cSt. See Figure: Viscosity vs. Temperature Graph of Different Lubricants.

Figure: Viscosity vs. Temperature Graph of Different Lubricants



Oil Pressure

The factory sets normal pressure on the downstream side of the oil filter at 60 psig (4.1 barg) for compressors tested mechanically complete (inspector tag displayed). If factory tested as mechanically incomplete (no inspector tag), the packager sets normal oil pressure at initial start-up to 60 psig (4.1 barg) at the lower of the frame or cylinder rated speed, or driver speed at normal operating temperature. Adjust the oil pressure after the filter to 60 psig at minimum operating speed and normal operating temperature.

Use the pump relief valve to adjust pressure into the compressor.If oil pressure into the compressor at minimum operating speed and normal operating temperature does not read about 60 psig (4.1 barg), see "Oil Pump & Regulating Valve" for pressure adjustment procedure. The compressor requires a 45 psig (3.1 barg) low oil pressure shutdown for protection. Oil pressure must be measured downstream of the oil filter at the auxiliary end for proper protection.

NOTE:

If oil pressure drops below 50 PSIG (3.4 barg) when crankshaft speed equals or exceeds minimum rated operating speed, find the cause and correct it.

Oil Temperature

Maintain frame inlet oil temperature as close to 170°F (77°C) as possible. Minimum lube oil operating temperature is 150°F (66°C) to drive off water vapor. Maximum allowable oil temperature into the compressor frame is 190°F (88°C). Higher temperatures increase the oxidation rate of mineral oil. Every 18°F (10°C) within the operating range doubles the oxidation rate of mineral oil.

Oil Sampling Valve

Ariel equips KBE:K:T:C:D:F frames with an oil sampling valve.

Oil System Cleanliness

Clean the compressor frame oil piping system and components of all foreign matter such as sand, rust, mill scale, metal chips, weld spatter, grease, and paint. Use proper cleaning procedures with proper cleaners, acids, and/or mechanical cleaning to meet cleanliness requirements. Ariel recommends flushing all oil-piping systems with an electric or pneumatic driven pump and filtered, clean production oil. Ariel thoroughly cleans all compressor frame cavities prior to assembly and test runs compressors with a filtered closed loop lube system.

Table: Maximum Angle from a Horizontal Plane Allowed in Transient Motion without Dry Sump

Frame[a]	1 or 2 Throw	4 Throw	6 Throw
KBE	3.5°	2°	N/A

NOTE:

Do not disturb lube oil piping downstream of the installed oil filter without approval from Ariel Technical Service. Contaminants that enter this piping or open ports flush into the bearings causing catastrophic damage. If contamination is suspected in the supply piping downstream of the oil filter or in the oil gallery, remove the supply piping and thoroughly clean both the piping and gallery. Before performing this work, contact Ariel Technical Service for best practices and procedures. To remove or alter piping, cover the inlets to the oil gallery, the ends of the piping, and the filter outlet so no contaminants enter. Before reinstallation, chemical and mechanical cleaning is required. Flush the pipe in accordance with Ariel cleanliness requirements (see Table: Oil Flush Cleanliness Requirements).

Prior to start-up, flush all compressors installed with an electric or pneumatic powered pre-lube pump and less than 50 feet (15 m) of oil piping as outlined below. Include cooler oil passages in the flushing loop. While oil systems for compressors without an electric or pneumatic powered pre-lube pump and less than 50 feet (15 m) of oil piping must be clean, oil flushing is desirable, but not required.

For all compressors with oil piping systems greater than 50 feet (15 m), cleaning and flushing must result in a cleanliness level to ISO-4406, Grade 16/14/11 and/or NAS-1638, Class 5 (see Table: Oil Flush Cleanliness Requirements), prior to start-up.

1. Prior to assembling lube oil piping, remove scale, weld slag, rust and any other matter that could contaminate lube oil. Confirm:
   - Complete and closed lube oil system.
   - Crankcase filled to the correct level with appropriate oil.
   - Proper and correctly installed lube oil filters.
   - Operational and viewable oil pressure transducer or gauge, oil filter differential-pressure transducers or gauges, and oil temperature RTD or indicator.
2. Start pre-lube pump. Record oil pressure, oil filter differential-pressure, and oil temperature. Minimum oil pressure is 30 psig (2.1 barg) for effective flushing. Do not exceed 90 psig (6.2 barg).
3. Flush continuously for one hour. Oil filter differential pressure must not increase more than 10% of measured oil pressure into the filter. Record the oil pressure, oil filter differential pressure, and oil temperature every 15 minutes. Oil temperature increases of more than 10°F (5.5°C) during an hour of flushing invalidate the system cleanliness test, due to oil viscosity change.

Table: Oil Flush Cleanliness Requirements

ISO-4406 GRADE 13/10/9		
Grade Requirement	**Particle Size µm/ ml Oil Sample**	**Particles Allowed**
/13	Greater than 4	40 to 80
/10	Greater than 6	5 to 10
/9	Greater than 14	2.5 to 5

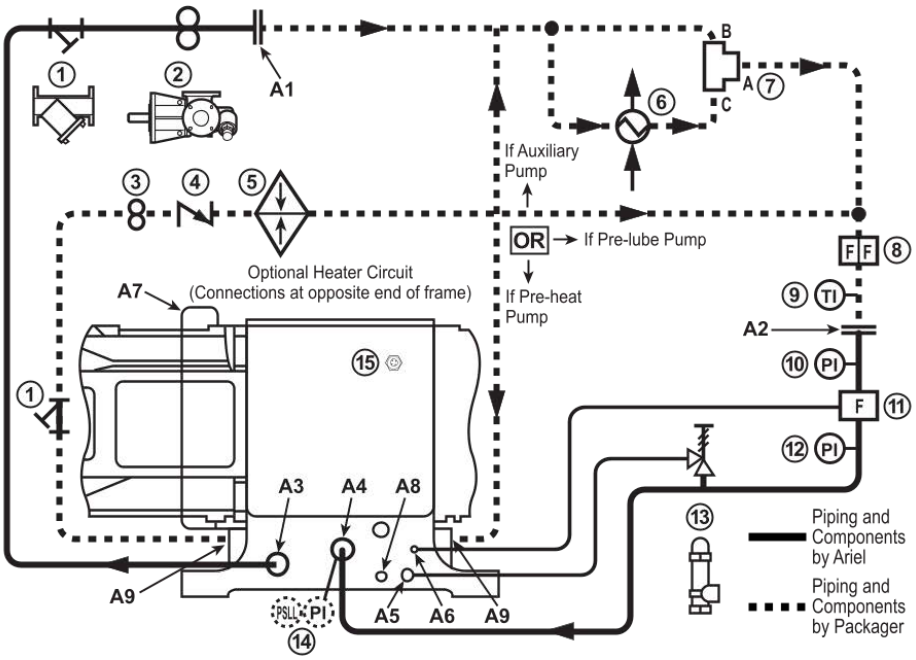
NAS-1638 GRADE 5	
Particle Size Range µm/ 100ml Oil Sample	**Grade 5 Maximum Particles Allowed**
5 to 15	8,000
15 to 25	1,424
25 to 50	253
50 to 100	45
Over 100	8

4. After one hour of pre-lube flushing, if differential pressure or temperature increases exceed the limits above, continue flushing. If the lube oil filter differential pressure exceeds change filter limits, stop the pre-lube pump and change the oil filter. To ensure system cleanliness, re-set time and continue flushing until the compressor achieves a continuous hour of flushing within differential pressure and temperature increase limits.

See ISO-4406 "International Standard - Hydraulic fluid power - Fluids - Method for Coding Level of Contamination by Solid Particles" and/or NAS-1638 "National Aerospace Standard, Aerospace Industries Association of America, Inc. - Cleanliness Requirements for Parts Used in Hydraulic Systems" for complete information. Use a competent oil lab for sample testing. To flush the entire oil system see “Flushing Procedure”.

Frame Oil System Components

Figure: Standard Frame Lube Oil Schematic



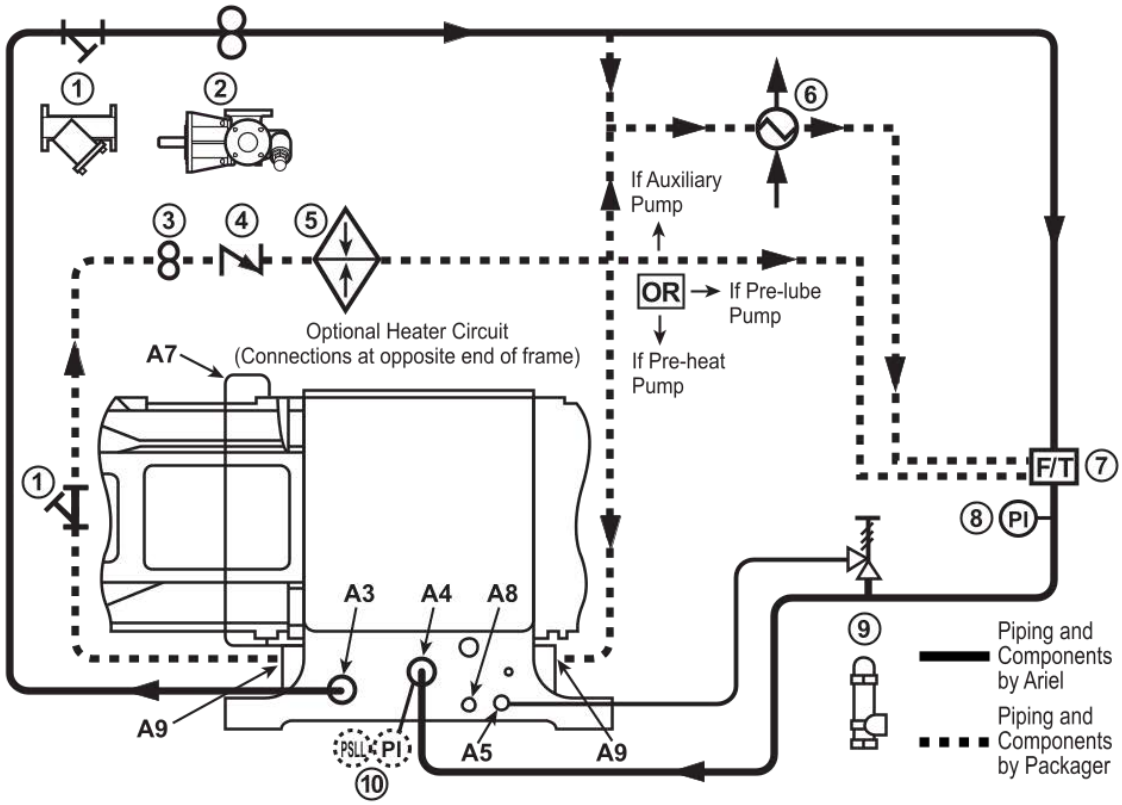
Oil Connections (see Ariel outline drawing for details)

- **A1** Packager connection from oil pump
- **A2** Packager connection to oil filter
- **A3** Oil connection from compressor crankcase (oil sump)
- **A4** Lube oil compressor inlet connection to gallery tube -Oil flows to crankshaft main bearings, connecting rod bearings, crosshead pins, and bushings.
- **A5** Pressure regulating valve return connection to oil sump (when applicable)
- **A6** Filter vent return connection to oil sump (when applicable on some models)
- **A7** Oil tubing connections from frame gallery tube to top and bottom of crosshead guides to lubricate crossheads.
- **A8** Compressor crankcase oil drain (oil sump drain)
- **A9** Pre-lube/recirculation/heater connections

System Components - For specific information on each system component, see applicable section below.

- **1** Y-Strainer
- **2** Compressor driven oil pump (with relief valve for pressure regulation, or in models with a separate regulating valve (7), for relief)
- **3** Pre-lube oil pump - required (shown with oil heating circuit, when applicable), or auxiliary pump.
- **4** Check valve (minimize cracking pressure, not to exceed 5 psi)
- **5** Heater (when applicable).
- **6** Oil cooler - required
- **7** Thermostatic control valve, 170°F (77°C) nominal rating - required
- **8** Optional duplex oil filter
- **9** Temperature indicator
- **10** Pressure indicator (dirty side)
- **11** Oil filter
- **12** Pressure indicator (clean side)
- **14** Pressure indicator/shutdown connection
- **15** Lube Oil Sightglass

Figure: Frame Lube Oil Schematic with Mounted Integral Thermostatic Valve



Oil Connections (see Ariel outline drawing for details)

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- **4** Check valve (minimize cracking pressure, not to exceed 5 psi)
- **5** Heater (when applicable).
- **6** Oil cooler - required
- **7** Oil filter and mounted integral thermostatic valve - Includes connections for temperature indicator and pressure indicator (dirty side).
- **8** Pressure indicator (clean side)
- **9** Pressure regulating valve with overflow return to oil sump (Z:U:B:V only)
- **10** Pressure indicator/shutdown connection

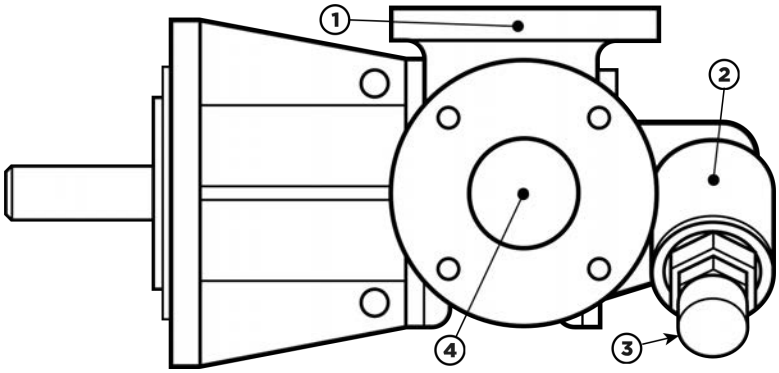
Oil Strainer

An oil strainer installed upstream of the pump prevents debris from entering the pump and damaging it. Ariel supplies a 30 mesh (595 microns) strainer for KBE compressors. It is located on the auxiliary end of the crankcase below oil level. Remove the strainer basket and wash it in an appropriate solvent whenever lubricating oil is changed.

Oil Pump & Regulating Valve

The oil pump constantly supplies oil to all journal bearings, bushings, and crosshead sliding surfaces. The crankshaft drives it by a chain and sprocket to provide adequate oil flow to bearings when the compressor operates at the minimum speed rating (typically half of maximum rated speed).

Figure: Typical KBE Lube Oil Pump



1. Discharge
2. Relief Valve
3. Dust Cap
4. Suction
   Smaller compressor models have threaded suction/discharge connections.

KBE compressors maintain oil pressure with a spring-loaded relief valve within the pump (see Figure: Typical Lube Oil Pump). With the unit running at minimum operating speed and the frame lube oil at operating temperature, 170°F (77°C), remove the dust cap to expose the relief valve adjustment screw and set it so the pressure gauge on the clean side of the oil filter reads 60 psig (4.1 barg).

When installing a new oil pump on a compressor or after maintenance on an existing oil pump, Ariel recommends priming the pump (wetting the gears with oil) before start-up to ensure proper operation. Failure to wet pump gears before operation could result in damage to internal seals. Upon compressor start-up after oil pump replacement, oil pressure should immediately reach 50-60 psi. If oil pressure fails to reach 50-60 psi immediately, shut down the compressor and correct the problem. Dry oil pump gears sometimes contribute to the problem.

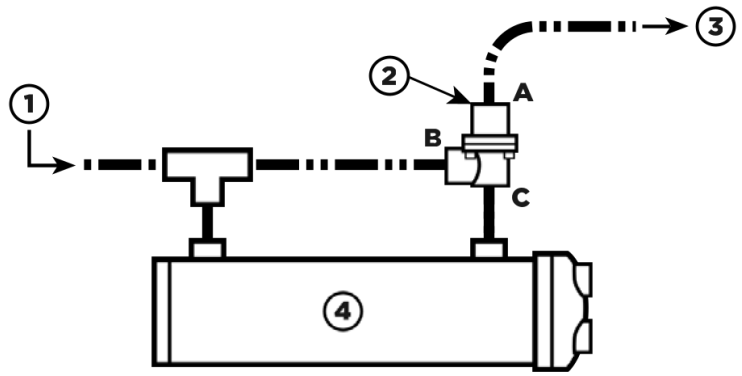
Oil Cooler

An oil cooler is required to remove heat from the frame lube oil. When sizing an oil cooler, consider temperature and flow rate of both cooling medium and lube oil. Insufficient cooling water flow rate is the primary cause of high oil temperatures. Ariel recommends mounting oil coolers below compressor frame oil level and as close to the compressor as possible with piping of adequate size to minimize pressure drop of both lube oil and cooling medium. If mounted above frame oil level, check valves may be necessary to prevent oil draining back to the frame. Configure oil cooler circuit so air can be manually vented at high points in the system during commissioning and after maintenance. Consider a continuous high point vent back to crankcase where possible.

Oil Temperature Control Valve

The lube oil system requires a thermostatic valve to control compressor oil temperature. A thermostatic valve is a three-way valve with a temperature sensitive element. As the oil heats, the sensing element opens the third port in the valve.

Figure: Thermostatic Valve in Mixing Mode



Thermostatic control valve configuration may vary from this schematic depending on valve size. Valve connections A-B-C are marked on the valve.

1. Lube oil from main oil pump
2. Thermostatic Control Valve
3. Lube oil to main oil filter
4. Lube Oil Cooler

Ariel recommends a thermostatic valve with a 170°F (77°C) element. Install the valve in mixing mode to more directly control oil temperature into the frame (see Figure: Thermostatic Valve in Mixing Mode) when a mounted integral thermostatic valve is not applied. For proper thermostatic control valve operation, differential pressure between the hot oil supply line and the cooled oil return line must not exceed 10 psid (0.7 bard). Systems utilizing duplex filter housings require a thermostatic valve upstream of the duplex filter. KBE:K:T:C:D:F compressors have a thermostatic valve integrated into the lube oil system as standard.

Oil Filter

All compressor frames require oil filters to remove particle contamination that can damage equipment and oil. Contaminants that damage equipment include wear particles from equipment, airborne particles such as dust or sand, and particulates in new oil. Contaminants that damage oil include oxidized oil components and air bubbles.

- Ariel filters are not designed for reverse flow often caused by pumping oil out of the compressor through the filter. This can invert and tear the filter media, sending dirty oil to crankshaft bearings.

KBE compressors ship with simplex, spin-on, non-bypassing, resin-impregnated filters as standard. Spin-on filters carry a 5 micron nominal and 17 micron absolute rating. The Beta ratings are ß5 = 2 and ß17 = 75. Many spin-on filters fit an Ariel compressor, but very few meet filtration ratings of Ariel filters. **Use only Ariel OEM filters.**

Pressure gauges monitor pressure drop across the filter. High differential pressure indicates a plugged filter. Ariel recommends lube oil filter replacement every 6 months of operation or when oil filter differential pressure at normal operating temperature reaches 10 psid (0.7 bard) for spin-on filters, whichever comes first. On start-up, differential pressure may exceed the filter replacement limit until the oil reaches operating temperature.

Oil Maintenance

Proper oil maintenance is vital to the efficient operation of the compressor. The severity of compressor service affects the frequency of oil and filter changes.

Oil Filter Replacement

Replace oil filters or elements with only genuine Ariel parts. See parts list or name plate for part numbers.

CAUTION:

Before oil filter replacement, use the frame-mounted pressure gauges to confirm system depressurization. Be careful - hot oil may cause burns! Wear proper personal protection equipment.

Oil Sampling

A consistent oil analysis program is the most effective way to determine when to change frame oil on the basis of need rather than a scheduled interval. Depending on service, oil analysis can significantly extend oil change intervals.

The packager installs an oil sampling valve on most frames between the oil pump and filter at an easily accessible location. Ariel equips KBE frames with an oil sampling valve. Minimize dirt or debris that can collect around it.

Collect and analyze oil samples to verify suitability for continued service. Oil analysis should include:

- Viscosity testing at 104°F (40°C) and 212°F (100°C). This verifies that oil has not mixed with cylinder oils or process gas.
- Particle counting to the latest version of ISO 4406.
- Spectroscopy to determine wear metals, contaminants, and additives.
- FTIR (Fourier Transform Infrared Spectroscopy) to check for oxidation, water or coolant contamination, and additive depletion. This is more important if the sump oil is not consumed by the force feed system.

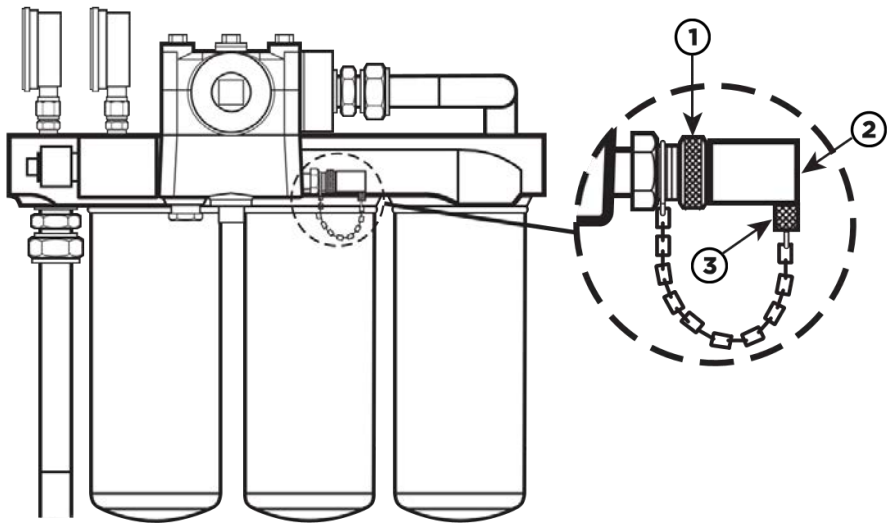
Collect oil samples with the compressor running at operating temperature (see Figure: Ariel Oil Sampling Valve). To collect an oil sample with an Ariel oil sampling valve:

1. Remove dust cap.
2. Loosen lock ring to release button.

> **CAUTION:**
>
> A loosened lock ring allows valve outlet rotation. Aim outlet carefully to avoid spilling hot oil.

3. Press button to purge a small amount of oil into a waste container.
4. Press button to collect oil sample in a new sample bottle.
5. Tighten lock ring against button.
6. Re-install dust cap.

Figure: Ariel Oil Sampling Valve



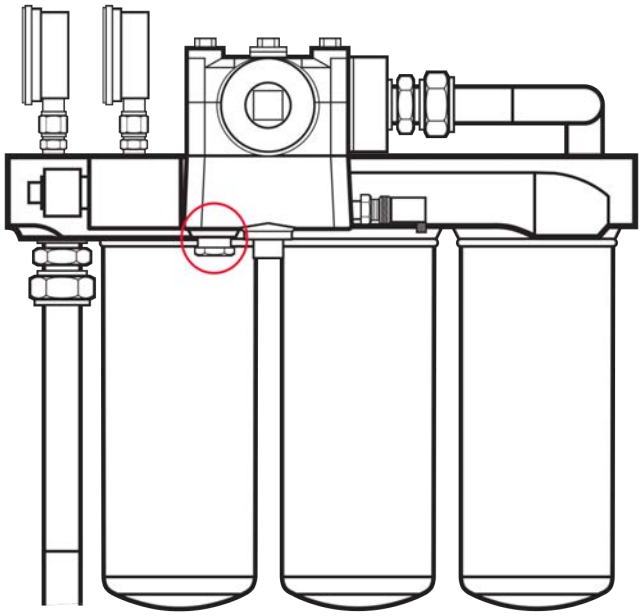
1 Lock Ring
2 Button
3 Dust Cap

Simplex Spin-on Filter Replacement

When filter differential pressure reaches 10 psid (0.7 bard) or once every six months, whichever comes first, replace oil filter with a genuine Ariel filter.

1. Remove old filter, clean filter head surfaces, and verify old gasket is removed.
2. Inspect new filter for damage or debris. Do not install damaged or dirty filters. **Never operate a unit with a damaged or leaking oil filter.**
3. Apply clean oil to the filter gasket and thread the filter with gasket onto the head.
4. Tighten the filter one turn after the filter gasket contacts the head. Install 1/2 inch NPT drain plug.
5. Use the compressor pre-lube pump to fill the oil filter with oil and purge air from the oil system. Manually filling the filter with oil can introduce debris on the clean side of the filter. This debris can damage the compressor to the point of bearing failure.
6. After starting the unit, check for leaks, and retighten if necessary.

Figure: Mounted Thermostatic Valve Drain Plug



Flushing to Change to a PAG Lubricant

Cylinder and frame lubrication systems require flushing when changing to or from a PAG lubricant or when applying a PAG lubricant to newly installed equipment.

Recommended Flushing Oil

The preferred flushing oil is soluble in both the original oil and PAG lubricant. A linear diester is preferred, such as Shell Corena DE 150 or Mobil Rarus 829. If this is unavailable, use the desired PAG lubricant fill oil as the initial flushing oil.

Flushing Procedure

Below is a general flushing procedure recommended to change from mineral or PAO oils to a synthetic PAG lubricant.

1. Drain old oil from frame while the oil is still warm.
2. Drain all supply and return lines upstream of the frame lube oil filter, filter housing (if applicable), coolers, and other peripheral equipment in the lubricant supply loop.
3. Remove standing oil in the frame and guide compartment low points. Remove as much oil as possible with a vacuum hose. If heavy deposits are visible, remove them as best as possible by hand using clean, dry, lint-free rags.
4. Install new frame lube oil filters.
5. Remove and either clean or replace the Y-strainer element.
6. Reconnect oil piping and install drain plug in frame.
7. Connect a diaphragm pump upstream of the frame mounted oil filter.

8. Feed the diaphragm pump flushing oil (see "Recommended Flushing Oil") from a clean reservoir and prime the lube oil system. Prime until old oil is purged from oil passages and flushing oil steadily flows from all bearings and crosshead bushings.
9. Drain frame. Remove standing oil in the frame and guide compartment low points. Remove as much oil as possible with a vacuum hose.
10. Remove diaphragm pump and plug connection upstream of frame mounted oil filter.
11. Install drain plug in frame.
12. Refill the frame with flushing oil (see "Recommended Flushing Oil").
13. Circulate flushing oil for at least four hours, and up to 24 hours if needed, to loosen all debris and existing deposits. Monitor lube oil filter differential pressure during flushing. The filter may need to be taken off line or changed several times during flushing, as loosened deposits tend to plug the filter quickly.
14. Drain the first flushing oil from the frame while the oil is still warm. Take care to also drain all supply and return lines, filter housing, and other peripheral equipment in the lubricant supply loop. If standing oil remains in the frame, remove as much as possible with a vacuum hose.
15. Refill the frame with the PAG lubricant of the recommended viscosity for the application. At this time, as much as 2-3% of the initial oil may still remain in the system. Circulate this fill for at least four hours and up to 24 hours if possible. Replace filters as needed.

> **NOTE:**
>
> If the PAG lubricant is clear and bright after circulating and no debris or emulsion is visible, it may be possible to skip the final steps and simply begin normal operation. However, a second flush is preferred (see steps 16 and 17).

16. Drain second flushing (PAG lubricant) oil from the frame while the oil is still warm.
17. Refill frame with PAG lubricant of the recommended viscosity and circulate for at least four hours. At this point, the initial oil remaining in the system should be less than 0.5%. Sample the oil and examine it for any emulsion or debris. If the oil is clear and bright (a slight haze is acceptable), then continue normal operation and top off as needed. If significant emulsion or debris remains, then conduct one more drain and flush.

    If the frame uses the same oil as the force-feed cylinder lubrication system, the overflow line on the lubricator box can remain attached to the frame auxiliary end cover. If the frame oil and cylinder oil are incompatible or of different ISO grades, disconnect overflow line on the lubricator box (see “Independent Oil Supply”). Fill the force-feed lubricator box with the same oil used to lubricate the compressor cylinders/packings. To change the lubricator box lubricant from a mineral oil or synthetic PAO lubricant to a PAG synthetic lubricant:

    a. Drain old oil from lubricator box.
    b. Rinse lubricator box with flushing oil (see "Recommended Flushing Oil").
    c. Fill lubricator box with PAG lubricant to be used for the cylinders.

Flushing the Force Feed Lubrication System - Common Oil Supply

1. Drain the lube line upstream of the force feed pump inlet.
2. Drain tubing/manifold feeding the force feed pump(s).
3. Remove the purge/vent connection on the outlet of each force feed pump.
4. Use a purge pump filled with clean/filtered flushing oil (see "Recommended Flushing Oil" to prime each force feed pump by forcing oil through the pump inlet and out the purge/vent connection.
5. Reconnect the purge pump to the force feed pump purge/vent outlet connection. If pumps are paired together at their outlets, close one of the purge/vent connections.

6. For the zone fed by the force feed pump being primed, disconnect lube lines at the inlet of the double-ball check valves.
7. Place paper towels or suitable collection containers beneath each disconnected lube line to catch flushing oil during priming.
8. Prime flushing oil from the force feed pump purge/vent connection to downstream tubing, divider blocks, and check valves.
9. Prime until no air bubbles appear in the flushing oil at all disconnected lube lines in the zone.
10. Reconnect lube lines to double ball check valves in the primed zone.
11. Continue to prime 2 fluid ounces of oil per throw through the primed zone.
12. Repeat steps 5 thru 11 for remaining zones.

Flushing the Force Feed Lubrication System - Independent Oil Supply

1. Drain daytank, oil filter, and lube lines upstream of the force feed pump inlet. Oil trapped in piping can be vacuumed out or flushing oil can be pumped through these lines to displace the old oil.
2. Drain tubing/manifold feeding the force feed pump(s).
3. Remove the purge/vent connection on the outlet of each force feed pump.
4. Use a purge pump filled with clean/filtered flushing oil (see "Recommended Flushing Oil") to prime each force feed pump by forcing oil through the pump inlet and out the purge/vent connection.
5. Reconnect the purge pump to the force feed pump purge/vent outlet connection. If pumps are paired together at their outlets, close one of the purge/vent connections.
6. For the zone fed by the force feed pump being primed, disconnect lube lines at the inlet of the double-ball check valves.
7. Place paper towels or suitable collection containers beneath each disconnected lube line to catch flushing oil during priming.
8. Prime flushing oil from the force feed pump purge/vent connection to downstream tubing, divider blocks, and check valves.
9. Prime until no air bubbles appear in the flushing oil at all disconnected lube lines in the zone.
10. Reconnect lube lines to double ball check valves in the primed zone.
11. Continue to prime 2 fluid ounces of flushing oil per throw through the primed zone.
12. Disconnect the purge pump and install cap for purge/vent connection.
13. Repeat steps 5 thru 11 for remaining zones.
14. Install a new oil filter.
15. Reconnect lube lines between daytank and pump inlets.
16. Place a collection container beneath the disconnected lube line near the force feed pump inlet connection.
17. Add enough flushing oil to the daytank to prime the system up to the disconnected line in step 16.
18. Collect oil until no air bubbles appear in the flushing oil.
19. Reconnect feed line to force feed pump.
20. Bleed trapped air by loosening tube connections nearest each pump inlet.

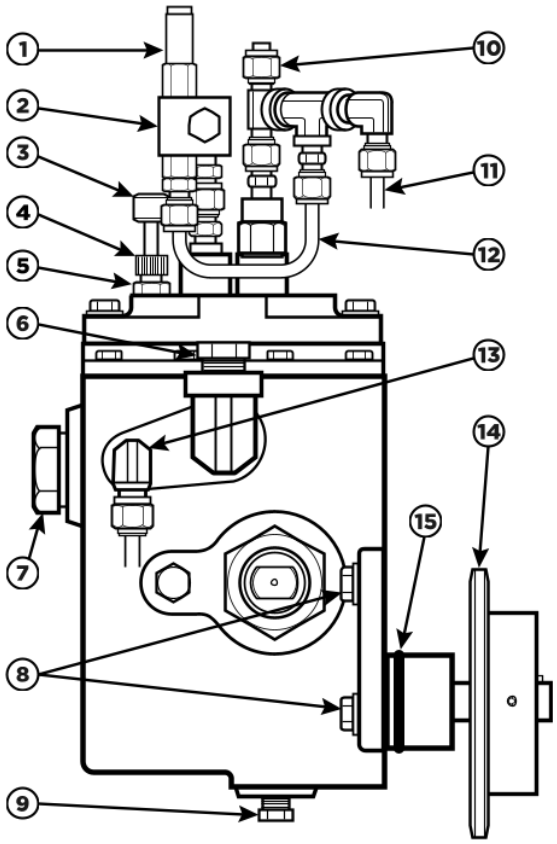
Cylinder and Packing Lubrication

The force feed lubrication system injects lubrication into cylinder bores and packings. The system consists of an oil supply, an oil filter, a force feed lubricator pump, distribution blocks, pressure indicators, check valves, and flow monitoring shutdown devices. The system is progressive; each piston in the distribution block must complete a cycle before the system cycles again. This type of system offers better lubrication precision and eases monitoring.

Force Feed Lubricator

The force feed lubricator provides pressurized lubricant to the distribution system. Oil flows to the suction side of the force feed lubricator pump directly from the pressure side of the frame lubrication system, or from an overhead tank. A self-contained oil reservoir lubricates the force feed lubricator worm gear and cam. The force feed lubricator sight glass shows the reservoir oil level. Fill the lubricator box with the highest viscosity oil that will be used by the force feed system. Flush the lubricator box before filling with a synthetic PAG lubricant.

Figure: KBE Force Feed Lubricator



1. Auto-Relief Valve (ARV)
2. ARV Manifold
3. Priming Stem
4. Pump Plunger Stroke Adjustment Screw
5. Lock Nut
6. Lubricator Oil Fill Connection
7. Sight Glass
8. Mounting Flange Capscrews (4)
9. Drain Plug
10. Purge/Vent Connection
11. Pump Outlet to Distribution Block
12. Inlet to ARV
13. Lubricator Overflow
14. Force Feed Lubricator Sprocket
15. O-ring

Pressure Relief

This equipment stays pressurized until pressure is manually relieved. To prevent serious injury from pressurized fluid, such as skin injection, splashing fluid, and moving parts, follow this procedure before cleaning, checking, or servicing the equipment:

1. Stop lubricator pump.
2. If installed, close oil supply valve located upstream from pump.

- If the supply valve cannot be closed, de-energize and relieve pressure of pressurized oil supplies feeding the pump.

3. Slowly crack open fluid line fittings to relieve pressure.

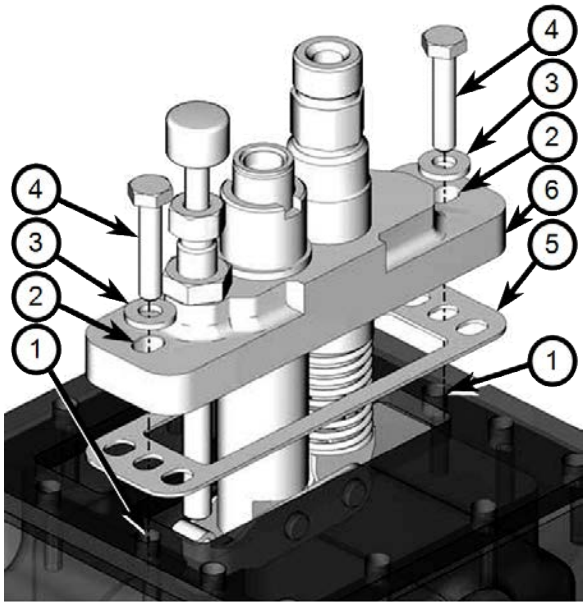
Pump Installation

CAUTION:

Shut down the compressor package and depressurize cylinders before performing any work, including the force feed lube system.

Before installation, verify the lubricator box and lube system are sized appropriately for the selected pressure. Verify a pressure relief device suitable for the application is installed in the system. Refer to Figure: Pump Installation during installation.

Figure: Pump Installation



1 Reservoir Cover Mounting Hole
2 Pump Mounting Hole
3 Washer
4 Mounting Bolt
5 Pump Gasket
6 Pump Body

1. Clean pump body.
2. Remove protective film from gasket and adhere gasket to pump body.
3. Orient pump so the roller bearing on the bottom of the pump aligns with the cam inside the reservoir.
4. Align the pump mounting holes with the reservoir cover mounting holes.
5. Install washers and bolts. Tighten the bolts securely, then torque each bolt to 50 Lb x In (5.65 N•m).

Pump Setup

Before setup, relieve all pressure from the force feed system (see "Pressure Relief").

The pump has an internal discharge check valve that has a cracking pressure of 55 psi (3.79 bar). Any oil pressure greater than this cracking pressure allows unrestricted oil flow through the pump, even in a pump at rest or adjusted to zero stroke. Pumps fed 60 psig (4.13 barg) oil from the compressor frame must pass through a 14 psi (0.96 bar) check valve installed at the lube oil gallery connection.

Specifications

- Inlet Size: 1/4” - 18 NPTF
- Outlet Size: 1/8”- 27 NPTF
- Strokes/Minute: Max. 50; Min. 3
- Stroke Length: Max. 1/2 inch (12.7 mm)
- Weight: 2.7 lbs (1.22 kg)
- Wetted Parts: carbon steel, stainless steel, aluminum, PTFE, fluoroelastomer

Piston Dia. In. (mm)	Max. Outlet Pressure psi (bar)	Max. Flow[a] In.^{3} (cm^3)
3/16 (4.76)	7500 (517)	0.012 (0.196)
1/4 (6.35)	6000 (414)	0.021 (0.344)
3/8 (9.53)	3500 (241)	0.048 (0.786)

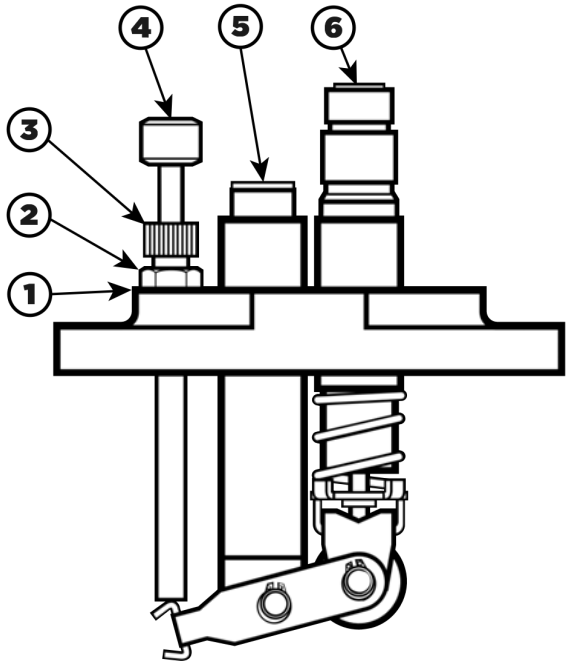
- Max. allowable inlet pressure for priming: 150 PSI (10.34 bar)
- Allowable viscosity range independent of temperature: 80-5000 SUS (16-1100 cSt)

Priming the Pump

Use filtered oil when priming the pump and force feed system (minimum 5 micron nominal filter). Refer to Figure: Pump Priming and Adjustment during priming.

1. Remove and discard the plastic plugs in the pump inlet and outlet.
2. Apply thread sealant (user supplied) to inlet and outlet line fittings.
3. Install the inlet and outlet fittings on the pump.
4. Connect the inlet line to the inlet fitting.
5. Introduce oil into the inlet line using a clean priming pump filled with cylinder oil connected to the purge/vent connection.
6. Wipe the pump priming stem clean and lubricate with oil.
7. Manually prime pump by using a priming pump until clean, air-free oil flows from the outlet fitting.
8. Connect outlet line to outlet fitting.
9. Manually prime pump by using a priming pump until clean, air-free oil flows from the outlet line at the cylinder and packing lube points.
10. Connect the outlet line to the lube point.
11. After start-up, adjust individual pumps to meet required flow rates. See "Pump Adjustment".

Figure: Pump Priming and Adjustment



1 O-ring
2 Flow Adjustment Screw Locknut
3 Flow Adjustment Screw
4 Pump Priming Stem
5 Pump Inlet
6 Pump Outlet

Pump Adjustment

The flow adjustment screw controls pump flow rate; turning it changes the length of the piston stroke which changes the pump discharge volume. Refer to "Figure: Pump Priming and Adjustment" during adjustment.

1. Loosen the flow adjustment screw locknut.
2. Lubricate the o-ring beneath the adjustment screw locknut with oil.

3. Turn the flow adjustment screw. Counterclockwise increases flow and decreases cycle time; clockwise decreases flow and increases cycle time.
4. Check force feed lubricator data plate or Ariel lube sheets for correct cycle time. Adjust flow until desired cycle time is achieved with the force feed system operating.
5. Repeat Steps 2 and 3 as needed.
6. After adjustment to the proper cycle time, tighten the adjustment screw locknut.

> **NOTE:**
>
> Inconsistent flow can occur if pump stroke is set below 20% of maximum stroke. A dust seal inside the flow adjustment screw holds the priming stem stationary during operation.

Pump Troubleshooting

Relieve all pressure from the system before cleaning, checking, or servicing the equipment.

Pump Problem:

- No flow
- Low flow or erratic operation
- Low discharge pressure
- Fluid flows from pump outlet when not running

Pump No flow

Problem

Pump no flow.

Possible Cause

Stroke adjusted to zero.

Solution

1. Adjust pump stroke for desired output. See "Pump Adjustment".

Possible Cause

Plugged pump outlet.

Solution

1. Clean/flush pump outlet.

Possible Cause

Damaged outlet check valve.

Solution

1. Replace pump.

Possible Cause

Damaged or worn inlet check ball and/or seat.

Solution

1. Replace pump.

Possible Cause

Stuck piston.

Solution

1. Replace pump.

Possible Cause

Stuck pump priming stem.

Solution

1. Clean and lubricate the priming stem. Replace pump if priming stem is corroded.

Possible Cause

Insufficient oil supply.

Solution

1. Check for line and filter restrictions upstream of pump. Verify pump inlet is free of debris.

Possible Cause

Air bubbles in the oil.

Solution

1. Prime pump. See "Priming the Pump".

Possible Cause

High oil viscosity.

Solution

1. Decrease flow restrictions upstream of pump.
2. Heat oil to maintain viscosity ‹1100 cSt.
3. Consult oil supplier for temperature.

Low Flow or Erratic Pump Operation

Problem

Low flow or erratic pump operation.

Possible Cause

Stroke adjusted to minimal travel.

Solution

1. Adjust pump stroke for desired output. See "Pump Adjustment".

Possible Cause

Damaged or worn inlet check ball and/or seat.

Solution

1. Replace pump.

Possible Cause

Damaged or worn outlet check ball and/or seat.

Solution

1. Replace pump.

Possible Cause

Sticking piston or priming stem.

Solution

1. Replace pump.

Possible Cause

Insufficient oil supply.

Solution

1. Check for line and filter restrictions upstream of pump.
2. Verify pump inlet is free of debris.
3. Check/inspect lube box cam gear for wear.

Possible Cause

Air bubbles in the oil.

Solution

1. Prime pump. See "Priming the Pump".

Possible Cause

High oil viscosity.

Solution

1. Decrease flow restrictions upstream of pump.
2. Heat oil to maintain viscosity ‹1100 cSt.
3. Check/inspect lube box cam gear for wear.

Low Pump Discharge Pressure

Problem

Low discharge pressure.

Possible Cause

Worn piston and bore.

Solution

1. Replace pump.

Possible Cause

Damaged or worn inlet check ball and/or seat.

Solution

1. Replace pump.

Fluid Flows From Pump Outlet

Problem

Fluid flows from pump outlet when not running.

Possible Cause

Inlet pressure exceeds cracking pressure of pump outlet check valve.

Solution

1. Reduce inlet pressure to below the cracking pressure. See "Pump Setup".

Possible Cause

Damaged or worn outlet check ball and/or seat.

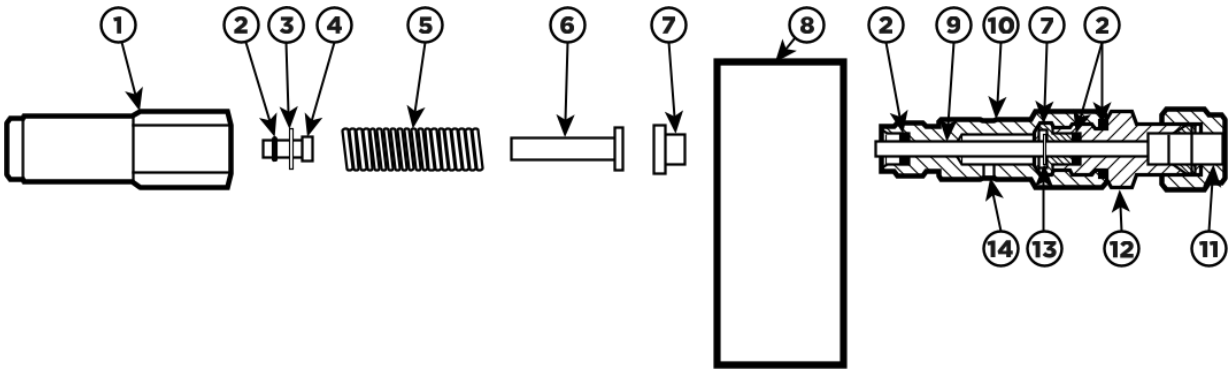
Solution

1. Replace pump.

Auto-Relief Valve (ARV)

An auto-relief valve (ARV) is a mechanical valve that protects force-feed lubrication systems from over-pressure conditions. Unlike a rupture disk assembly, the ARV is reusable and does not expel oil onto the compressor skid.

Figure: Auto-Relief Valve Components



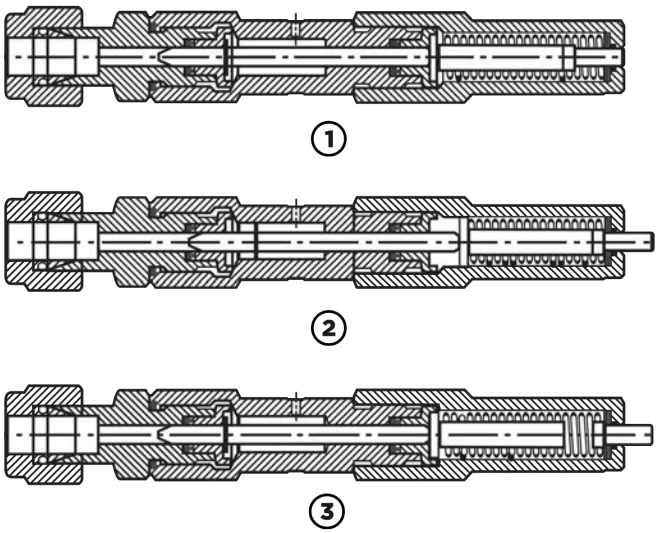
1	Spring Barrel	**6**	Spring Piston	**11**	ARV Inlet
2	O-ring	**7**	O-ring Retainer	**12**	Adapter
3	Washer	**8**	Manifold	**13**	E-Clip
4	Indicator Pin	**9**	Pressure Pin	**14**	ARV Outlet (Bypass Port)
5	Spring	**10**	Intermediate Barrel		

ARV Operation

When a force-feed pump encounters a downstream restriction, pump discharge pressure increases. If discharge pressure exceeds ARV cracking pressure, the ARV bypasses excess oil back to the low-pressure inlet side of the pump (Figure: ARV Operation – High Pressure). Bypassing oil limits the pressure created by the pump. The pump will operate in bypass condition until either the divider valve no-flow device times out due to lack of flow or pressure downstream of the pump decreases allowing the ARV to close (Figure: ARV Operation– Pressure Relieved).

When the ARV begins bypassing, an indicator pin protrudes from the top of the ARV indicating an over-pressure condition. The indicator pin is a “tattle tale” and remains protruding after the ARV bypasses oil (Figure: ARV Operation– Pressure Relieved). Two bypass conditions are possible:

Figure: ARV Operation



1 Normal Pressure (pin does not protrude)
2 High Pressure (pin protrudes)
3 Pressure Relieved (pin not reset)

Intermittent blockage downstream of the pump - A brief condition or one-time event that prevents the shutdown device from timing out and allows the compressor to continue to operate.

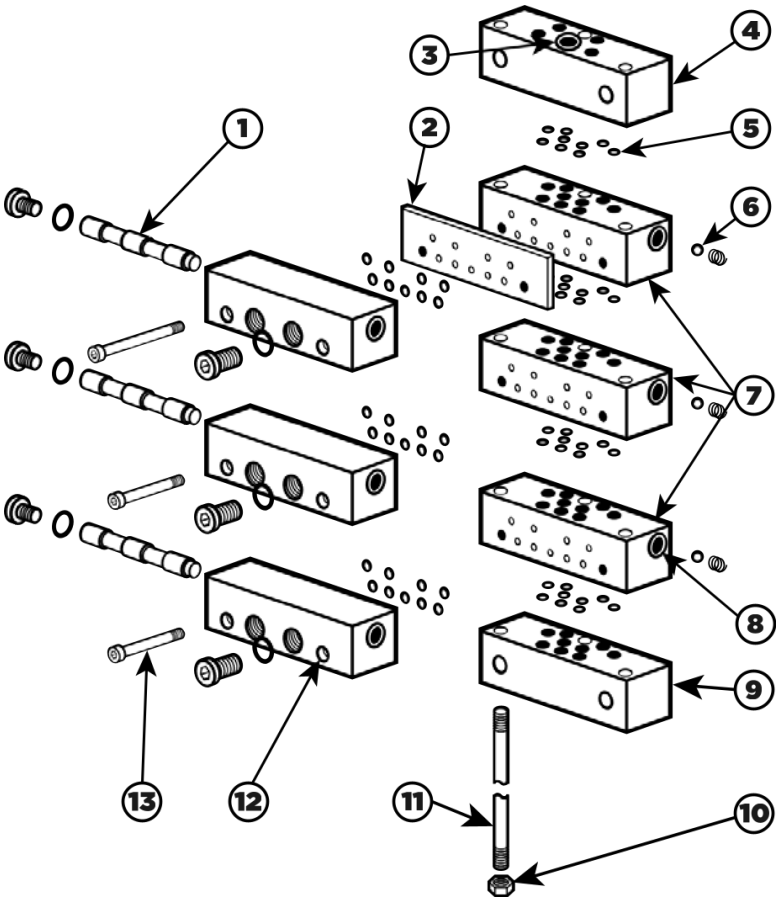
Significant blockage downstream of the pump - A condition of long enough duration to cause the shutdown device to timeout and shutdown the compressor.

After downstream blockage/restriction removal, the indicator pin can be manually reset by depressing the indicator pin until it is flush with the ARV housing.

Distribution Blocks

Distribution blocks consist of three to seven divider valves and an optional bypass block fastened to a segmented baseplate. Viton O-rings (90 Durometer) seal between the divider valves and baseplate and between baseplate segments. Check valves are installed at all lube port outlets.

Figure: Typical Distribution Block



1 Piston
2 Crossport Plate
3 Inlet Port
4 Inlet Block
5 O-Rings
6 Check Valve
7 Intermediate Blocks
8 Outlet Port (correct position is off-center toward inlet block)
9 End Block
10 Tie Rod Nut
11 Tie Rod
12 Divider Valve
13 Valve Screw

Divider valves contain metering pistons that discharge a predetermined amount of oil with each cycle in a single line, progressive lubrication system.

> **NOTE:**
> Pistons are matched to a specific divider valve. Do not try to place a different piston into a divider valve.

Single or twin valves may be externally singled or cross-ported. Plug unused outlets when singling or cross-porting.

Bypass blocks can replace a divider valve section to eliminate the lubrication points. Plug both outlets below the bypass block. A bypass block can mount in any baseplate location, but it requires at least three working divider valves in the block assembly (i.e. bypass blocks cannot be used on 3-section baseplates).

The baseplate contains the divider valve inlet and outlet connections, interrelated passageways, and built-in check valves. All lubricant piping to and from the distribution block connects to the baseplate. The baseplate consists of one inlet block, three to seven intermediate blocks, one end block, and three tie rods. The number of baseplate intermediate blocks determines the number of divider valves allowed. Each distribution block requires a minimum of three divider valves.

Distribution Block Assembly

NOTE:

See specific frame Parts Book for available divider valve assembly drawings, parts lists, and repair kits.

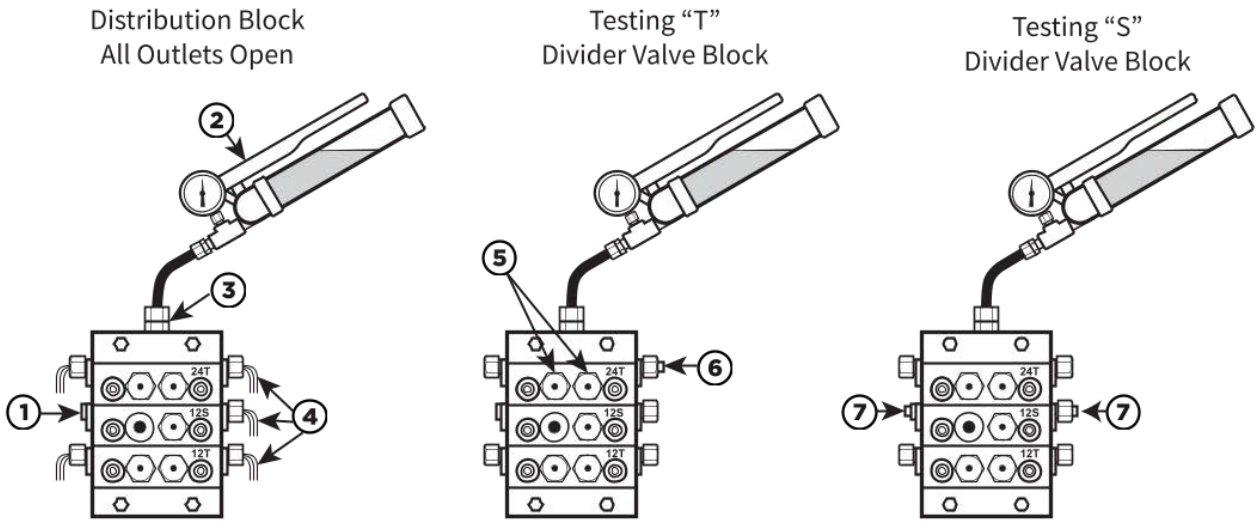
1. Thread three tie rods into inlet block until the ends flush with the top surface of the block.
2. Slide Intermediate blocks onto tie rods. Verify all O-rings are installed and discharge ports are off center toward the inlet block.
3. Slide end block onto tie rods.
4. Lay base plate assembly on a flat surface. Use "Recommendations for Torque Accuracy" to tighten the tie rod nuts to the correct torque listed in Appendix A.
5. Mount the divider valves with O-rings onto the base plate. Use "Recommendations for Torque Accuracy" to tighten the tie rod nuts to the correct torque listed in Appendix A.

Divider Valve Bypass Pressure Test

Lubricant can leak (by-pass) from a divider valve when the divider valve piston becomes worn. Pressure test or replace all divider valves every twelve months. Testing verifies the piston fits tightly enough in the divider valve bore for adequate pressure to force oil into the injection point. **The supplier pressure tests all new Ariel divider valves.**

To test distribution blocks for valve by-passing requires a manual purge pump with a pressure gauge capable of developing 5000 psig (350 barg). Ariel offers an optional force feed lubrication hand purge pump with a pressure gauge (see Section 1). Pressure test each divider valve one at a time, complete with pin indicators installed.

Figure: Pressure Testing Divider Valves



1 "S" Divider Valve - one side plugged with 1/8" NPT pipe plug
2 Purge Pump
3 Divider Block Inlet
4 Base outlets open - oil flows freely
5 Optional Resettable Pin Indicators
6 One side Plugged to test a "T" valve - all other outlets open
7 Both sides plugged to test an "S" valve - all other outlets open

1. For new divider valves, verify working piston section fastener torque is 75 lb x in (8.5 N•m). Loosen the section fasteners, then step-torque them first to 40 lb x in (4.5 N•m), and then to 75 lb x in (8.5 N•m). Used divider valves can be tested “as is”, with approximate section fastener torque verified after testing.
2. Place the distribution block in an open container with all base outlets open. Connect purge pump filled with room temperature [65°F (18°C)] 10 wt. (ISO 32) mineral oil to the divider block inlet. Pump the purge pump to purge air from the divider block assembly. Verify that oil flows freely from all outlets (see Figure: Pressure Testing Divider “Divider Block All Outlets Open”). Testing with 10 wt. mineral oil at 65°F (18°C) simulates divider block operation at 120°F (49°C) with 40 wt. (ISO 150) mineral oil. If 10 wt. (ISO 32) mineral oil is unavailable, use 40 wt. (ISO 150); however, the pressure test will be less sensitive in detecting a bypassing divider valve.
3. The divider block assembly should cycle at less than 300 psi (2068 kPa) while purging at a steady rate. Cycle pressures above 300 psi (2068 kPa) indicate inhibited piston movement, possibly caused by the piston rubbing the bore, oil contamination, part geometry, or bore distortion due to over-tightening the section fasteners. If re-torquing the valve section fails to improve cycle pressure, replace the divider valve section. Do not attempt to alter the piston/bore fit by polishing components
4. For divider valves stamped with a “T” (for “twin”), use a 1/8 inch (3.2 mm) pipe or tubing plug to plug only one base outlet when testing that side of the piston, and leave all other outlets open. Plug and test each base outlet of divider valves stamped with a “T” one side at a time. See Figure: Pressure Testing Divider “Testing ‘T’ Divider Valve”. Individual testing of each outlet ensures both sides of the piston build adequate pressure. For all divider valves stamped with an “S” on the front (for “single”), plug the working outlet as well to test both sides of the piston for by-passing simultaneously. See Figure: Pressure Testing Divider “Testing ‘S’ Divider Valve”.
5. Pump the purge pump until the pressure gauge indicates 3000 psi (20684 kPa). Hold this pressure for 5 seconds, then increase it to 3500 psi (24132 kPa). Stop pumping at 3500 psi (24132 kPa).
6. Start timer and monitor the pressure drop from 3500 psi (24132 kPa) for 30 seconds. Check discharge outlet(s) plug(s) to confirm no external leaks. Also check O-rings for external leakage. If the valve bypasses rapidly, repeat the test to ensure the bypass wasn’t due to trapped air. New valves should not exceed a 400 psi (2758 kPa) pressure drop in 30 seconds. Used valves should not exceed 1000 psi (6895 kPa) pressure drop in 30 seconds. If pressure drops exceed these limits, the divider valve fails the test.
7. Repeat Steps 4 thru 6 for the remaining outlets on the divider block assembly.

> **NOTE:**
>
> Test distribution blocks at higher pressures if the application dictates higher system operating pressure.

If a divider valve fails, replace it. Discard worn divider valves. If the tested valve passes the test, relieve the pressure, move the plug to the next outlet, and repeat the test for all divider valves. When all divider valves either pass this pressure test or are replaced, reassemble the distribution block, purge it with the proper force feed lubricant, and put it into service.

Ariel recommends periodic tests for proper cylinder bore lubrication rates and/or aftermarket devices that measure flow.

Force Feed System Design, Operating Parameters, and Conditions

See specific packager data to confirm designed gas composition, operating pressures, and speed. If operating outside design conditions, system may need to be reviewed. To optimize force-feed lubrication system operation, Ariel uses these general guidelines:

1. Maintain lube rates within prescribed limits and cycle times as low as possible (normally 10 sec minimum) to lubricate each point as frequently as possible.
2. Do not operate pumps below 20% of full stroke or they become inconsistent. When two or more pumps are manifolded to feed one distribution block, adjust them equally. Start with pumps wide open, and adjust them together so that when break-in cycle time is set, the pumps stroke about the same. After break-in period, adjust

the pumps in the same manner, provided the final pump stroke is not too short. Try to keep the stroke greater than 20% of maximum; a shorter stroke produces unreliable pump output. If needed, close one pump to stop its flow and open the other(s) to make the “normal” cycle time, and maintain a stroke greater than 20% in the functioning pump(s).

3. In some applications, one divider valve delivers lubrication to both a packing and a cylinder to optimize lubrication system operation. In some applications, cross-ported divider valves deliver the proper proportion of lubricant to a given point.
4. During operation, verify oil fills the lubricator reservoir sight glass to the overflow line. Reservoir oil lubricates the worm gear and cam; it does not flow through the system. The reservoir also catches force feed lube pump leakage. Add oil only if needed to raise reservoir oil level. See Table: Force Feed Lubricator Reservoir Oil Capacity.

Table: Force Feed Lubricator Reservoir Oil Capacity

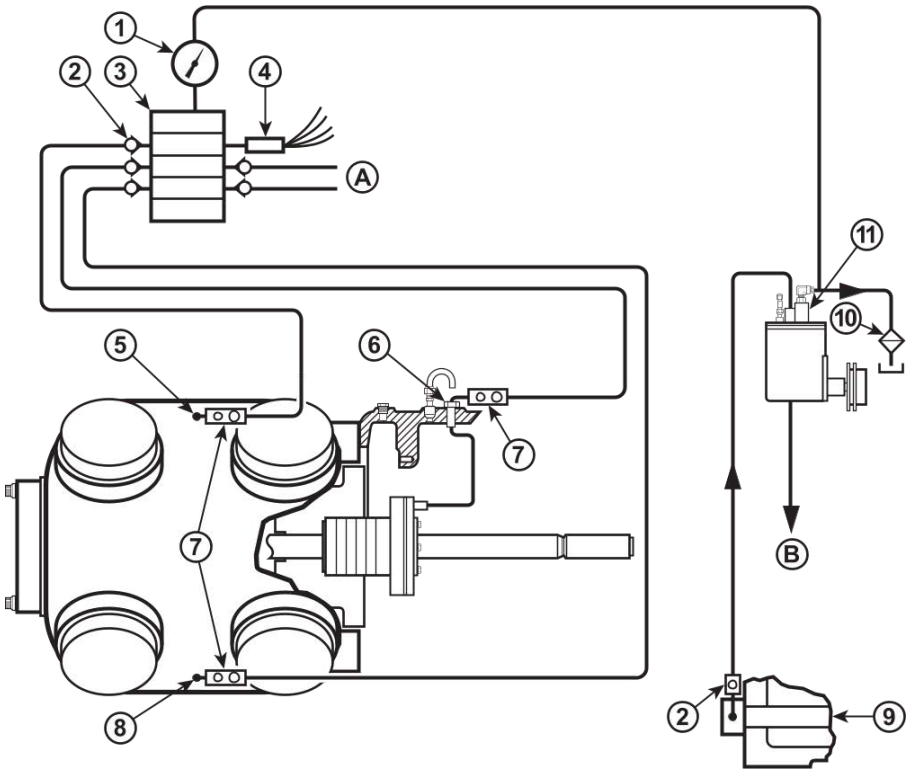
Lubricator Type Ariel Designs	US Gallons (Liters)
Single Pump	0.25 (1)
Dual Pump	0.5 (1.75)
Four Pump	0.8 (3)

5. Ariel primes the force feed system with oil. If tubing is missing, or if the system is drained, fill and prime the system through a 1/8 inch plug on the discharge end of the force feed lubricator pump.
6. Prime the force feed lubrication system with a proper priming pump just prior to starting the compressor (see "Force Feed Lubricator" for pump illustration). Use clean force feed lubricating oil of same type and grade as used in service. Do not use any other fluid type or grade to prime.
7. For recently overhauled units, adjust force feed cycle time for maximum delivery. Loosen adjusting screw locknut. Turn plunger stroke adjustment screw to the full up position. Tighten adjusting screw locknut. After two hours of operation or before leaving site, reset to “normal” cycle time.

Common Oil Supply

Figure: Force Feed Lubrication System Common Oil Supply shows the force feed lube system installation when compressor frame lube oil is also used for cylinder and packing lubrication.

Figure: Force Feed Lubrication System Common Oil Supply



- **A** To other cylinders and packings
- **B** Drain to crankcase
- **1** Pressure Gauge
- **2** Single-Ball Check Valve
- **3** Divider Valves/ Distribution Block
- **4** Fluid Flow Monitor No-Flow Timer Shutdown Switch
- **5** Top Cylinder Injection Point
- **6** Packing Injection Point
- **7** Double-Ball Check Valve
- **8** Bottom Cylinder Injection Point
- **9** Frame Oil Gallery
- **10** ARV or Rupture Disk
- **11** Force Feed Lubricator Pump

Independent Oil Supply

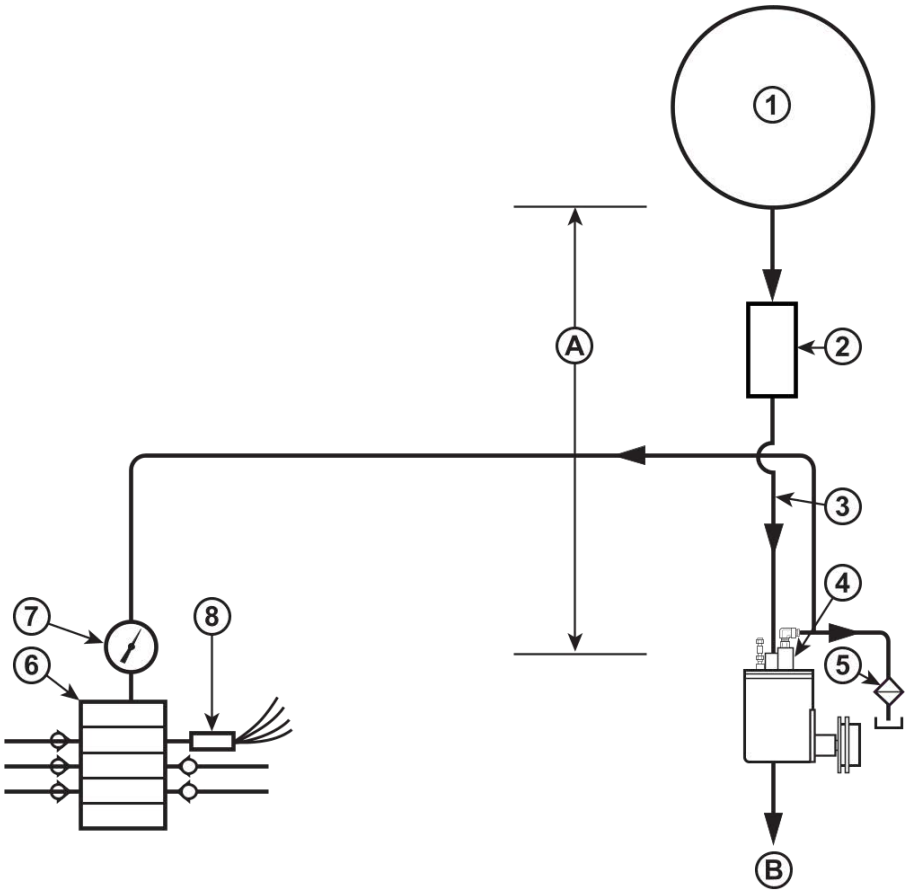
When cylinders and packing require oil different from frame oil, the force feed lubricator system requires an independent oil supply system (see Figure: Force Feed Lubrication System Independent Oil Supply). An elevated tank must supply pressurized force feed lube oil at a minimum of 1 psig (0.07 barg) to the force feed pump inlet manifold. Oil piping downstream of filter must be stainless steel.

To prevent force feed oil from contaminating compressor frame oil, disconnect lubricator box overflow tubing from compressor frame and direct it to a suitable drain system.

Independent force feed lube systems require oil with a viscosity below 1100 cSt at the lubricator pump inlet (e.g. ISO 320 at 73°F (23°C), contact lube supplier about specific lubricants). Possible ways to ensure force feed pump is fed oil during suction stroke:

- Increase pipe and fitting size from tank to force feed pump to deliver a minimum of four times the maximum oil consumption of the force feed lube system to the force feed pump manifold inlet.
- Heat the oil.
- Heat trace and insulate the line from the tank to the force feed pumps, including any inline filters.
- Pressurize the supply tank (tank must be designed for pressure).

Figure: Force Feed Lubrication System Independent Oil Supply



- **A** NOTE: System must provide positive pressure to force feed lubricator pump.
- **B** 1/4-inch NPT (female) overflow. Pipe or tube to appropriate isolated customer connection. Do not drain to crank case.
- **1** Oil Supply Tank
- **2** Oil Filter (max. 5 micron)
- **3** System Inlet 1/4-inch NPT (female) customer connection
- **4** Force Feed Lubricator Pump
- **5** ARV or Rupture Disk
- **6** Divider Valves Distribution Block
- **7** Pressure Gauge
- **8** Fluid Flow Monitor No-Flow Timer Shutdown Switch

Force Feed Lubrication Conditions

CAUTION:

To prevent personal injury, verify driver or compressor cylinder gas pressure cannot turn compressor crankshaft during maintenance. Before any maintenance or component removal, relieve all pressure from compressor cylinders. See Packager's instructions to completely vent the system. After maintenance, purge the entire system with gas prior to operation to avoid a potentially explosive air/gas mixture.

Cycle Time

Lubricant flow rates are measured in seconds per cycle of the distribution block. The calculated break-in and normal cycle times are stamped on the lubricator box data plate. These cycle times are calculated based on the gas analysis, operating conditions, and applied speed specified in the compressor order. If gas conditions were not supplied with the compressor order, the data plate/lube sheet rates default to clean, dry, 0.65 specific gravity, sweet gas and cylinder MAWP. If the applied speed is not specified, the cycle time is based on maximum rated speed of the frame or cylinders, whichever is less. The lube sheets in the Ariel Parts Book state gas conditions and list the base rate multiplier at each lube point.

Break-in Rate - Set the break-in lube rate about twice the recommended daily rate (150% minimum); i.e. set the break-in cycle time about half the normal cycle time (67% max.) to increase lube rate. Maintain break-in rate for

200 hours of operation for new equipment. Contact Ariel if existing pump is incapable of minimum flow rate required.

Lube Rate and Speed - Recommended lube rates for break-in or normal operation, in seconds per cycle, are calculated at the frame rated speed (RPM) stamped on the frame data plate or the cylinder rated RPM stamped on the cylinder data plate, whichever is lower. Lube rate decreases with speed, (as compressor actual running speed decreases, cycle time increases, reducing lube rate):

(RPM_{max} ÷ RPM_{actual}) x cycle time seconds from lube plate = cycle time seconds at actual running speed.

See the Ariel Parts Book Lubrication Sheets for the Cycle Time (seconds) vs. RPM (compressor speed) table at various running speeds at stated gas operating conditions and lubricant.

Adjusting the cycle time changes the lubrication rate for all the compressor cylinders and packings supplied by a particular pump. To change the lubrication rate for only one cylinder or packing requires individual divider valve changes. Contact your packager or Ariel for information.

To set proper force-feed lubricator pump flow rate, read or measure the cycle time from the installed device. For a magnetic cycle indicator assembly, time the cycle from initial indicator pin movement at the fully retracted position to the time when the pin returns to the fully retracted position and just begins initial movement out again. Adjust the lubricator pump to provide the required cycle time.

> **NOTE:**
>
> Pump output can become inconsistent if flow rate is set too low, oil contains air, or pump has insufficient oil supply. See “Force Feed System Design, Operating Parameters, and Conditions”.

Changes in operating conditions (such as gas properties or pressures, temperatures, flow requirements, or cylinder re-configuration) require lubrication rate re-calculation and possible force-feed lubrication system alteration. Consult the Ariel Packager Standards, the Packager, and/or Ariel.

Under/Over Lube

Inadequate lubrication causes premature wear of piston and packing ring materials. Indications of under lubrication are black, gummy deposits in the distance piece, packing case, cylinder, and valves. When symptoms indicate under lubrication:

- Verify proper operation of force feed lubricator pumps.
- Confirm distribution block cycle time matches the lube sheet or force feed lubricator data plate.
- Verify all tubing and fitting tightness; check for leakage. Check fittings inside cylinder gas passages.
- Pressure test or replace divider valves to ensure they do not bypass.

Over lubrication can result in excessive oil carryover into the gas stream, and increased valve and gas passage deposits. Valve plate breakage and packing failure may also indicate over-lubrication.

Cylinder Lubrication Paper Test

1. Remove head end head and position piston for the desired cylinder at inner dead center.
2. Using light pressure, wipe the cylinder bore with two layers of regular unwaxed cigarette paper together. Begin at the top and wipe downward about 20° along the bore circumference. The paper against the bore surface should be stained (wetted with oil), but the second paper should not be soaked through.
3. Repeat the test at both sides of the bore at about 90° from the top, using two clean papers for each side. Paper against the bore surface not stained through may indicate under-lubrication; both papers stained through may

indicate over-lubrication. In either case, Ariel normally recommends changing lubrication rate accordingly and repeating all paper tests until passed.

4. Repeat this procedure for all cylinders. If the test indicates a lubrication rate reduction or increase for a cylinder, change in 5% increments by adjusting cycle time at the force feed lube pump (see "Pump Adjustment"). Repeat oil film testing, for affected cylinders, after 24 hours of operation. For a formal lube rate reduction program, see Ariel ER-125.

> **NOTE:**
>
> The paper test indicates only oil film quantity. Aftermarket devices exist that measure flow. Neither method indicates viscosity quality. Oils diluted with water, hydrocarbons, or other constituents may appear to produce an adequate film or flow, but dilution will reduce lubricant effectiveness below requirements.

Section 4 - Part Replacement

Major frame assembly components include the crankcase, crankshaft and bearings, connecting rods, chain drive system, crossheads, and guides/distance pieces. Removable end covers, an aluminum top cover, and crosshead guide side covers provide easy access to internal components. Below are general guidelines to follow during maintenance.

1. Cleanliness during part replacement:
   - Use lint-free wiping cloths.
   - When not working on a frame with access covers removed, cover it to protect the interior from rain, dust, and other airborne debris.
   - Place removed components in a safe place away from work area to prevent damage. Take precautions to keep parts from rusting.
2. Gasket maintenance:
   - When dismantling the machine, carefully inspect gaskets at non-pressure locations. If damaged, replace them.
   - Always replace pressure-retaining gaskets.
   - Always apply an anti-seize lubricant to both sides of metallic gaskets.
3. For major overhauls, drain and clean the crankcase.

If replacing a connecting rod assembly, piston, piston and rod assembly, crosshead-balance nuts, or crosshead, weigh component parts and compare to the Compressor Balancing Record included in the parts manual with each compressor. If there are weight changes, recalculate opposing throw reciprocating weight differential. See "Opposed Throw - Reciprocating Weight Balancing" in Appendix C.

Relieving Gas Pressure

Before conducting maintenance on the compressor, it is vital to ensure the cylinders contain no gas pressure. During maintenance, unrelieved gas pressure could cause equipment damage, personal injury, or even death.

Isolation

Isolate the compression system from all energy sources prior to any maintenance. To ensure the system remains isolated during maintenance, a lock out procedure is required. This is a requirement of most corporate policies and occupational safety regulatory agencies. The extent of the isolation depends on the system.

Depressurization

Fully depressurize the compression system prior to any maintenance. If the application involves a corrosive or poisonous gas, purging is necessary to remove the hazard. There are many different configurations of compression systems, so refer to the information provided by the system provider for depressurization. In general, the following steps are required:

1. Isolate inlet gas from entering the system.
2. Isolate discharge gas from entering the system.
3. Vent and isolate scrubber dump lines, vent and drain lines, flare vent lines, and any common blow-down lines to prevent backflow into the system.
4. Open the system blow-down line and reduce the system pressure to atmospheric.
5. If so equipped, open the indicator valves on the individual cylinders to remove any residual pressure in the cylinder(s).

WARNING:

Process gas from these valves can be under high pressure if the system is not depressurized properly and can be flammable and/or poisonous depending on the application.

Positioning a Throw

WARNING:

To decrease the risk of injury or death, only trained and approved personnel can do maintenance work. Approved personnel must read and know the applicable Ariel Technical Manual and Packager-supplied information.

Before you start maintenance:

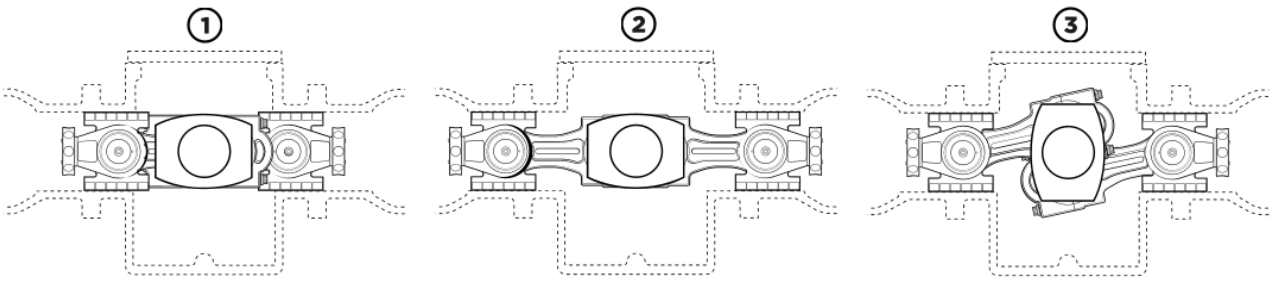
- Always use personal protection equipment (PPE) and obey your company policies for safety and operation.
- Do not tighten or loosen fasteners or fittings while the unit is in operation or is pressurized.
- Make sure the driver and compressor cannot turn during maintenance.
- Obey your company and packager instructions to isolate and vent the system.
- Make sure there is no remaining pressure in the cylinders. If equipped with “Kiene” valves on the indicator ports, open them to release residual pressure after blowdown.

  WARNING: RELEASED GAS MAY BE POISONOUS AND/OR FLAMMABLE.

Before operation, purge the system. A pressurized hydrocarbon and air mixture can explode and cause injury or death.

Component replacement often requires manually turning the crankshaft to position a throw in one of two positions: inner dead center or outer dead center. During normal operation, crossheads slide back and forth in the crosshead guides. In the inner dead center position, the crossheads slide toward the crankcase as far as possible. In the outer dead center position, the crossheads slide away from the crankcase as far as possible (see figure below). Some procedures require a dial indicator with magnetic base to locate the precise inner or outer dead center. At other times, a procedure may require turning the crankshaft so a throw occupies its highest or lowest position. In the figure below, the throw on the left is in its highest position and the throw on the right is in its lowest position.

Figure: Throw Positioning



1 Inner Dead Center Position

2 Outer Dead Center Position

3 Highest/Lowest Throw Position

Variable Volume Clearance Pocket (VVCP)

A VVCP changes the clearance volume of the head end of a cylinder. The amount of clearance depends on the position of the clearance pocket piston. Users turn the piston/stem assembly counterclockwise to increase clearance, and clockwise to decrease it. In the event that gas begins to leak from the VVCP vent, replace the V-packing or seal within the VVCP.

VVCP Removal

WARNING:

To decrease the risk of injury or death, only trained and approved personnel can do maintenance work. Approved personnel must read and know the applicable Ariel Technical Manual and Packager-supplied information.

Before you start maintenance:

- Always use personal protection equipment (PPE) and obey your company policies for safety and operation.
- Do not tighten or loosen fasteners or fittings while the unit is in operation or is pressurized.
- Make sure the driver and compressor cannot turn during maintenance.
- Obey your company and packager instructions to isolate and vent the system.
- Make sure there is no remaining pressure in the cylinders. If equipped with “Kiene” valves on the indicator ports, open them to release residual pressure after blowdown.

  WARNING: RELEASED GAS MAY BE POISONOUS AND/OR FLAMMABLE.

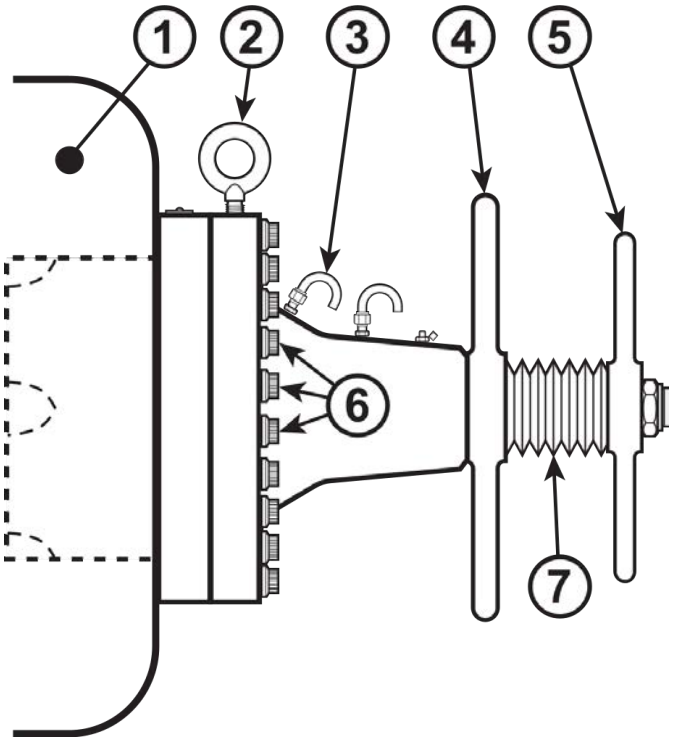
Before operation, purge the system. A pressurized hydrocarbon and air mixture can explode and cause injury or death.

1. After venting the cylinder, loosen the locking handle so the adjustment handle can turn freely. Turn the adjustment handle.

> **CAUTION:**
>
> A hard to turn adjustment handle or piston stem may indicate pressurized gas trapped within the VVCP. See VVCP Disassembly caution.

2. Tighten the locking handle.
3. Disconnect the VVCP vent.
4. Remove plastic plug from eyebolt hole and thread an eyebolt into the eyebolt hole.
5. Use the Ariel supplied eyebolt to support the weight of the VVCP.
6. Remove all flange bolts and slide the VVCP out from the cylinder. The Ariel Performance Program contains approximate VVCP weights.
   If the VVCP does not slide out easily, loosen the valve cap bolts. If it still does not slide out easily:
   a. Remove head end valves.
   b. Clean the area between the VVCP head and the cylinder.
   c. Loosen discharge bottle supports.
7. Discard the head gasket and transport the VVCP to a suitable work area for disassembly.

Figure: Typical VVCP



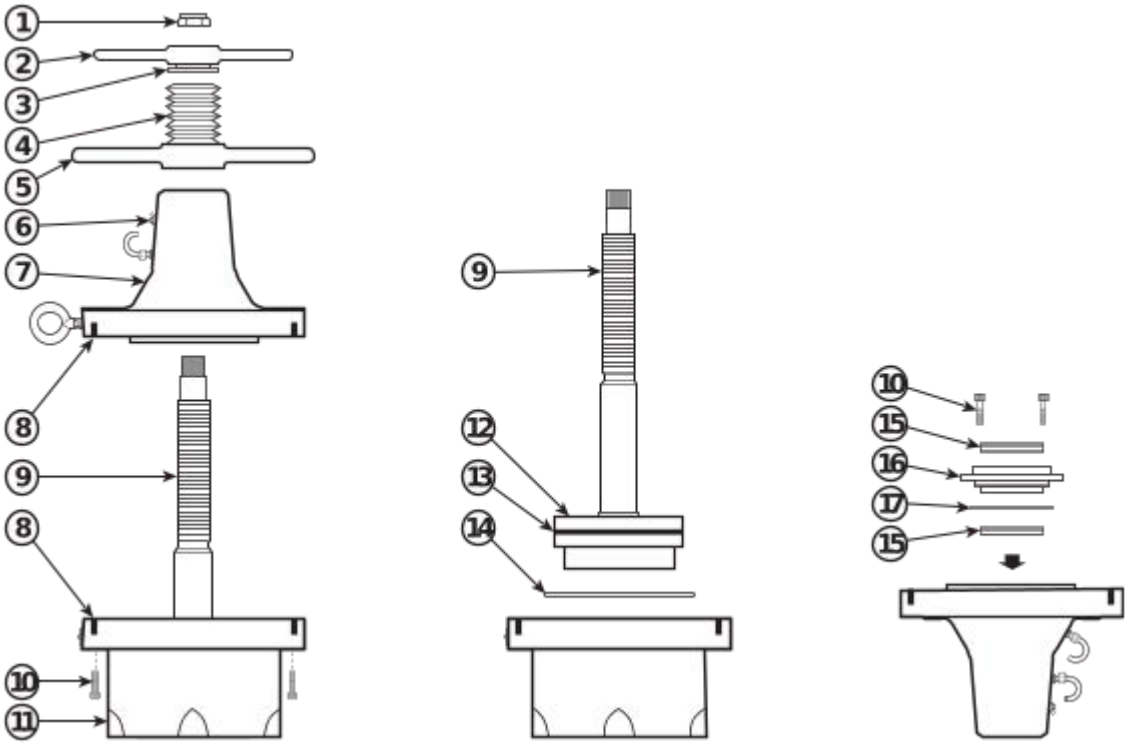
1	Cylinder	5	Adjustment Handle
2	Eyebolt	6	Flange Bolts
3	VVCP Vent	7	Bellows Thread Protector
4	Locking Handle		

VVCP Disassembly

The VVCP piston ring is not gas-tight to allow a nearly balanced gas pressure, which eases VVCP adjustment with the cylinder pressurized. Gas pressure behind the VVCP piston normally vents when the cylinder vents. Process debris or rust around the piston ring can form a seal that traps gas within the VVCP assembly, sometimes at substantial pressure.

> **CAUTION:**
>
> Trapped gas pressure can present a personal safety hazard when servicing the VVCP. Work in a well-ventilated, non-sparking area, particularly with sour gas applications. Do not breathe gas emission from VVCP when venting trapped gas.

Figure: VVCP Disassembly/Assembly



1 Locknut
2 Adjustment Handle
3 Adjustment Handle Lip
4 Bellows Thread Protector
5 Locking Handle
6 Grease Fitting
7 Adapter
8 Match Mark
9 Piston Stem
10 Socket Head Bolt
11 Head
12 Piston
13 Piston Ring
14 O-Ring - fits into a groove machined into either the head (11) or adapter (7).
15 Seal
16 Seal Carrier
17 O-Ring

1. Place the VVCP on a table, adapter up.
2. The bellows thread protector grips a lip on the adjustment handle. Use pliers to pull the bellows from the adjustment handle lip. Do not tear the bellows.
3. Remove locknut.
4. Remove adjustment handle. The adjustment handle is a tapered fit onto the piston stem; use a soft hammer or puller to break the fit.
5. Thread the locking handle off of the piston stem. Leave the bellows thread protector attached to the locking handle.
6. Depending on size, a VVCP may have two to four socket head bolts. Use a marker to draw match marks on the side of the VVCP where the adapter and head join at each socket head bolt location. Loosen the socket head bolts evenly and incrementally to release any potentially high-pressure gas. **Do not breathe gas emission from VVCP.**
7. Remove socket head bolts and turn the adapter counterclockwise to thread it off of the piston stem. Place adapter flange up next to the head.
8. Remove the piston stem and piston assembly from the head. The piston stem and piston are permanently joined; do not attempt to disassemble them.
9. Remove the piston ring from the piston.

10. Depending on VVCP size, Ariel machines a groove for the o-ring into either the face of the head or the base of the adapter. Remove and discard the o-ring.
11. Using an appropriate puller tool, remove the first seal from the seal carrier bore. Be careful not to damage the seal housing surfaces.
12. Remove the four socket head bolts, seal carrier, and second seal from the adapter bore. Remove the O-ring from the seal carrier.

VVCP Reassembly

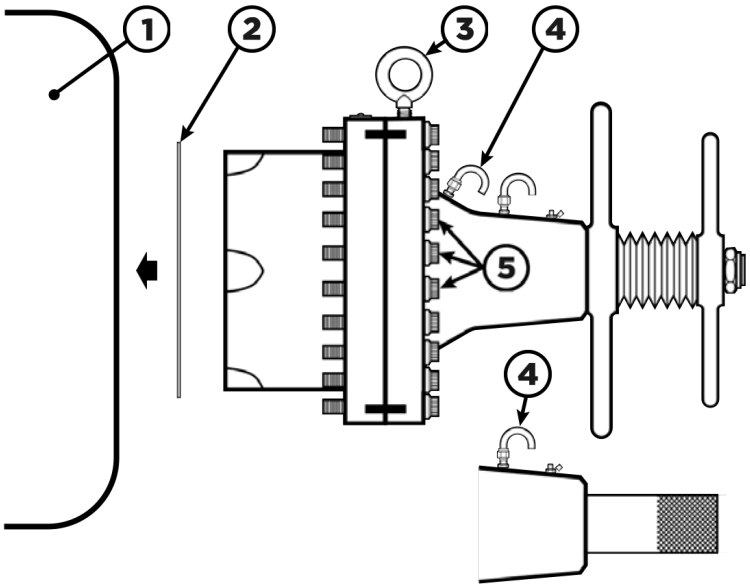
Clean all VVCP parts of all debris, rust, etc. Replace worn or damaged parts. Rebuild kits are available through your Ariel Parts Distributor.

1. With flange up, place the head on a table. With flange up, place the adapter beside the head.
2. Depending on VVCP size, Ariel machines a groove for the O-ring into either the face of the head or the base of adapter. Oil and insert a new O-ring into O-ring groove.
3. Install the piston ring onto the piston.
4. Oil the head bore, then insert stem and piston assembly into the head cavity.
5. Lightly oil and install a new seal in the VVCP adapter bore with the spring facing up. Lightly oil and install a new O-ring on the seal carrier.
6. Install the seal carrier in the VVCP adapter, tightening the four socket-head bolts to the torque listed in Appendix A
7. Lightly oil and install a new seal in the seal carrier with the spring facing up.
8. Grease piston stem threads and install adapter onto piston stem. Install adapter as far as possible on the piston stem, then turn the adapter counter-clockwise to align the socket head bolt match marks on the adapter flange with the match marks on the head.
9. Install the socket head bolts. Tighten socket head bolts to the torque listed in Appendix A.
10. Use the adjustment handle to turn the piston stem counter-clockwise to tighten the seal between the adapter and head, then lock the locking handle.
11. Apply 3-4 pumps of all-purpose petroleum grease with a grease pump at the grease fitting.

VVCP Installation

Install the VVCP prior to installing valves and torqueing valve caps. Use the Ariel supplied eyebolt to support the weight of the VVCP.

Figure: VVCP Installation



1 Cylinder
2 Head Gasket
3 Eyebolt
4 VVCP Vent
5 Flange Bolts

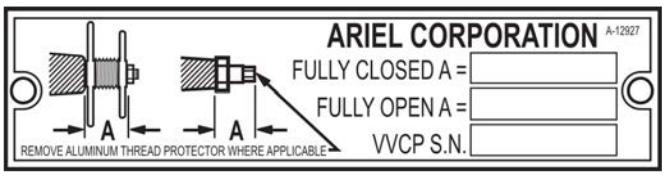
1. Apply an anti-sieze lubricant to both the new head gasket and the cylinder seating surface. Stick the new head gasket to the cylinder seating surface.
2. Slide the VVCP into the cylinder and align the adapter flange bolt holes with the cylinder bolt holes.
3. Lubricate and torque flange bolts per Appendix A.
4. Reconnect VVCP vent. Verify piston end clearances. See Appendix B.
5. Install valves and torque the valve caps. See "Compressor Valves - Removal and Installation".

VVCP Adjustment

Users may change VVCP clearance volume with the compressor running or stopped. The expected change in compressor flow and absorbed power depends on compression ratio and properties of the compressed gas. Consult packager instructions regarding where to set the VVCP. Also see the VVCP data sheet in the Ariel Parts Book. To open VVCP to a desired percentage:

1. Loosen the locking handle or adjustment locknut so the piston stem can turn freely.
2. Find the VVCP dimension plate.
3. Subtract fully closed dimension from fully open dimension. The Ariel Performance program also lists these dimensions if the VVCP dimension plate is missing.

Figure: VVCP Dimension Plate



4. Multiply the step 3 result by the desired percentage expressed as a decimal.
5. Add the step 4 result to the fully closed dimension.
6. Turn the adjustment handle until the measurement from the base of the locking handle, when locked, to the top of the adjustment handle equals the step 5 result.

   If you turn the adjustment handle or piston stem too fast to fully close the VVCP, the piston can hit the VVCP head. If this happens, the adjustment handle or piston stem will lock and no longer turn. To correct:

   a. Remove the VVCP (refer to VVCP Removal).

b. Loosen the socket head bolts (refer to VVCP Disassembly).
c. Turn the adjustment handle 1 turn counterclockwise.

> **CAUTION:**
> **Do not breathe gas emission from VVCP.**

d. Tighten socket head bolts to the correct torque per Appendix A.

The handle should now turn freely. If not, contact Ariel.

Compressor Valves - Removal and Installation

Before servicing any valve, see the correct valve assembly drawing, parts list, and service literature in the Ariel Parts Book. The cylinder cover sheet in the Ariel Parts Book lists the valve originally supplied with each cylinder. Changes in operating pressures, temperatures, RPM, or gas composition may require alternate valve configuration including springing and lift. Contact your Packager and/or Ariel for help in valve selection.

Required Tools and Materials

- Appropriate Ariel parts list
- Appropriate Ariel Maintenance and Repair manual
- Calibrated torque wrench with the appropriate range for target values
- Lubricant specified in Appendix A
- Appropriate personal protection equipment
- Bottle brush
- Pressurized liquid solvent
- Valve Tool

Valve Cap Removal

WARNING:

To decrease the risk of injury or death, only trained and approved personnel can do maintenance work. Approved personnel must read and know the applicable Ariel Technical Manual and Packager-supplied information.

Before you start maintenance:

- Always use personal protection equipment (PPE) and obey your company policies for safety and operation.
- Do not tighten or loosen fasteners or fittings while the unit is in operation or is pressurized.
- Make sure the driver and compressor cannot turn during maintenance.
- Obey your company and packager instructions to isolate and vent the system.
- Make sure there is no remaining pressure in the cylinders. If equipped with “Kiene” valves on the indicator ports, open them to release residual pressure after blowdown.

  WARNING: RELEASED GAS MAY BE POISONOUS AND/OR FLAMMABLE.

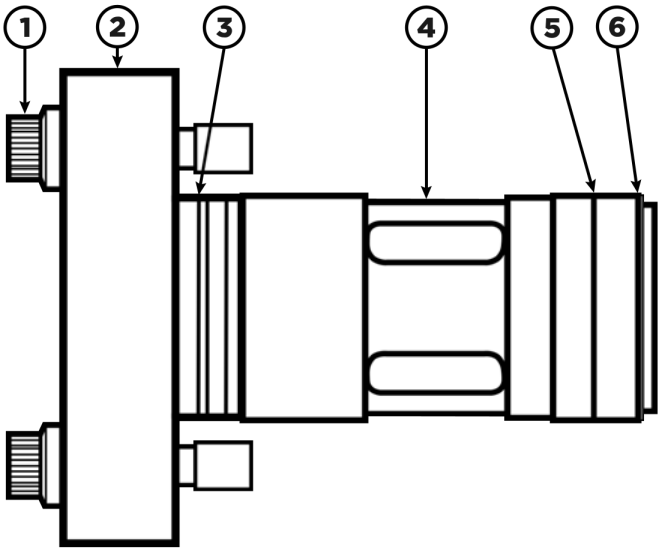
Before operation, purge the system. A pressurized hydrocarbon and air mixture can explode and cause injury or death.

1. Slightly loosen all valve cap fasteners in steps and evenly; the cap should stay in its original position. If it pushes outward, **STOP!** Completely vent the cylinder. See Warning above.
2. Continue to turn the cap screws out, for a total of three complete 360° turns. Pry the valve cap out, observing cap movement to ensure there is no gas pressure. Perform steps 1 and 2 for a suction and discharge valve cap on each acting end of the cylinder.
3. Remove cap screws and valve cap. Take care to prevent damage to fasteners.

WARNING:

Even with the above precautions, the cylinder bore could still be pressurized if the valve assemblies are fouled with deposits from the gas stream. Continue to use caution until at least one complete valve assembly has been removed from each acting end of the cylinder, which completely assures there is no gas pressure.

Figure: Valve Cap Assembly



1	Cap Screw	4	Valve Retainer
2	Valve Cap	5	Valve Assembly
3	O-Ring	6	Valve Seat Gasket

CAUTION:

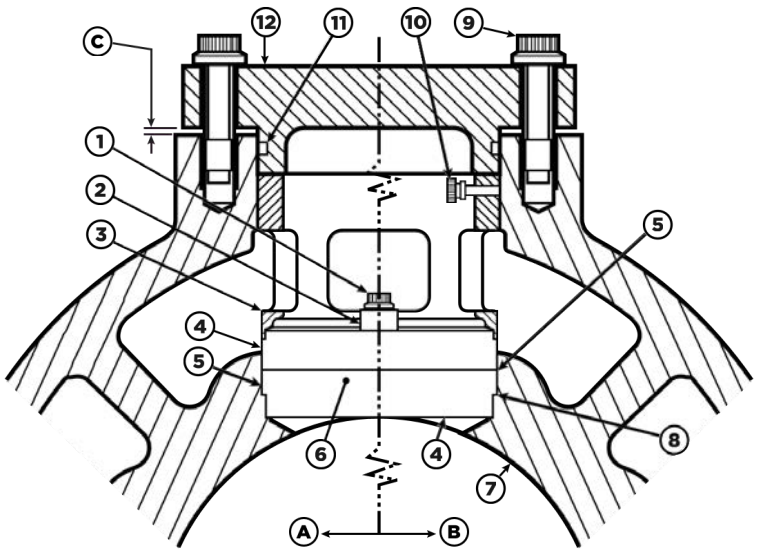
Valve cap, retainer, and valve may fall out of bottom pockets if not supported. Support them after cap screw removal to prevent personal injury.

Valve Removal

1. With the valve cap removed and the valve retainer still in place, install a valve tool onto the valve center bolt or threaded washer. See Section 1 for an illustration of the tool.
2. It may be necessary to loosen the plastic thumb screws in the valve retainers. In some cylinder classes, the discharge valve retainer is held in place with an O-ring - simply pull the retainer out to remove it.
3. Remove the valve and retainer together.
4. In most cases, the flat metal gasket remains in the pocket. It is difficult to see. A flashlight and a small mirror on an adjustable rod are the best tools to see the gasket clearly. On cylinders with horizontal valves, the gasket may fall into the gas passage. Use a small magnet on a flexible extension rod to remove gasket from cylinder.
5. Head end tandem cylinders require removal of discharge piping and the cylinder head to access the concentric valve. A concentric valve combines suction and discharge valves in one assembly.

Valve Installation

Figure: Typical Valve Assemblies



- **A** Suction Valve
- **B** Discharge Valve
- **C** 1/8 in. (3 mm)
- **1** 12-Point Capscrew (Center Bolt)
- **2** Threaded Washer
- **3** Valve Retainer
- **4** Seat
- **5** Guard
- **6** Valve
- **7** Cylinder Bore
- **8** Valve Seat Gasket
- **9** Valve Cap Bolt
- **10** Retainer-Keeper (Thumbscrew)
- **11** Valve Cap O-Ring
- **12** Valve Cap

1. Coat valve seat gasket with Never Seez. Either insert it into the valve pocket or stick it on the valve. In either case, do not allow the gasket to fall into the gas passage. Replace valve seat gaskets anytime a valve is removed from a cylinder, if the unit has been in service.
2. Install suction valves only in suction pockets and discharge valves only in discharge pockets. The valve pockets have identification plates. Install all valves with valve fastener(s) positioned away from the cylinder bore (see Item 1, Figure: Typical Valve Assemblies). If a valve is not marked for suction or discharge, manually depress the valve plate to verify the type. Preferably, use a tool softer than the valve plate material or exercise care to prevent damage to the plate. A suction valve plate may be depressed only from the valve fastener (bolting) side of the valve; a discharge valve plate may be depressed only from the side of the valve that faces the cylinder bore.

3. Use the valve tool to insert the valve and retainer into the pocket together (see tools in Section 1). Verify the valve seats properly in the pocket. When installed correctly, the valve rotates freely by hand.
4. Plastic thumbscrews or o-rings hold valve retainers in position in bottom valve pockets. Tighten thumbscrews just enough to provide friction so retainers and valves in bottom pockets stay in position during valve cap installation. **Do not replace plastic thumbscrews with metal.**
5. Lightly lubricate a new O-ring with oil and install it in the groove on the valve cap nose.

Valve Cap Installation

Depending on the application, Ariel compressor cylinders utilize different styles of valve caps. Cast iron cylinders utilize valve caps in which an O-ring on the valve cap seals gas within the cylinder.

Ensure all threads are clean and free of debris or burrs. Use a bottle brush and pressurized liquid solvent to clean threaded holes thoroughly. Avoid sending debris into cylinder gas passage. Bolts should thread into hole and bottom out on the valve cap by hand without resistance or use of a wrench. If bolts do not thread into hole freely, contact ARC for thread repair instructions.

See Appendix A for critical information on torque accuracy. Valve cap bolt failures can occur if the cap is torqued improperly. Draw bolting to full torque in even, gradual steps, without bias on one bolt or cocking the valve cap in the bore. Such bias or cocking can unevenly crush the gasket, which may cause leakage and/or bolt failure. In correct assemblies with new valves, the distance from the underside of the cap to the surface of the cylinder is a uniform 1/8 inch (3 mm). The distance is slightly less with re-worked valves. Proper tightening of gas containment fasteners is critical to ensure safe and reliable operation.

> **CAUTION:**
>
> Severe personal injury and property damage may result from improperly torqued valve cap bolts. See Appendix A for correct torque and detailed recommendations.

Gas Containment Fastener Torque Checks

Per Section 5 (start-up checklists) and Section 3 (Recommended Maintenance Intervals), Ariel recommends checking all gas containment fastener torques by the schedule below. Gas containment components include cylinders, valve caps, cylinder heads, unloaders, gas passage caps, flanges, and packings. Include this schedule in your computerized maintenance management system (CMMS):

1. Prior to start-up
2. 24 hours after start-up
3. 750 hours after start-up
4. Every 8,000 hours after start-up

Per Section 3, re-check any fasteners found loose in any of these intervals after an additional 750 hours. If loosening continues, contact your packager immediately.

Piston and Rod

Most piston and rod assemblies consist of separable pistons that attach to the piston rod with a crank-end collar and a head-end nut. Some cylinder classes require other designs, including one-piece (integral) piston/piston rods, and there are some multi-piece assemblies that do not use a crank-end collar.

Piston Rod Inspection

At the specified recommended maintenance interval, inspect piston rods for damage and excessive wear.

- For condemning limits, see Section 3.

Piston and Rod Removal

WARNING:

To decrease the risk of injury or death, only trained and approved personnel can do maintenance work. Approved personnel must read and know the applicable Ariel Technical Manual and Packager-supplied information.

Before you start maintenance:

- Always use personal protection equipment (PPE) and obey your company policies for safety and operation.
- Do not tighten or loosen fasteners or fittings while the unit is in operation or is pressurized.
- Make sure the driver and compressor cannot turn during maintenance.
- Obey your company and packager instructions to isolate and vent the system.
- Make sure there is no remaining pressure in the cylinders. If equipped with “Kiene” valves on the indicator ports, open them to release residual pressure after blowdown.

  WARNING: RELEASED GAS MAY BE POISONOUS AND/OR FLAMMABLE.

Before operation, purge the system. A pressurized hydrocarbon and air mixture can explode and cause injury or death.

NOTE:

See Section 1 for illustrations of tools mentioned below.

1. Loosen cylinder head bolts. Remove cylinder head and all crosshead guide side covers.
2. Move crosshead to inner dead center. Back off, but do not remove, the crosshead balance nut set screws. Loosen crosshead balance nut with the special slugging wrench. Use the open end wrench for hex nuts or the peg wrench for round nuts, or use the separately purchased hydraulic crosshead balance nut torque tool.
3. In tandem cylinders, where the outboard cylinder bore is smaller than the inboard bore, remove the outboard cylinder. Support such cylinders during removal and installation, to avoid excessive weight on the piston and rod assembly that may bend them.
4. Move crosshead to outer dead center, then use the piston nut spanner to turn the piston and rod assembly out of the crosshead. The two dowels on the tool fit the piston nut holes. Remove the crosshead nut from the piston rod.
5. Slide the piston rod entering sleeve over the piston rod threads. With extreme care, slowly slide piston rod through packing so as not to damage wiper or packing rings.
6. For one-piece piston rings, slide the piston rod assembly from the cylinder, then carefully remove the piston rings. For two-piece piston rings (in high pressure applications), slide the piston rod assembly out of cylinder until a fraction of the first ring clears the cylinder. Hold the first ring in the piston groove by hand, or use a band for larger sizes, until the ring clears the cylinder, then carefully remove the ring. This prevents the ring from popping out of the groove and damaging the ring. Remove succeeding two-piece rings and wear band in the same way. Handle all piston rings carefully with clean tools and hands to protect them from nicks, marring, and bending. Despite their toughness in service, some rings are fragile when removed.

7. The weight of the piston and rod assembly is stamped on the head end. Support the piston either manually or with a crane as appropriate as it slides from the cylinder bore. The piston rod must remain parallel with the bore until it leaves the bore completely. Take care not to ding any surfaces.

Hydraulic Tensioner Tool

Proper preload is essential for piston nuts. Ariel offers a hydraulic tensioner tool as an option to assemble and disassemble only KBE piston rod assemblies that require the tool.

Hydraulic Tensioner Tool Setup

Set up tensioner tool with correct adapter foot plate, dowel drive pins, and puller screw insert for the particular piston and rod (per Tables: Hydraulic Tensioner Tool A-24512 Setup and Hydraulic Tensioner Tool A-28362 Setup). The tool comes with multiple puller screw inserts, two pairs of different size dowel drive pins to match each puller screw, and three adapter foot plates. A change in puller screw size requires a change in dowel drive pins. Puller screws with wrench flats and dowel drive pins with recessed Allen sockets facilitate removal and installation.

Table: Hydraulic Tensioner Tool A-24512 Setup

Frame	Thread Size	Foot Plate	Puller Screw	Dowel Pin	Thread Projection, In.
KBE	7/8-9	A-24531	A-24524	A-24529	15/16

Table: Hydraulic Tensioner Tool A-28362 Setup

Frame	Thread Size	Foot Plate	Puller Screw	Dowel Pin	Thread Projection, In.
KBE	7/8-9	A-28363	A-24524	A-28367	15/16

Piston & Rod Disassembly with Hydraulic Tensioner Tool

CAUTION:

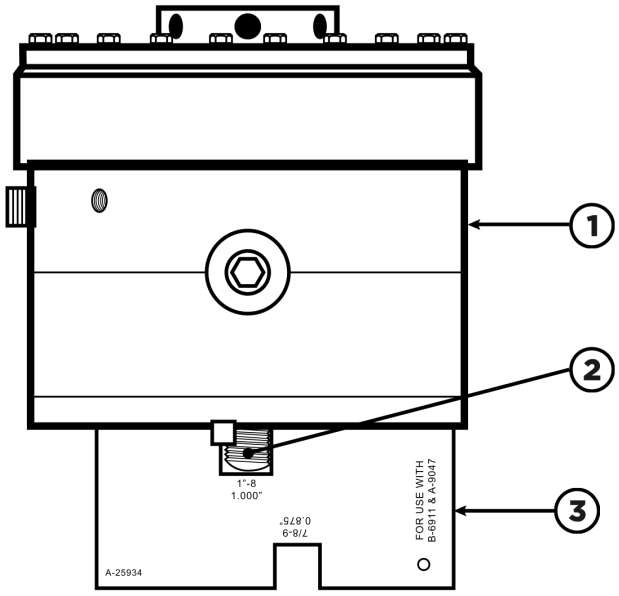
Trapped gas pressure can present a personal safety hazard when disassembing the piston and rod. Work in a well-ventilated, non-sparking area, particularly with sour gas applications. Do not breathe gas emission when venting trapped gas.

1. Remove the two Allen set screws from the piston nut and discard.
2. Clean and lubricate threaded hole in piston rod.
3. Ensure the tool hydraulic cylinder is de-energized and fully retracted. Use the inspection gauge to verify the puller screw has the proper projection (see Figure: Typical Inspection Gauge).
4. Position tensioner tool with the two dowels inserted into the piston nut. Tighten puller screw until tensioner tool is completely tight against the piston rod assembly, then back off 1/2 turn.

> **NOTE:**
>
> The puller screw of the tensioner tool has a 3/4 inch hex to tighten or loosen the puller screw.

> **CAUTION:**
>
> If tool fails to completely tighten against piston assembly, STOP. Inspect tool to find the cause. Correct the problem and try again. Configure tool properly.

Figure: Typical Inspection Gauge



1 Tensioner Tool
2 Puller Screw
3 Inspection Gauge

5. Apply hydraulic pressure to the tensioner tool to stretch piston rod (see Appendix A). Use clean ISO 32 hydraulic fluid in pump/tool system. Ariel offers an appropriate 10,000 psi manual pump kit for use with the tool.

> **CAUTION:**
>
> Do not overpressure torque tool. It can cause tool failure and/or excessive piston rod pre-load, which may cause piston rod failure and/or personal injury.

6. Loosen piston rod nut. Release hydraulic pressure, remove tensioner tool, and then remove the piston nut.

Piston & Rod Reassembly with Hydraulic Tensioner Tool

1. Clean all piston and rod assembly parts thoroughly.

> **NOTE:**
>
> Any reassembly of used parts requires re-cleaning and re-lubrication of threads and seating surfaces.

2. Inspect collar (or collar area on rod, when collar is integral and not separable), piston, and piston nut for nicks, burrs, or scratches. Replace if surfaces are damaged, worn, or galled. Inspect piston ring grooves. If damaged, or ring groove width is out of tolerance, replace the piston (see Appendix B for tolerances). Dress surfaces with a fine grit stone as required. Fit collar and nut into piston to verify the outside diameter fits and turns freely by hand in the piston.
3. Inspect piston rod threads and collar shoulder. Clean and de-burr threads. Install collar and nut onto piston rod to verify inside diameter fits and rotates freely. Turn piston nut until rod threads protrude to verify freedom of thread engagement. Remove nut and collar.

4. Apply a thin coat of Never-Seez Regular Grade to piston rod shoulder, rod collar locating band, and collar face in contact with piston, then slide collar onto rod.
5. Apply a thin coat of Never-Seez Regular Grade to piston rod threads at the piston end, then slide piston onto rod and collar.

> **NOTE:**
>
> If one end of the piston has an undercut, and the piston has the same number of piston ring grooves on each side of the wearband, assemble with the undercut toward the head end. For pistons with a different number of piston ring grooves on each side of the wearband, assemble with the side of fewer piston ring grooves toward the head end. See Figure: Typical Inspection Gauge.

6. Apply a thin coat of Never-Seez Regular Grade to piston nut threads and piston mating face. Install nut and hand tighten to make up the piston rod assembly.
7. Position tensioner tool with the two dowels inserted into the piston nut.

> **CAUTION:**
>
> If tool fails to completely tighten against piston assembly, STOP. Inspect tool to find the cause. Correct the problem and try again. Configure tool properly.

8. Apply hydraulic pressure listed in Appendix A to tensioner tool to stretch the piston rod. To tighten piston rod nut, use a calibrated torque wrench to torque the spanner ring to the torque listed in Appendix A. Keep the torque wrench perpendicular to the tool housing.

> **CAUTION:**
>
> Maintain clearance between the torque wrench and the tensioner tool window opening when torquing spanner ring. Proper seating torque will not be applied to the spanner ring if the torque wrench hits the window opening. Do not overpressure torque tool. It can cause tool failure and/or excessive piston rod pre-load, which may cause piston rod failure and/or personal injury.

> **NOTE:**
>
> Periodically calibrate hydraulic pressure gauges for the required pressure. Use optional Ariel hydraulic hand pump kit.

9. Release hydraulic pressure. Re-apply hydraulic pressure, use a breaker bar to loosen piston nut without disassembling, and then re-tighten to the recommended torque. This double-torquing ensures piston assembly integrity.
10. Release hydraulic pressure and remove tensioner tool. Inspect the piston rod at the end of the piston; it should not protrude more than 0.010 inch (0.25 mm) past the piston face. The nut should be flush or recessed. Verify piston nut has turned a minimum of 95° to achieve proper pre-load by comparing the marks made earlier across the piston and nut faces.
11. Apply a thin coat of Never-Seez Regular Grade to two new Allen set screws. To install a set screw, tighten it 15° past the Allen wrench yield point. Discard the deformed Allen wrench, and use a new Allen wrench to tighten the other set screw in the same way. If Allen wrenches are unavailable, use the torques in Appendix A.

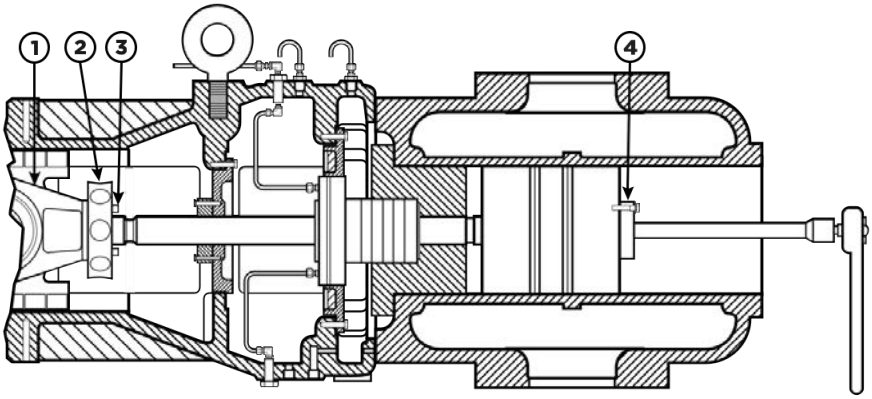
12. Use a calibrated scale to weigh piston rod assembly with piston rings and wear band. Stamp weight on piston head end. Flatten any raised lips to avoid clearance measurement errors. Record weight on a new balance sheet and send to Ariel. See Appendix C for approximate component weights.

Figure: Staking a Set Screw



Piston and Rod Installation

Figure: Typical Piston and Rod Installation



1 Crosshead
2 Crosshead Nut
3 Set Screw
4 Piston Nut Spanner

1. Install piston rings in cylinder main bore without piston to measure end gaps. Ariel recommends replacing piston rings when the end gap equals three times its original specification. See Appendix B for specification.
2. Except for non-lube cylinders, liberally lubricate the cylinder bore, piston rod, piston rings, wearbands and piston with compatible cylinder oil. Install rings on piston with ring gaps staggered.
   a. For cylinders requiring two-piece piston rings or wear bands, verify that they are a set by the match marks. Two-piece rings use dots as match marks - verify the dots face the pressure side of cylinder. Two-piece piston rings and wear bands are manufactured as matched sets; mixing ring halves may result in incorrect end gaps.
   b. For all other rings, see "Piston Rings".
3. Slide the piston rod entering sleeve over the piston rod threads. Damage to the packing rings is likely if an entry sleeve is not used.
4. Slide piston rod with piston rings and wear band into cylinder bore. A worn wear band can be used as a ring compressor for new piston rings:
   a. Place the worn wear band over the piston.
   b. Position the wear band onto the crank end rings and begin sliding the piston into the cylinder. Ensure the wearband covers piston ring end gaps.
   c. As the piston slides further in, continue shifting the wear band further back until all piston rings are inside the cylinder.
   d. Remove the worn wear band and continue the installation.
5. Using a dial indicator, position the crosshead at the inner dead center position.
6. Verify the balance nut and crosshead mating surfaces are flat and free from dings or other disparities. If necessary use a flat stone to smooth the surfaces.
7. Lubricate piston rod and crosshead threads with Never-Seez Regular Grade. Apply a thin coat of oil to either the balance nut or crosshead mating surface. Do not apply Never-Seez to the mating surfaces. Install crosshead-balance nut on piston rod.
8. Turn the piston rod into the crosshead. Visually center the packing flange around the piston rod and torque packing bolts per Appendix A.

9. Set piston end clearance (see Appendix B):
   a. Using the piston nut spanner, turn the piston rod into the crosshead.
   b. Insert a feeler gauge, equal to the required crank end clearance, through an open valve pocket. For 13 in. (330 mm) and larger cylinders, insert feeler gauge through a bottom valve pocket. Place the feeler gauge as close to the piston rod as possible. Tighten the piston rod into the crosshead until piston is tight against the feeler gauge, and the feeler gauge cannot be easily removed.
   c. Tighten the crosshead nut by the slugging method or by hydraulic torquing device. See "Crosshead Installation" for hydraulic torquing device. After tightening, the feeler gauge may be removed, usually with slight resistance.
   d. Tighten the balance nut set screws and be sure to remove the piston nut spanner.
10. See "Crosshead Installation" for details to re-attach piston rod to crosshead, check piston end clearance, piston rod runout and crosshead clearances, and re-assemble to close cylinder and crosshead guide. See Appendix B and "Piston Rod Runout".

CAUTION:

Ariel recommends leaving distance piece covers off after any piston rod or rod packing ring maintenance. Verify no ignition sources exist in the area, pressurize the unit, and check for gas leaks before distance piece cover installation. This protects against crosshead guide over-pressure and possible ignition due to incorrect installation of packing case or components. When checking for leaks, take proper precautions in process gas applications, such as H_2S.

Piston Rod Runout

Check piston rod runout after new unit installation, unit relocation, or maintenance that may affect rod runout.

Table: KBE Max. Piston Rod Runout, Inch (mm)[a]

Vertical	0.0020 (0.051)
Horizontal	0.0010 (0.025)

a Machines with readings falling outside the limits require disposition by Ariel.

Position the stem of a 0.0001 inch (0.001 mm) increment calibrated dial indicator against the piston rod, close to the packing case. Set the indicator to zero with the piston toward the crank end. Bar over crankshaft and record indicator readings with the piston at mid-stroke and H. E. positions, in both vertical and horizontal directions. Record upward vertical rod movement as positive and downward as negative. Record horizontal rod movement toward the auxiliary end of the frame as positive and toward the drive end as negative. Copy the table below to record readings and compare to Table: KBE Max. Piston Rod Runout, Inch (mm) values.

Table: Piston Rod Runout, Inches

Throw		1	2	3	4
Vertical (0°)	Piston @ C. E.	0.0000	0.0000	0.0000	0.0000
	Mid-Stroke				
	Piston @ H. E.				
Horizontal (0°)	Piston @ C. E.	0.0000	0.0000	0.0000	0.0000
	Mid-Stroke				
	Piston @ H. E.				

If a vertical reading is greater than the maximum acceptable, use this procedure to determine component acceptability: Break the balance nut loose. Looking from the head end, turn the piston 90° counter-clockwise, and re-torque the balance nut. Record both vertical and horizontal readings in the table below. If readings are within 0.0003 inch of the original runout readings above, the rod is acceptable - break the balance nut loose. Re-set C.E. clearance and re-torque the balance nut. If rod runout is not within 0.0003 inch of the original reading above, contact Ariel.

Table: Alternate Piston Rod Runout, Inches

Throw		1	2	3	4	5	6
Vertical (90°)	Piston @ C. E.	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	Mid-Stroke						
	Piston @ H. E.						
Horizontal (90°)	Piston @ C. E.	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	Mid-Stroke						
	Piston @ H. E.						
Vert. Runout Difference[a]							
Horiz. Runout Difference[a]							

a Piston @ H. E., original minus alternate.

Piston Rings

Most cylinders use one-piece angle-cut filled PTFE piston rings. High-pressure cylinders use two-piece thermoplastic rings. Ariel recommends replacing rings when the end gap increases to three times the new dimension. To measure end gaps, insert rings in the cylinders without pistons (see Appendix B for minimum end gap dimensions).

Seal-Joint Piston Rings

Seal-joint piston rings are directional and require correct installation.

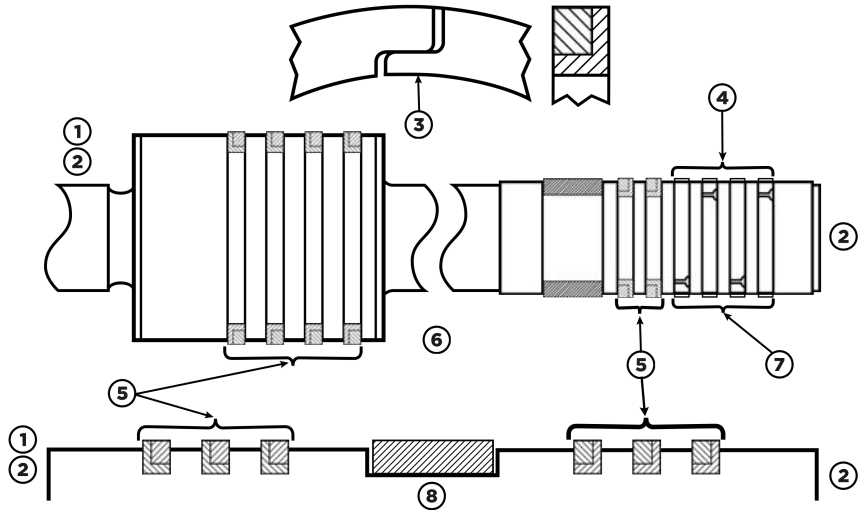
NOTE:

On tandem cylinders with a combination of seal-joint rings and other ring styles, always position the seal-joint rings farthest from the pressure side. See Figure: Seal-Joint, Straight-Cut, and Angle-Cut Piston Ring Installation Orientation.

Straight-Cut Chamfered Piston Rings

Straight-cut chamfered piston rings are directional and require correct installation. Install them with chamfers or dots toward the pressure and with the end gaps staggered.

Figure: Seal-Joint, Straight-Cut, and Angle-Cut Piston Ring Installation Orientation

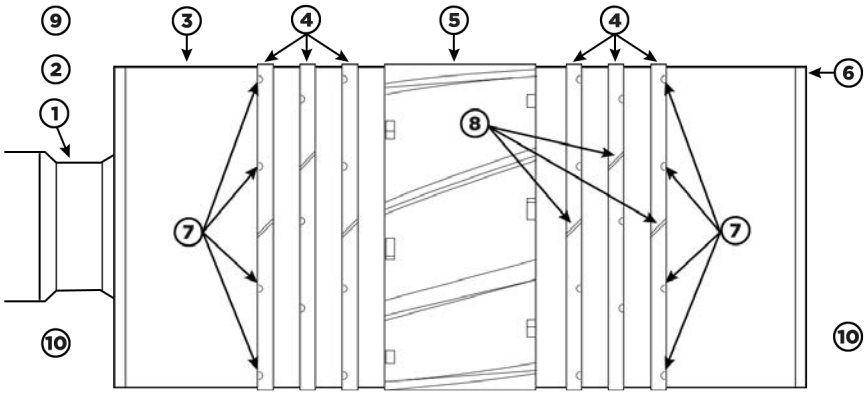


1 Frame Side
2 Pressure
3 Seal-Joint Ring
4 Chamfers or Match Marks toward Pressure, End Gaps Staggered
5 Match Mark Side toward Pressure
6 Typical for Tandem Cylinder
7 Angle Cut or Straight Cut Rings
8 Typical for Double Acting Cylinder - with or without Wearband

High-Pressure Face-Cut Piston Rings

Face-cut piston rings are typically used in cylinders with 2500 psi MAWP or higher, and generally with notched and fluted wear bands. **The rings install with the face cuts toward the highest pressure, or toward the nearest head.** Gas pressure in the wear band area escapes past the piston rings via the face cuts during the suction stroke to prevent loading the wear band and increase service life. Also, do not align the ring end gaps; stagger them to minimize gas leakage.

Figure: High-Pressure Face-Cut Piston Ring Orientation

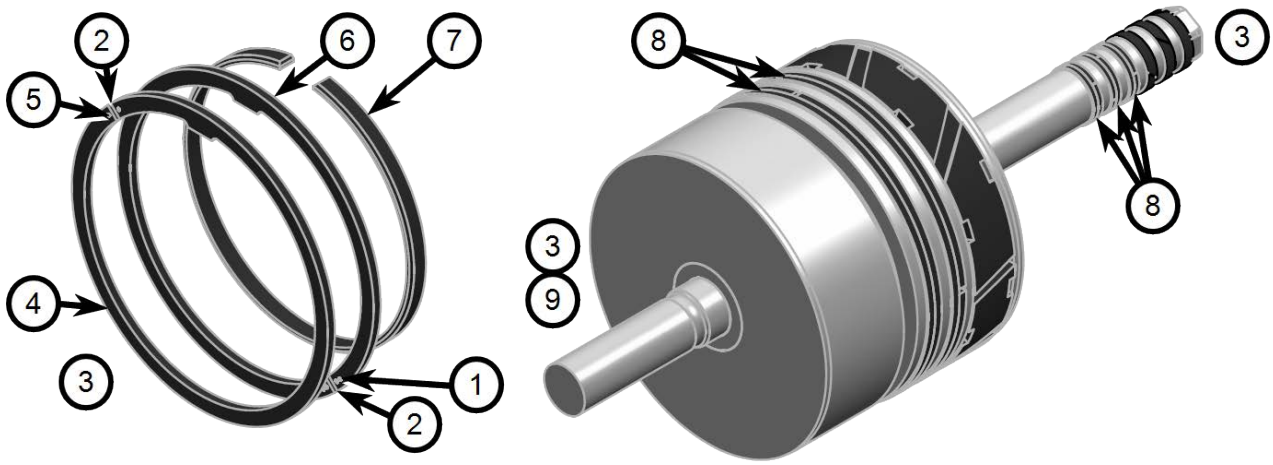


1 Piston Rod
2 Crank End
3 Piston
4 Piston Rings
5 Wear Band
6 Head End
7 Notches
8 Staggered End Gaps
9 Frame Side
10 Pressure

Three-Piece Piston Rings

This ring style consists of two “outer” rings installed over an “inner” ring. The first outer ring has double match marks and the second outer ring has single match marks. The inner ring rests in the bottom of the piston ring groove; install with expander chamfer facing the nearest head or highest pressure (see Figure: Three-Piece Piston Ring Installation Orientation). Install outer rings with match marks facing the nearest head or highest pressure and end gaps staggered to prevent leakage.

Figure: Three-Piece Piston Ring Installation Orientation

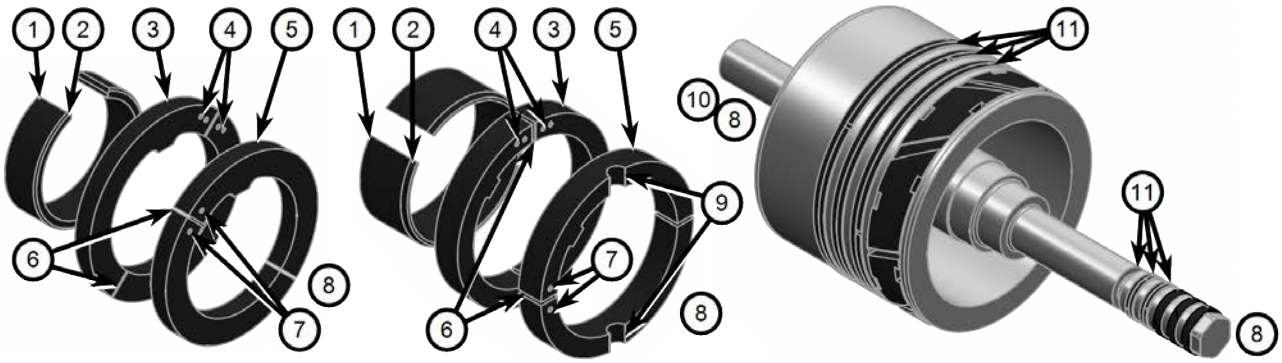


1 Double Match Marks
2 Staggered End Gaps
3 Pressure
4 Second Outer Ring
5 Single Match Marks
6 First Outer Ring
7 Inner Ring
8 Match Marks toward Pressure
9 Frame Side

Five-Piece Piston Rings

A five-piece ring set consists of two outer rings cut in half and an inner ring with a single cut. In the five-piece design, Ariel uses two different styles: face-cut (notched) and non-face-cut. Ring orientation is critical to proper function.

Figure: Five-Piece Piston Ring Installation Orientation



Non-Face-Cut Rings
1 Inner Ring
2 Chamfer (where applicable)
3 First Outer Ring
4 Double Match Marks

Face-Cut Rings
5 Second Outer Ring
6 Staggered End Gaps
7 Single Match Marks
8 Pressure

Piston Rod Assembly
9 Face-Cut
10 Frame Side
11 Match Marks toward Pressure

Face-Cut (Notched) Rings: When the inner ring is chamfered, the chamfer faces the nearest head or highest pressure. The first outer ring has no face-cut (notch) and double match marks, and the second outer ring has a face-cut (notch) and single match marks. Install both outer rings with match marks facing the nearest head or highest pressure. When the match marks are correct, the end gaps of the outer rings do not align.

Non-Face-Cut Rings: When the inner ring is chamfered, the chamfer faces the nearest head or highest pressure. The first outer ring has double match marks and the second outer ring has single match marks. Install both outer rings with match marks facing the nearest head or highest pressure. When the match marks are correct, the end gaps of the outer rings do not align.

Wear Bands

Most cylinders use one-piece angle-cut filled PTFE wear bands. High-pressure cylinders use two-piece thermoplastic bands (see Figure: High-Pressure Face-Cut Piston Ring Orientation).

During installation, check and record wear band end gap and projection. To measure end gap, insert one or two-piece wear band into cylinder without the piston then measure the total gap while the band maintains full contact with the cylinder bore. See Appendix B for end gap dimensions.

To measure wear band projection, install the entire piston assembly or remove the head to check during maintenance. Measure the gap between the piston and cylinder bore at the bottom of the cylinder. Compare the wear band projection to the as-new condition (recorded at new wear band installation or found in Appendix B). Replace the bands if there is any indication the next planned run time may fully degrade the wear band and allow the piston to contact the cylinder bore.

Piston Rod Packing

The piston rod packing prevents gas from entering the crosshead guide. Piston rod rings within the packing wear over time and need replacement.

Piston Rod Packing Removal

WARNING:

To decrease the risk of injury or death, only trained and approved personnel can do maintenance work. Approved personnel must read and know the applicable Ariel Technical Manual and Packager-supplied information.

Before you start maintenance:

- Always use personal protection equipment (PPE) and obey your company policies for safety and operation.
- Do not tighten or loosen fasteners or fittings while the unit is in operation or is pressurized.
- Make sure the driver and compressor cannot turn during maintenance.
- Obey your company and packager instructions to isolate and vent the system.
- Make sure there is no remaining pressure in the cylinders. If equipped with “Kiene” valves on the indicator ports, open them to release residual pressure after blowdown.

  WARNING: RELEASED GAS MAY BE POISONOUS AND/OR FLAMMABLE.

Before operation, purge the system. A pressurized hydrocarbon and air mixture can explode and cause injury or death.

1. Remove piston and piston rod. See "Piston and Rod Removal".
2. Remove packing diaphragm (if applicable) and oil wiper packing.

3. Disconnect all tubing and instrumentation (if applicable) from packing flange. Ensure there is no debris in the tubing between the check valve and packing flange. Remove the twelve-point cap screws that hold the pressure packing gland to the crank end of the cylinder/head.
4. The small nuts are not removed from the studs as they hold the packing case together for removal as an assembly.Do no remove the small nuts from the studs. They hold the packing case together for removal as an assembly. Slowly slide the pressure packing down through the oil wiper packing diaphragm opening, and then through the crosshead guide opening.
5. Remove the pressure packing and take it to a clean place for disassembly.
6. Set pressure packing on a clean surface on its nose cup or cylinder end. Packing cups are marked to represent the assembly order starting with the cup closest to the cylinder as number 1. Long tie studs hold the pressure packing together. The stud holes are unequally spaced to prevent misalignment of the stack of parts. Remove the nuts and unstack the pressure packing. Replace these nuts each time the pressure packing is serviced.
7. To check ring wear, place assembled rings (note match-marks) on the piston rod. Check end gap clearance. If the ends butt, or nearly butt, replace the rings. See "Types of Piston Rod Packing Rings" for correct ring orientation.
8. Pry loose the metal or spiral-wound gasket on the end cup with a sharp awl. Do not scratch the sides of the gasket groove. Discard the used gasket.
9. Before reassembly, clean all parts thoroughly.

Piston Rod Packing Assembly and Installation

1. Refer to the pressure packing assembly in your parts book. Ariel supplies parts books with each unit. Contact your distributor to obtain a parts book. Each pressure packing re-build kit includes a pressure packing assembly drawing.
2. Take care not to scratch mating surfaces of the cups. Cup surfaces must be clean and dry for re-assembly.
3. To install a new set of rod rings in an existing packing case, inspect case parts for wear. Cups should be smooth and flat on the back side where the rod rings must seal. If cups or grooves are concave or tapered, regrind or relap them. Contact Ariel for appropriate rework thickness dimensions. It is rarely necessary to alter the crosshead side of cups, but if necessary, take care not to destroy the correct side clearance for the renewal rings.

> **NOTE:**
>
> If premature wear is suspected, see “Force Feed Lubrication Conditions”.

4. Before a packing case installation, disassemble and thoroughly clean it in an appropriate solvent for the intended service.
5. Verify proper position of each rod ring and cup and, unless non-lube, coat rings liberally with new, clean lubricant before reassembly. Use only the same lubricant in the force feed lube system. Ensure tie studs are completely threaded into end cup. Examine all parts for unusual nicks or burrs which may interfere with the free floating of the rod rings in the cups. Take particular care with rod rings made of soft materials, such as bronze or PTFE. It is extremely important to handle and install wiper rings carefully to prevent damage to scraping edges.
6. Lay out parts on a work bench for progressive installation, with each part in its correct position and the rod rings with their proper faces toward the pressure. Long tie studs hold the pressure packing together. The stud holes are unequally spaced to prevent misalignment of the stack of parts. Note that all rod ring segments are carefully lettered; assemble them accordingly. This is most important for proper sealing. Center side-loaded VR1 and PR1 rings prior to tightening tie stud nuts. Install tie stud nuts and tighten to the torques in Appendix A. Manually verify all rings move freely, radially, in their grooves. Side-loaded rings are tight, but should still move manually. Center these rings.

   Packing assembly:

a. With no gasket installed, install the studs into the end cup (cup # 1).
b. Using cylinder oil, lubricate the cup and the pressure breaker ring or ring set, then install them into cup (see pressure packing assembly drawing for correct orientation).
c. Install cup # 2 and repeat Step b for the seal ring sets.
d. Repeat steps above for cups #3, #4, etc, until packing case is reassembled per the pressure packing assembly drawing.
e. Using a clean oil pump filled with cylinder oil, squirt oil into the lube fitting(s). If the packing is assembled correctly, the oil will come out the lube cup. If not, disassemble the packing and correct the error.
f. Using pressurized air, repeat the above process for vent and purge lines.

7. For new installations, carefully clean all accumulated dirt in the lines and compressor. Any foreign material lodges in the packing and becomes destructively abrasive.
8. Verify the gasket surface in the packing counter bore on the crank end of the cylinder/head is clean and not scratched. Install a new end cup gasket. Round wire gaskets will fit tightly in the gasket groove.
9. Reinstall complete packing case assembly with oil supply point on top. Use rod packing bolts to pull packing into place.
10. Install piston and rod. Refer to "Piston and Rod Installation".
11. Tighten rod packing bolts evenly in a criss-cross pattern to the recommended torque in Appendix A. This procedure squares the pressure packing on its nose gasket.

> **NOTE:**
> Repeat final torque for rod packing bolts until the bolts no longer turn. Re-check torque on these fasteners at the next service interval.

12. Install tubing connections and instruments (if applicable). Take care not to cross-thread tubing nuts. Tubing nuts must be tight.

> **NOTE:**
> After pressure packing installation, see “Force Feed Lubricator” in Section 3 to prime the force feed lube system and obtain recommended lubrication rates for new machine break-in. Priming may be necessary after extended down time. Break-in lube rates are approximately twice the normal rates, or half the normal cycle time.

> **CAUTION:**
> Ariel recommends leaving distance piece covers off after any piston rod or rod packing ring maintenance. Verify no ignition sources exist in the area, pressurize the unit, and check for gas leaks before distance piece cover installation. This protects against crosshead guide over-pressure and possible ignition due to incorrect installation of packing case or components. When checking for leaks, take proper precautions in process gas applications, such as H_2S.

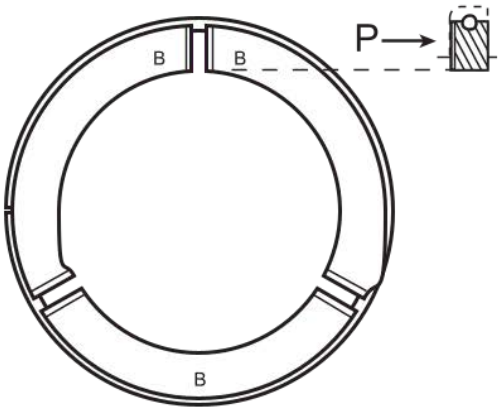
Types of Piston Rod Packing Rings

There are several types of piston rod packing rings depending on the application. Below are those used in Ariel KBE compressors.

PB1 Pressure Breaker

This single ring is cut radially into three equal segments. It breaks down or slows gas flow without sealing it completely. This ring uses a solid wire spring.

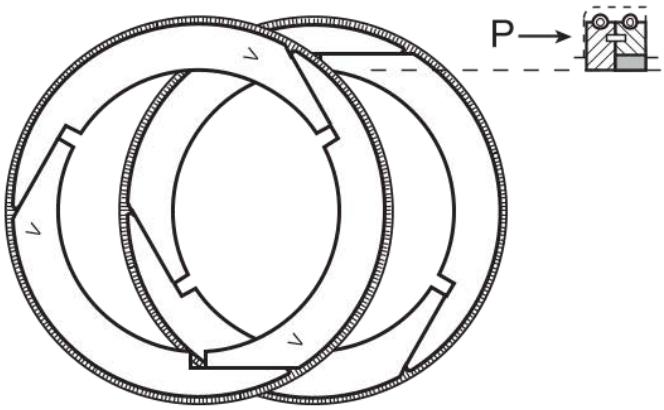
Figure: PB1 Pressure Breaker



VR Double-Acting Seal Set

This two-ring set consists of tangentially step cut rings doweled to stagger the tangential cuts from one ring to the other. This double acting set seals in either direction. In cylinders operating near atmospheric pressure, it prevents air from entering the cylinder. Install with the match mark letters facing towards the cylinder.

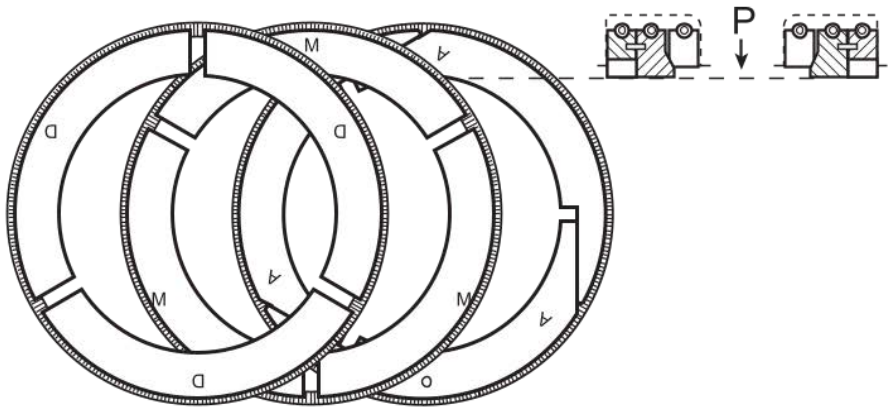
Figure: VR Double-Acting Seal Set



Double VR1 Double-Acting Seal Set

In this three-ring PTFE set, the first two rings (pressure side) are radially cut; the third is tangentially step cut. The last two rings are doweled to stagger the cuts from one ring to the other. The first ring, along with the center ring, forms a wedge that overcomes rod friction and holds the ring set against both groove faces during either direction of rod travel. Use this ring set primarily for low pressure applications. VR1 rings must be centered when installed in the packing case.

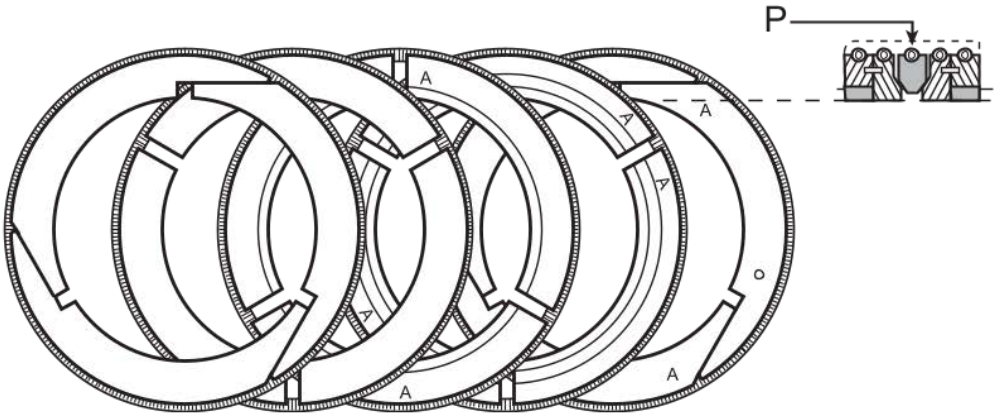
Figure: Double VR1 Double-Acting Seal Set



PR1 Double-Acting Seal Set

This five-ring PTFE set functions like a double-ended VR1 ring set. PR1 rings must be centered when installed in the packing case.

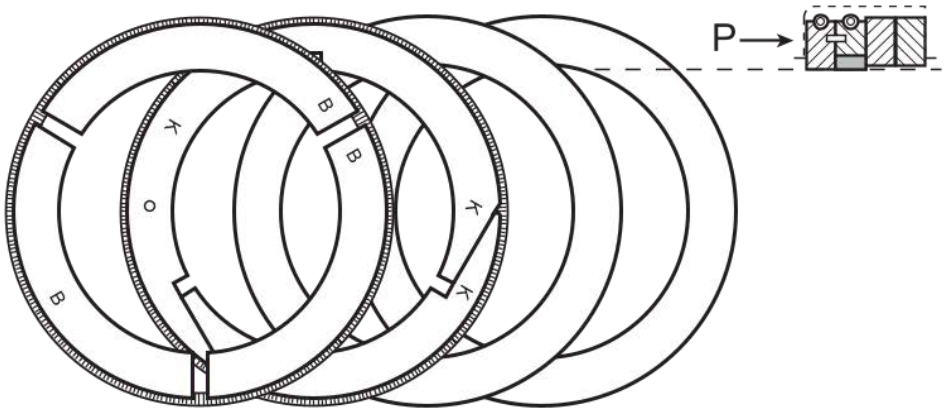
Figure: PR1 Double-Acting Seal Set



RS4 Single-Acting Seal Set

This four-ring set seals in one direction only. The first ring (pressure side) is radially cut; the second is tangentially step cut. These rings are doweled to stagger the cuts from one ring to the other. The third and fourth are uncut one-piece rings with clearance over the rod, allowing the joints of the first two rings to form a gas-tight seal. In many applications, the uncut rings will deflect under pressure and seal against the piston rod.

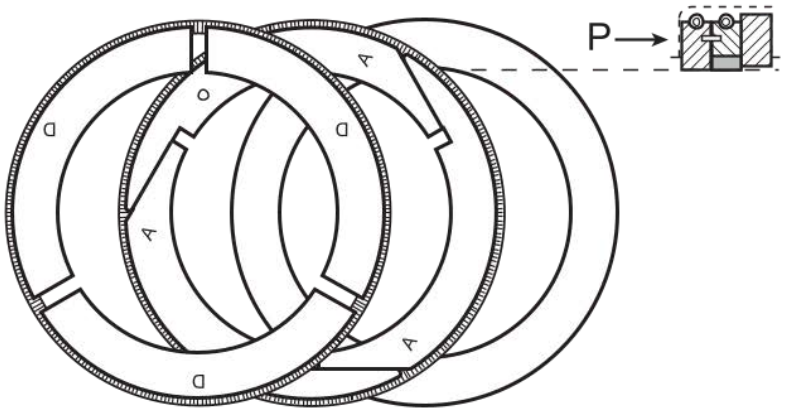
Figure: RS4 Single-Acting Seal Set



RS3 Single-Acting Seal Set

This three-ring set seals in one direction only. The first ring (pressure side) is radially cut. The second ring is tangentially step cut. These first two rings are doweled to stagger the cuts from one ring to the other. The third ring is called a back-up ring. It is a solid one-piece ring with clearance over the rod. It holds the first two rings perpendicular to the rod, allows their joints to form a seal, and prevents extrusion of the tangent ring.

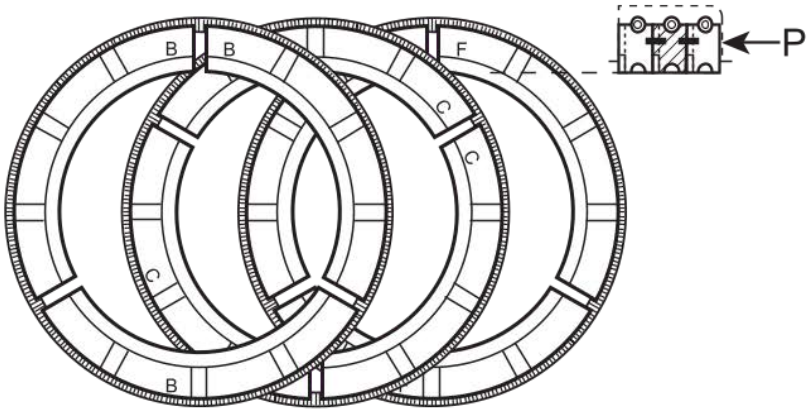
Figure: RS3 Single-Acting Seal Set



OW3 Oil Wiper Set

The three rings in this set are radially cut and doweled to stagger the cuts from one ring to the other. They keep crankcase oil out of the packing and cylinder. Assemble with the blank face towards the oil (crankcase) and the slotted side towards the pressure packing.

Figure: OW3 Oil Wiper Set



Crossheads

The crosshead assembly consists of the crosshead and crosshead pin bushings, where required. Most crossheads use a layer of soft babbitt material on the shoes, which are the surfaces that slide in the crosshead guide bore.

Take care during maintenance to avoid damaging the babbitt or the bushings.

Crosshead Removal

WARNING:

To decrease the risk of injury or death, only trained and approved personnel can do maintenance work. Approved personnel must read and know the applicable Ariel Technical Manual and Packager-supplied information.

Before you start maintenance:

- Always use personal protection equipment (PPE) and obey your company policies for safety and operation.
- Do not tighten or loosen fasteners or fittings while the unit is in operation or is pressurized.
- Make sure the driver and compressor cannot turn during maintenance.
- Obey your company and packager instructions to isolate and vent the system.
- Make sure there is no remaining pressure in the cylinders. If equipped with “Kiene” valves on the indicator ports, open them to release residual pressure after blowdown.

  WARNING: RELEASED GAS MAY BE POISONOUS AND/OR FLAMMABLE.

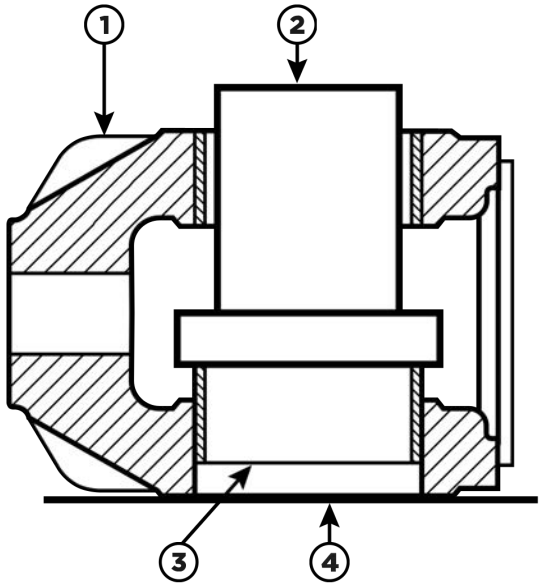
Before operation, purge the system. A pressurized hydrocarbon and air mixture can explode and cause injury or death.

NOTE:

See Section 1 for illustrations of tools mentioned below.

1. Remove crosshead guide side covers and head end cylinder.
2. Move crosshead to its inner dead center position. Back off, but do not remove, the crosshead nut set screws.
3. Loosen crosshead nut with the special slugging peg or open end wrench (depending on nut type), or use separately purchased hydraulic crosshead nut torque tool.
4. Use the piston nut spanner to turn the piston rod out of the crosshead. Remove crosshead nut from piston rod. Install the piston rod entry sleeve and push the rod end forward towards the cylinder.
5. With crosshead in its outer dead center position, remove crosshead pin thru-bolt, lock nut, end-plates and pin. Discard old lock nut.
6. Turn crankshaft to its inner dead center position.
7. Move crosshead to its outer dead center position, free of the connecting rod. **Support connecting rod so it does not drop and damage the crosshead guide surface.**
8. Push crosshead onto crosshead installation/removal tool and rotate crosshead 90°.
9. Slide a metal plate into the gap between the crosshead and crosshead guide, 3/16 inch (5 mm) thick (see Figure: Crosshead Alignment Tool).
10. Remove diaphragm from crosshead guide and slide crosshead out of crosshead guide onto the plate.
11. Check crosshead pin to bushing clearance (see Appendix B). Determine pin wear by inspection.
12. Replace pin, if necessary. To replace bushings, hacksaw or file to within 1/32 inches (1 mm) of their thickness, then drift them out.
13. New bushing installation requires a press. To install a bushing in the crosshead, cool the bushing in 95% alcohol with dry ice solution. Leave the bushing in the solution long enough to reach the same temperature as the solution, about -110°F (-80°C). An alternate method of cooling is to submerge the bushing in liquid nitrogen. When the nitrogen stops bubbling, the temperature has reached equilibrium, and the bushing is ready to install.
14. Where applicable, verify oil holes in the bushing line up with the holes in the crosshead.
15. Inspect shoe surfaces. Except for light polishing, there should be no wear or de-lamination.

Figure: Crosshead Bushing Replacement



1	Crosshead	**3**	Bushing
2	About 3 Tons (27kN) Force Required	**4**	Support Here

Crosshead Installation

NOTE:

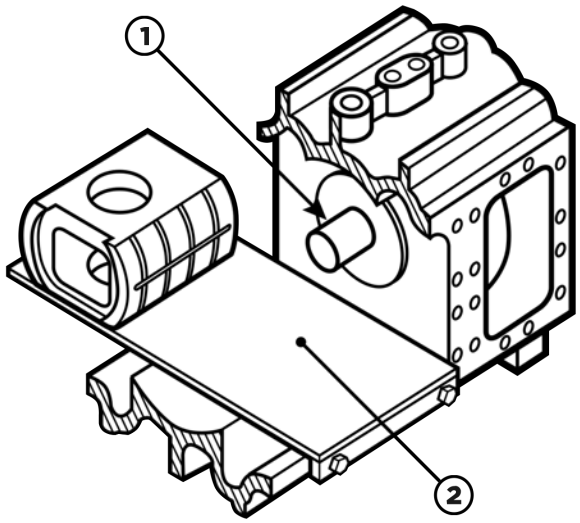
Return crossheads to their original throw location. Use frame oil for lubrication where needed.

1. Lay a 3/16 inch (5 mm) thick installation plate in the bottom of the crosshead guide, and lay crosshead on its side (see Figure: Crosshead Alignment Tool). Oil crosshead and guide surfaces and push crosshead into position.
2. Mount crosshead installation tool onto crosshead guide diaphragm and oil tool bearing surfaces (see Figure: Crosshead Alignment Tool). Slide crosshead onto tool.

NOTE:

Long two-compartment crosshead guides supplied prior to November 2008 require a crosshead installation tool with a smaller pilot diameter. Contact Ariel for the proper tool.

Figure: Crosshead Installation

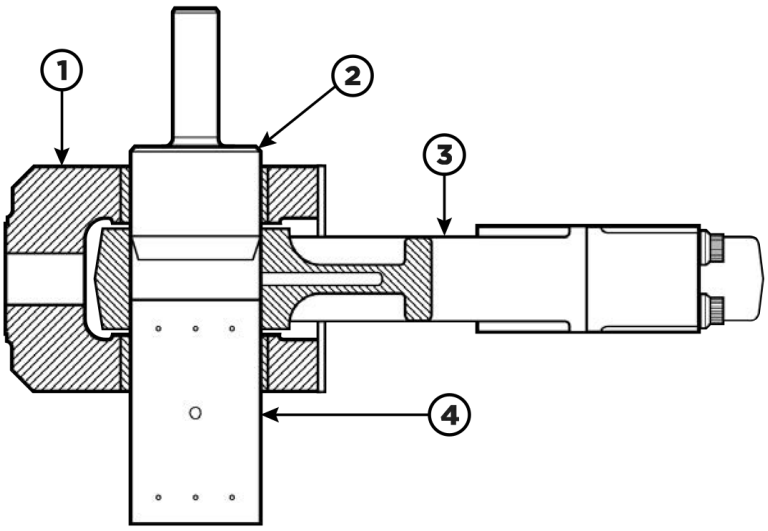


1 Crosshead Installation Tool

2 Metal Plate

3. Remove the plate. Rotate crosshead 90°. If crosshead becomes wedged, do not force it; it will damage the crosshead shoe surface. Use a box end wrench on the connecting rod bolt to lever the small end of the connecting rod into the centered position, then slide the crosshead into the guide under the connecting rod and off of the installation tool. Be careful not to damage crosshead shoe or crosshead slide surfaces during installation.
4. Remove crosshead installation tool and reinstall wiper packing.
5. Turn crankshaft to its outer dead center position. Deburr and clean the crosshead pin alignment tool for use on the opposite side of the crosshead to assist pin insertion. Oil crosshead pin, crosshead, crosshead pin alignment tool, and connecting rod bushing prior to installation.
6. Install crosshead pin alignment tool. Install crosshead pin and remove crosshead pin alignment tool. Install crosshead pin end-plates (be careful to align roll pins with holes in the end plates), thru-bolt, and a new lock nut. Tighten thru-bolt and lock nut to the torque listed in Appendix A.

Figure: Crosshead Alignment Tool



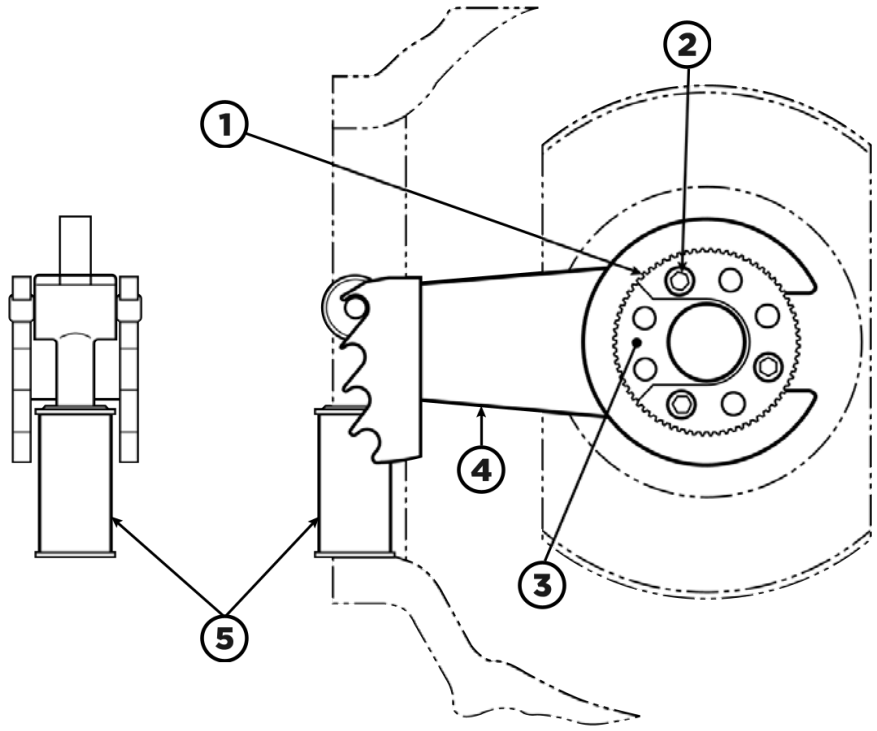
1 Crosshead
2 Alignment Tool
3 Connecting Rod
4 Crosshead Pin

7. Reposition throw to inner dead center and push the piston rod assembly toward the crosshead far enough to remove piston rod entry sleeve.
8. Lubricate piston rod threads with Never-Seez regular grade. Apply a thin coat of oil to either the balance nut or crosshead mating surface. Do not apply Never-Seez to the mating surfaces. Install crosshead-balance nut on piston rod.
9. Push the rod into the crosshead and turn the piston rod assembly until the required crank end piston clearance is reached.

> **NOTE:**
>
> At this point, set piston end clearance or serious damage may occur. See required piston crank end clearance on cylinder data plate or Appendix B.

10. Insert a feeler gauge, equal to the required crank end clearance, through an open valve pocket. For 13 in. (330 mm) and larger cylinders, insert feeler gauge through a bottom valve pocket. Place the feeler gauge as close to the piston rod as possible. Tighten the piston rod into the crosshead until piston is tight against the feeler gauge, and the feeler gauge cannot be easily removed.
11. Tighten crosshead-balance nut in one of two ways:
    a. Strike the special slugging peg or open end wrench with a dead blow semi-soft faced hammer until an audible difference can be heard or the wrench “bounces” indicating a tight nut. Some mechanics obtain desired tightness in 3 to 4 strikes; others may require more.
    b. Use a separately purchased crosshead-balance nut hydraulic torque tool and hand pump kit. See Appendix A for the hydraulic pressure to apply to the torque tool. When initially installing the tool, position the spanner wrench adapter open area toward the spanner wrench as shown in "Figure: Typical Crosshead Balance Nut Hydraulic Torque Tool". Apply hydraulic pressure to the ram to tighten the crosshead-balance nut. When the ram ends its travel, release pressure and index the ram to the spanner wrench slot and/or spanner wrench to adapter, until the nut stops moving at the specified hydraulic pressure (see Appendix A). During tightening, if the adapter open area turns 90° before the nut is tight, remove the spanner wrench. Remove the bolting in the spanner wrench adapter to the crosshead-balance nut and reposition the adapter so the open area again faces the spanner wrench.

Figure: Typical Crosshead Balance Nut Hydraulic Torque Tool - Tightening Position



1. Spanner Wrench Adapter mounted on the balance nut with three special bolts; position open side of adapter toward spanner wrench
2. Special High-Strength Shoulder Bolt (3)
3. Crosshead Balance Nut
4. Spanner Wrench
5. 10,000 psi (690 bar) cylinder with 1" (25mm) stroke, single-acting spring return

CAUTION:

Do not allow open areas of the spanner wrench and adapter to overlap, as the spanner wrench can be sprung. If the wrench becomes sprung, destroy and replace it.

When nut is tight, the feeler gauge may be removed, usually with slight resistance. Verify removal of all tools from cylinder and crosshead guide.

12. With the head end head or unloader properly installed (closed position) and fasteners tightened, rotate crankshaft 180° to outer dead center position of throw. Remove a top head end valve. Determine required piston head end clearance limits from cylinder data plate or Appendix B. Measure head end clearance at the top of the head end, using feeler gauges through the open valve pocket. Determine if measured clearance is within tolerance. Tighten the crosshead-balance nut set screws. Re-install valve assemblies and properly tighten fasteners. See "Valve Installation".
13. Check piston rod runout and crosshead clearances upon re-assembly, any time a piston rod is removed. Check crosshead guide to crosshead clearance at the top by inserting a standard 0.5 inch (13 mm) wide feeler stock from one side edge of the crosshead across to the opposite side - do this at both ends. Check bottom clearance with 0.0015 inch (0.038 mm) feeler stock at the four corners. If the feeler at the bottom inserts more than 0.5 inch (13 mm), the assembly is not acceptable. See Appendix B for crosshead clearances.
14. Examine top cover and side cover seals. If there is doubt about their condition, replace them. Replace top cover and crosshead guide side covers. Tighten all cap screws hand wrench tight.

Connecting Rods

A connecting rod assembly consists of a rod with bushing, a rod cap, rod cap bolts (4), and two half bearing shells. The bearing shells join together to form the connecting rod bearing.

Connecting Rod Removal

WARNING:

To decrease the risk of injury or death, only trained and approved personnel can do maintenance work. Approved personnel must read and know the applicable Ariel Technical Manual and Packager-supplied information.

Before you start maintenance:

- Always use personal protection equipment (PPE) and obey your company policies for safety and operation.
- Do not tighten or loosen fasteners or fittings while the unit is in operation or is pressurized.
- Make sure the driver and compressor cannot turn during maintenance.
- Obey your company and packager instructions to isolate and vent the system.
- Make sure there is no remaining pressure in the cylinders. If equipped with “Kiene” valves on the indicator ports, open them to release residual pressure after blowdown.

  WARNING: RELEASED GAS MAY BE POISONOUS AND/OR FLAMMABLE.

Before operation, purge the system. A pressurized hydrocarbon and air mixture can explode and cause injury or death.

1. Remove top cover from crankcase and side covers from crosshead guides.
2. Remove the middle frame spacer bar of the three over the desired pair of throws. Spacer bars are compressor and throw dependent. They are match-marked with the throw position and machined with a tight fit. After bolt removal, spacer bars should remove easily by hand with maybe a slight resistance. A very loose or tight fit indicates an unlevel frame, softfoot, incorrect crosshead guide shimming, improperly adjusted discharge bottle wedges, or pipe stress affecting the cylinder flange fit. Correct any of these conditions before returning the compressor to service. If a spacer bar is tight and will not easily come out, it may help to loosen one bolt on each spacer bar above the nearest main bearing caps.
3. Move throw to outer dead center position.
4. Remove crosshead as described in "Crosshead Removal".
5. Turn the crankshaft until the throw is at its highest point.
6. Loosen all connecting rod bolts approximately 3/4 in. (19 mm).
7. Use the connecting rod cap removal tool to jack against the bolt heads and pull the rod cap free from the dowels (see Section 1). If the cap does not separate easily from the rod, **STOP**; the rod cap may be in a bind, which can break the dowel pins. If binding occurs, re-tighten the connecting rod bolts and repeat the procedure.

   NOTE:

   Do not pry or chisel at joint of connecting rod to separate cap from rod; it damages the connecting rod and can break alignment dowel pins. Do not pry on any machined running surfaces to separate cap from rod.

8. Remove the cap removal tool.
9. Completely loosen the connecting rod fasteners while holding the rod and cap in position. Slide fasteners back to clear the rod. Use fasteners as a handle to hold cap in position.

10. While supporting the cap and fasteners, slide the rod toward the cylinder. When the rod is clear of the crankshaft, carefully rest it in the guide.
11. Rotate the cap up and remove all fasteners, then remove the rod cap and bearing. Hold the bearing in the cap while lifting it off the pin. **Do not remove cap with fasteners in place.**
12. Carefully remove the connecting rod through the crosshead compartment window.

Connecting Rod Bearing Removal

WARNING:

To decrease the risk of injury or death, only trained and approved personnel can do maintenance work. Approved personnel must read and know the applicable Ariel Technical Manual and Packager-supplied information.

Before you start maintenance:

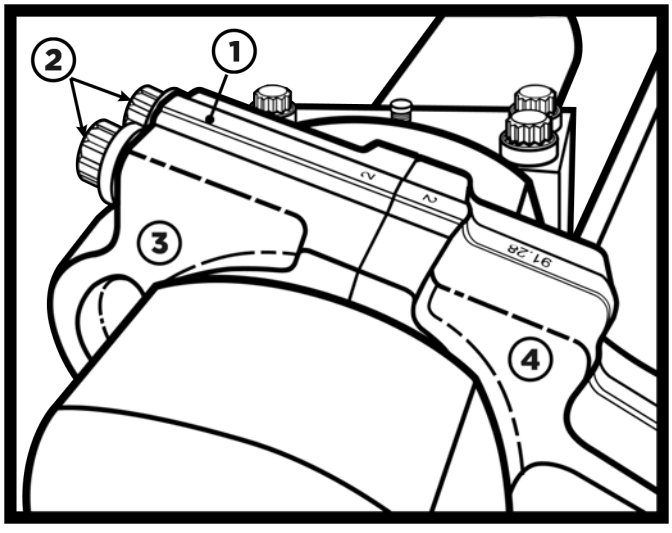
- Always use personal protection equipment (PPE) and obey your company policies for safety and operation.
- Do not tighten or loosen fasteners or fittings while the unit is in operation or is pressurized.
- Make sure the driver and compressor cannot turn during maintenance.
- Obey your company and packager instructions to isolate and vent the system.
- Make sure there is no remaining pressure in the cylinders. If equipped with “Kiene” valves on the indicator ports, open them to release residual pressure after blowdown.

  WARNING: RELEASED GAS MAY BE POISONOUS AND/OR FLAMMABLE.

Before operation, purge the system. A pressurized hydrocarbon and air mixture can explode and cause injury or death.

1. Remove frame top cover and all crosshead guide covers.
2. Remove all head end heads.
3. Rotate crankshaft to approximately 45º (see Figure: Crank Rotation)

Figure: Rod/Rod Cap Separation



1	Top Face	**3**	Rod Cap
2	Connecting Rod Bolts	**4**	Rod

4. Loosen all connecting rod bolts about 1/8 in. (3.2 mm).
5. Pry crosshead toward the cylinder until it bottoms out.

> **NOTE:**
>
> **Do not pry on any machined running surfaces.** Work rod cap away from rod as evenly as possible. Separating the two parts unevenly may break rod cap dowels or cause them to bind in the rod dowel holes. If the two parts of the conrod do not separate easily by hand, loosen all bolts an additional 1/8 in. (3.2 mm) and pry the crosshead again.

> **NOTE:**
>
> Do not pry or chisel at joint of connecting rod cap/rod to separate cap from rod; it damages the connecting rod and can break the alignment dowel pins. Do not pry on machined running surfaces when separating the cap from the rod. Separate caps by backing out all 4 bolts no more than 1/4 inch and prying behind the crosshead (see Figure: Prying Behind Crosshead)

6. Support rod and cap by hand while another person pries the crosshead toward the cylinder as much as possible. When the rod is clear of the crankshaft, carefully rest it in the guide.
7. Rotate the cap up and remove all bolts, then remove the rod cap and bearing. Hold the bearing in the cap while lifting it off the pin. **Do not remove cap with bolts in place.**
8. After removing bearings, clean any coking or debris from both the rod and rod cap surfaces with a non-metallic Scotch-Brite or similar scouring pad and solvent. Check for raised edges on machined surfaces. All surfaces should be as clean and dry as possible.

Figure: Crank Rotation

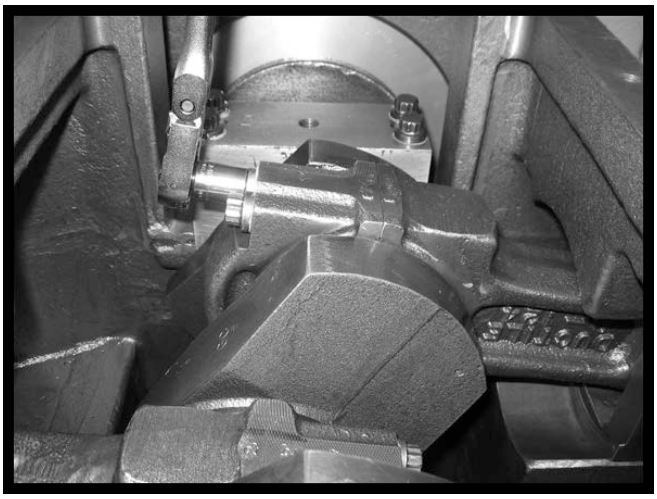
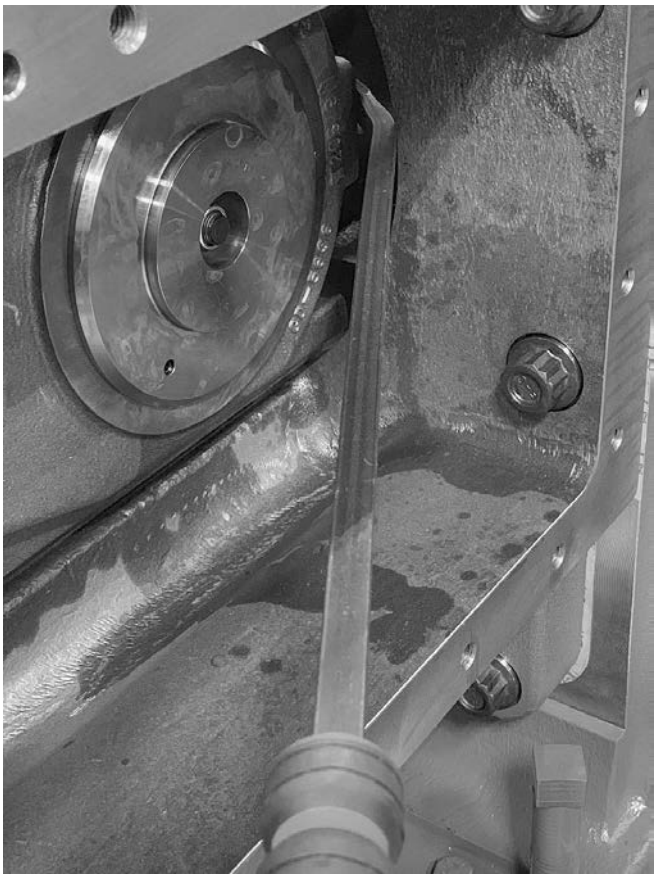


Figure: Prying Behind Crosshead



Crank Pin Assessment

Inspect bearings after removal. If bearings indicate scratches or dings on pin, inspect crank pin, and if needed, carefully stone the pin using an Arkansas stone (Ariel part number A-19676). Lightly slide the stone around the circumference of the crank pin surface to remove any high spots; do not slide it back and forth over the length of the pin. If the pin is scored around the circumference, use 600 grit emery cloth to remove high spots. After dressing, ensure crank pin is clean and dry. Use only lint-free, clean paper towels to clean crankshaft pin of any debris after dressing.

Connecting Rod Bearing Installation

1. Check new bearing shells for handling damage, scratches, burrs, and loose material at the tabs. **DO NOT RUB BEARING SURFACE WITH FINGERNAIL.** New bearing shells and crankshaft crank pin bearing surfaces must be absolutely clean. Clean new bearings with appropriate solvent and clean paper towel. Paper towels do not leave lint behind compared to rags. Fit the tab of a new, dry half bearing shell into the rod notch.

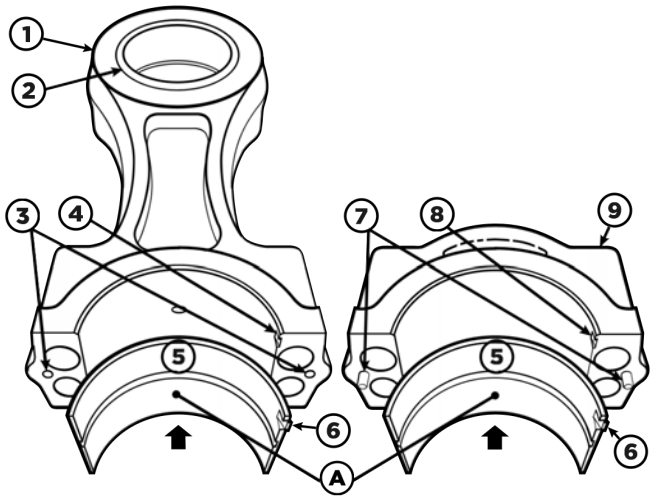
> **NOTE:**
>
> Where bearings are identified with an ink pictograph, connecting rod bearings and main bearings are not interchangeable. DO NOT put main bearings in a connecting rod. Do not mix part numbers on an individual main or connecting rod. Both bearing half shells must have the same part number.

2. Apply clean, new lube oil liberally to the crank pin. Verify oil is compatible with the oil used in the frame.
3. Fit the tab of the other bearing shell into the rod cap notch.
4. Use the following steps to attach the rod cap to the rod:
   a. Re-position crankshaft, if needed.
   b. Carefully lower the connecting rod cap **(WITHOUT BOLTS!)** onto the crank pin. Verify rod bolts are undamaged and free of dirt. Lubricate them per Appendix A, then place all four bolts into the rod cap.

> **NOTE:**
>
> Ensure notches on both the connecting rod and connecting rod cap match. Install with both notches on top, in the upward position.

   c. Rotate cap so the bolts align with the connecting rod bolt holes, then pull all four bolts back. Support the rod cap while another person carefully supports the connecting rod and guides it back onto the crankshaft until it bottoms out. Align the rod cap dowels with the rod dowel holes and push the rod cap onto the rod with even pressure. The rod cap fits onto the rod only one way. Do not force the rod cap onto the rod; the dowels should slide into the rod holes without difficulty. After ensuring the rod cap seats properly on the rod, tighten all four bolts and pre-torque according to Appendix A.
   d. Rotate crankshaft as needed and tighten all connecting rod bolts to the torque listed in Appendix A.
5. Repeat steps 1-4 for all remaining connecting rods.
6. Measure each connecting rod bearing jack and thrust clearance as detailed in this document. If clearance readings are not within tolerances of Appendix B, contact your packager or Ariel before proceeding.

Figure: Connecting Rod Bearing Installation



- **A** Apply lube oil liberally to this surface.
- **1** Rod
- **2** Rod Bushing
- **3** Dowel Holes
- **4** Rod Notch
- **5** Bearing Shell
- **6** Bearing Shell Tab
- **7** Dowels
- **8** Rod Cap Notch
- **9** Rod Cap

> **NOTE:**
>
> If replacing main bearings, continue to "Main Bearing Removal". If not replacing main bearings, continue to Step 7.

7. Account for all tools, equipment, supplies, and parts to ensure none are left inside the crankcase.
8. Examine top cover and side cover seals. If there is doubt about their condition, replace them. For O-ring seals, oil the O-ring and apply a light coating of corrosion inhibitor or marine grease to mating surface. Replace top cover and crosshead guide side covers. Tighten all cap screws hand wrench tight.
9. Reverse lockout procedures.
10. After replacing bearings, it is recommended to pre-lube compressor for at least 10-15 minutes at 30 psi (2.1 barg) to ensure bearing lubrication and to help remove foreign materials from the lube system.
11. Run the compressor according to the time intervals listed on the form in Appendix D. After each run, shut down and remove the frame top cover. Measure main and connecting rod bearing cap temperatures with a hand held thermocouple probe or infrared thermometer and record them on a copy of the form in Appendix D. Complete the form and file it in the unit records for future reference.
12. After each shutdown, visually inspect the crankcase for signs of excessive heat in bearing areas, and babbitt or other debris. If any are found, find and correct the problem before continuing.

Connecting Rod Bushing Removal and Installation

1. Check crosshead pin to bushing clearance (see Appendix B). Determine pin wear by inspection. Replace pin, if necessary.
2. To replace a bushing, file or hacksaw the existing bushing to reduce the tightness of the shrink fit. From the inside diameter, file or saw across the length of the bushing to within 1/32 inches (1 mm) of its radial thickness. It can then be easily drifted out. **DO NOT file or saw into the connecting rod; any bore damage renders the rod useless and requires rod replacement.**
3. Use a hydraulic press in a qualified machine shop to install the new bushing. Do not hammer bushing into place; it will distort the bushing bore.
4. Before installation, cool new bushing in a 95% alcohol and dry ice solution. Leave bushing in solution long enough to reach the solution temperature, about -110°F (-80°C). An alternate method of cooling is to submerge the bushing in liquid nitrogen. When the nitrogen stops bubbling, the temperature has reached equilibrium, and the bushing is ready to install.

> **CAUTION:**
>
> Do not touch cold surfaces without proper protection. Alcohol is flammable; use it only in open air or well-ventilated buildings. Avoid sparks and open flame. Avoid alcohol vapors which may cause injury to nose and eye tissue. Do not return solution to a closed container until it reaches room temperature or container may explode.

> **NOTE:**
>
> Thoroughly clean bushing and connecting rod to prevent dirt accumulation between them. Immediately assemble them so the bushing does not warm and stick before it is in place. If the bushing sticks, remove it by notching as in step 2 above.

5. Position connecting rod on the press table so the chamfered edge of the rod bushing bore is on top. Press bushing into rod bore.

Connecting Rod Installation

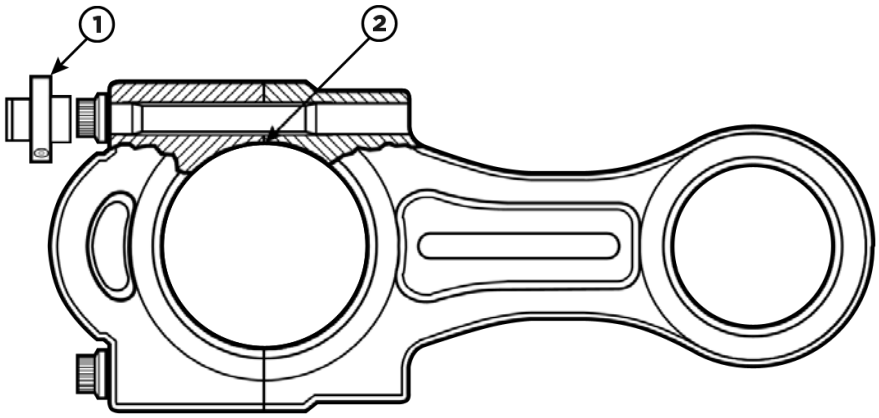
1. To install a new connecting rod, stamp match mark numbers matching the throw location on the tops of the connecting rod and bearing cap (with bearing notches up). See Figure: Typical Connecting Rod.
2. Check new bearing shells for handling damage, scratches, burrs, and loose material at the tabs. **DO NOT RUB BEARING SURFACE WITH FINGERNAIL.** New bearing shells and crankshaft crank pin bearing surfaces must be absolutely clean. Snap a new, dry half bearing shell into the rod and rod cap with the bearing tabs properly located in the rod and cap notches. Turn the connecting rod pin to the vertical position and oil crankshaft connecting rod pin bearing surfaces with new clean lubricating oil, the same type used in the frame. Carefully lower the connecting rod cap (without bolts) onto the pin.

> **NOTE:**
>
> Connecting rod bearings and main bearings are not interchangeable. Connecting rod bearings have a narrower groove or no groove at all. DO NOT put main bearings in connecting rod bearing locations.

> **NOTE:**
>
> Caps and rods are numbered by throw beginning with number one at the drive end. For throw numbering sequence, see Plate Locations. Always install rods with numbers up. Protect crank pin at all times. Be sure both bearing tabs are on top.

3. Verify rod bolts are undamaged and free of dirt, then lubricate them per Appendix A and insert all bolts into rod cap. Rotate the cap and fit the connecting rod to the rod pin and snug the bolts, pulling the cap and rod together evenly.
4. Reconnect rod and crosshead with pin. Install end plates, thru-bolt, and new lock nut. Tighten lock nut to the torque listed in the Appendix A.

Figure: Typical Connecting Rod



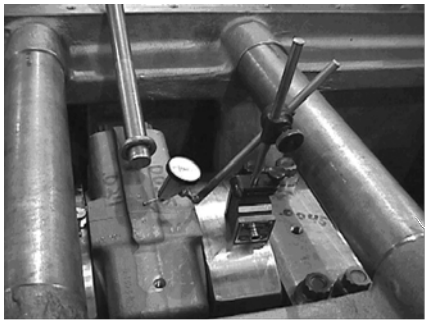
1 Turn Indicator
2 Correct rod orientation is with bearing notches on top joint. NOTE: Install joint match marks up.

5. Follow the "Recommendations for Torque Accuracy" in Appendix A to tighten connecting rod cap screws to the recommended torque.
6. Measure each connecting rod bearing to crankshaft jack clearance and connecting rod thrust (side) clearance (see "Connecting Rod Bearing Vertical Jack Clearance Measurement" and "Connecting Rod Thrust (Side) Clearance Measurement"). Record measurements on a copy of the form in Appendix D. If measurements are out of tolerance after installing new bearings, contact your packager or Ariel before proceeding.

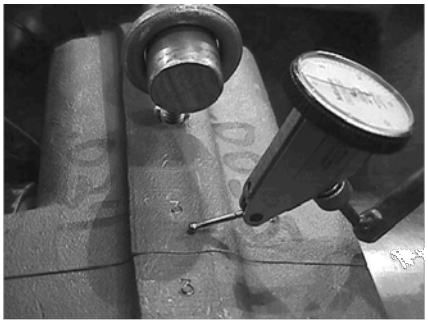
7. Examine top cover and side cover seals. If there is doubt about their condition, replace them. Replace top cover and crosshead guide side covers. Tighten all cap screws hand wrench tight.

Connecting Rod Bearing Vertical Jack Clearance Measurement

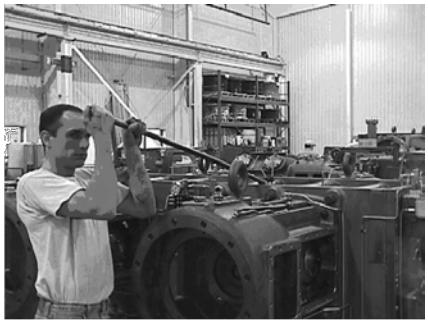
Figure: Typical Connecting Rod Vertical Jack Clearance Measurement



Magnetic stand placement on top of adjacent crankshaft web.



Dial indicator placement on connecting rod.



Use pry bar to check jack clearance.

1. Turn the crankshaft pin of the desired throw to its highest position.
2. To ensure measurement of only vertical movement between the crankshaft and connecting rod, attach the dial indicator magnetic base to the crankshaft web adjacent to the measured connecting rod.
3. Zero the dial indicator. Insert a pry bar into the eye bolt and pry against the frame to force the connecting rod up until the dial indicator needle stops moving. While maintaining upward pressure, note the reading.

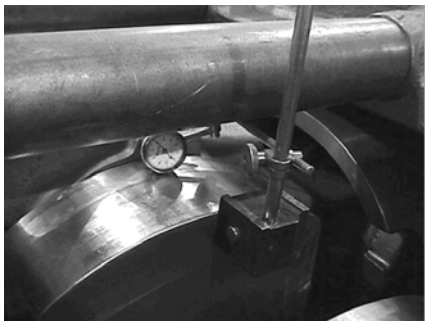
> **NOTE:**
>
> Too much force when prying on the connecting rod eyebolt may lift the crankshaft resulting in incorrect readings.

4. Release upward pressure and push downward on the rod until the indicator reads zero again. If a zero reading cannot be obtained, zero the indicator again and repeat steps 3 and 4. Repeat the measurement process until the measurement returns to zero twice in a row to verify accuracy.
5. Repeat measurement process for every rod. If a reading is outside the tolerances of Appendix B, contact your packager or Ariel.

> **NOTE:**
>
> Remove eye bolts, magnetic bases, dial indicators, and pry bars after measurement.

Connecting Rod Thrust (Side) Clearance Measurement

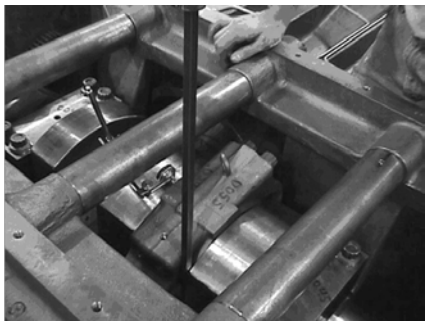
Figure: Typical Connecting Rod Thrust (Side) Clearance Measurement



Dial indicator magnetic stand placement.



Dial indicator placement.



Thrust rod tight toward and away from dial indicator.

1. Turn the crankshaft pin of the desired throw to its highest position.
2. Attach the magnetic base of a dial indicator to the crankshaft web adjacent to the connecting rod. Place the button of the dial indicator against the thrust surface of the connecting rod (see Figure: Typical Connecting Rod Thrust (Side) Clearance Measurement).
3. Use a pry bar to pry gently against the crankshaft web and thrust the connecting rod tight toward the dial indicator (do not pry on rod cap). Release pressure and zero the dial indicator.
4. Thrust the connecting rod tight in the opposite direction until the dial indicator needle stops moving. Release pressure and note reading. Gently pry the connecting rod in the opposite direction to ensure the dial returns to zero.
5. Repeat the measurement process for every rod. If a reading is outside the tolerances in Appendix B, contact your packager or Ariel.

> **NOTE:**
>
> Remove magnetic bases, dial indicators, and pry bars after measurement.

Crankshafts

The crankshaft is the heart of the compressor. With proper maintenance, it should provide years of trouble-free service. If replacing a crankshaft fitted with torsional detuners, contact the Ariel Response Center with the frame and crankshaft serial numbers and the part numbers and quantity of detuners to ensure compatibility.

Main Bearing Removal

WARNING:

To decrease the risk of injury or death, only trained and approved personnel can do maintenance work. Approved personnel must read and know the applicable Ariel Technical Manual and Packager-supplied information.

Before you start maintenance:

- Always use personal protection equipment (PPE) and obey your company policies for safety and operation.
- Do not tighten or loosen fasteners or fittings while the unit is in operation or is pressurized.
- Make sure the driver and compressor cannot turn during maintenance.
- Obey your company and packager instructions to isolate and vent the system.
- Make sure there is no remaining pressure in the cylinders. If equipped with “Kiene” valves on the indicator ports, open them to release residual pressure after blowdown.

  WARNING: RELEASED GAS MAY BE POISONOUS AND/OR FLAMMABLE.

Before operation, purge the system. A pressurized hydrocarbon and air mixture can explode and cause injury or death.

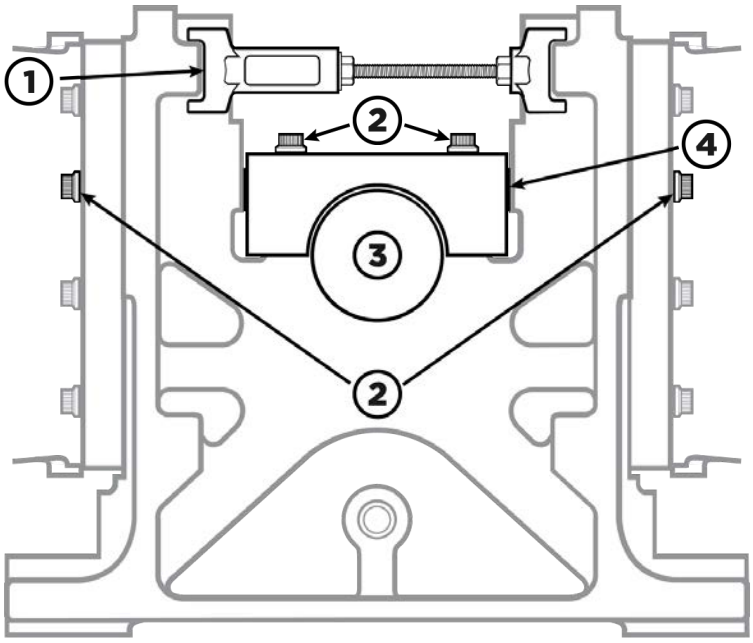
1. Remove top cover and spacer bars. Spacer bars are compressor and throw dependent. They are match-marked with the throw position and machined with a tight fit. After bolt removal, spacer bars should remove easily by hand with maybe a slight resistance. A very loose or tight fit indicates an unlevel frame, softfoot, incorrect crosshead guide shimming, improperly adjusted discharge bottle wedges, or pipe stress affecting the cylinder flange fit. Correct any of these conditions before returning the compressor to service.

   CAUTION:

   When the main cap is removed, the bearing can roll and damage the RTD. Remove the externally installed main bearing RTD before cap removal.

2. Remove the six main (journal) bearing cap screws, which includes the cross bolts at the guide mounting on the side of the frame. Use main bearing installation/removal tool to apply force on the frame rail as close as possible to main bearing cap being removed (refer to Figure: KBE:K:T Main Bearing Installation/Removal Tool). Do not exceed maximum torque or turn written on the tool. If cap is tight after applying force with the tool, ensure spacer bars are removed or loosened, adjacent main bearing cross-bolts are loosened, and check crosshead guide shimming for correct pre-load. Remove main bearing cap installation/removal tool. Remove shells from main bearing caps.

Figure: KBE:K:T Main Bearing Installation/Removal Tool



1 Main Bearing Installation/ Removal Tool
2 Main Bearing Cap Screw
3 Crankshaft
4 Main Bearing Cap

3. Inspect bearings. If bearings indicate scratches or dings on pin, inspect crankshaft journal, and if needed, carefully stone the journal using an Arkansas stone (Ariel part number A-19676). Lightly slide the stone around the circumference of the journal surface to remove any high spots; do not slide it back and forth over the length of the journal. If the crank is scored around the circumference, use 600 grit emery cloth to remove high spots. After dressing, ensure crank pin is clean and dry. Use only lint-free, clean paper towels to clean crankshaft journal of any debris after dressing. Rotate the crankshaft 180° to expose the other half of the journal and repeat.
4. To limit debris entering the system, work on only one main bearing at a time. Remove main journal bearing shells from under the crankshaft one at a time. If needed, attach clean nylon straps around the crankshaft and lift it 0.003 - 0.005 inch (0.076 - 0.127 mm) to lessen weight on the bearings and allow easier removal. To remove, rotate shell under the crankshaft tab side out first, by pushing or tapping with a non-metallic tool on the opposite side. Ariel provides an optional tool (B-3340) to push the shell out. Do not damage crankshaft bearing surfaces. Replace with new bearing shell, before rotating out the next main bearing shell.
5. If any coked oil is observed on bearing cap or frame saddle surfaces, contact Ariel for instructions. Check for raised edges on machined surfaces. Verify all surfaces are clean and dry as possible and no bolting is damaged.

Main Bearing Installation

1. Clean new bearings with appropriate solvent and clean paper towel. Paper towels do not leave lint behind compared to rags. Check new bearing shells for damage, scratches, burrs, and loose material at the tab. **DO NOT RUB BEARING SURFACE WITH FINGERNAIL.** Keep the backs of shells dry and clean; installation requires perfectly clean bearing shells. Install bearing shells in caps, and frame journals, properly located in the tab notch, (rotate in the un-tabbed end first).

> **NOTE:**
>
> Where bearings are identified with an ink pictograph, main and connecting rod bearings are not interchangeable. Do not put connecting rod bearings in main bearing locations. Do not mix part numbers on an individual main or connecting rod.

2. Apply clean, new lube oil liberally to the crank journal. Verify oil is compatible with oil used in the frame.
3. Use main bearing installation/removal tool to apply force on the frame rail as close as possible to the main bearing cap being installed (refer to Figure: KBE:K:T Main Bearing Installation/Removal Tool). Do not exceed maximum torque or turn written on the tool. If cap is tight after applying force with the tool, ensure spacer bars are removed or loosened and check guide shimming for correct pre-load.
4. Main bearing caps have position match-marks corresponding with frame spacer bars and spacer bar bosses. Install main bearing caps with new bearing shells in their correct locations. Verify proper alignment of main bearing cap dowels with the holes in the crankcase base. Seat the cap by tapping it with the handle of a hammer or screwdriver. It is a machined fit, not a press fit, and the cap should not rock in the jaw fit when it seats. Do not use bolting to force the cap to seat. Remove main bearing installation/removal tool. Tighten the cap screws. Tighten the main bearing cap screws in the pattern and torque listed in Appendix A.
5. Check crankshaft journal bearing jack clearances and crankshaft thrust clearance to values in Appendix B (see procedure below). Record readings on a copy of the form in Appendix D. If readings are out of tolerance after new bearing installation, contact your packager or Ariel before proceeding.
6. Install spacer bars so the match mark is up and next to the same match mark on the frame rail. Tighten all spacer-bar cap screws to the torque listed in Appendix A.
7. Account for all tools, equipment, supplies, and parts to ensure none are left inside the crankcase. Examine top cover and side cover seals. If there is doubt about their condition, replace them. For O-ring seals, oil the O-ring and apply a light coating of corrosion inhibitor or marine grease to mating surface. Replace top cover and crosshead guide side covers. Tighten all cap screws hand wrench tight. Bar compressor over 360° to ensure free rotation.
8. Reverse lockout procedures.
9. After replacing bearings, it is recommended to pre-lube compressor for 10-15 minutes at 30 psi (2.1 barg) to ensure bearing lubrication and to help remove foreign materials from the lube system.
10. Run the compressor according to the time intervals listed on the form in Appendix D. After each run, shut down and remove the frame top cover. After each shutdown, visually inspect the crankcase for signs of excessive heat in bearing areas, and babbitt or other debris. If any are found, find and correct the problem before continuing. Measure main and connecting rod bearing cap temperatures with a hand-held thermocouple probe or infrared thermometer and record them on a copy of the form in Appendix D. Complete the form and file it in the unit records for future reference.

Crankshaft Removal

WARNING:

To decrease the risk of injury or death, only trained and approved personnel can do maintenance work. Approved personnel must read and know the applicable Ariel Technical Manual and Packager-supplied information.

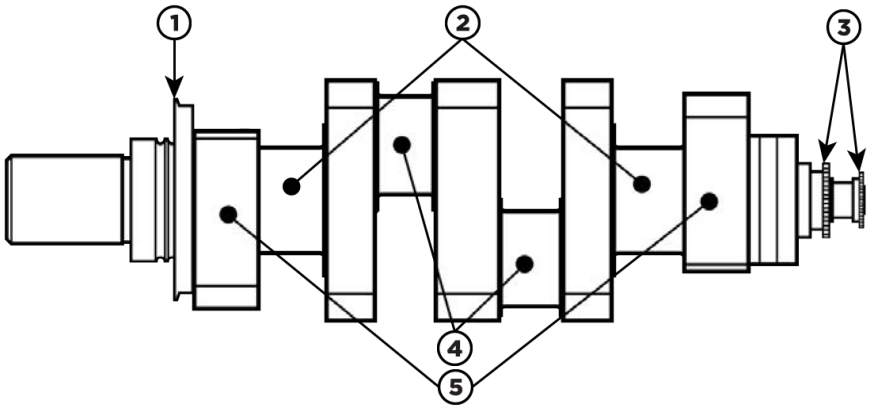
Before you start maintenance:

- Always use personal protection equipment (PPE) and obey your company policies for safety and operation.
- Do not tighten or loosen fasteners or fittings while the unit is in operation or is pressurized.
- Make sure the driver and compressor cannot turn during maintenance.
- Obey your company and packager instructions to isolate and vent the system.
- Make sure there is no remaining pressure in the cylinders. If equipped with “Kiene” valves on the indicator ports, open them to release residual pressure after blowdown.

  WARNING: RELEASED GAS MAY BE POISONOUS AND/OR FLAMMABLE.

Before operation, purge the system. A pressurized hydrocarbon and air mixture can explode and cause injury or death.

Figure: Typical Crankshaft



1 Oil Slinger
2 Main Bearing Journals
3 Chain Sprocket
4 Crank Pins
5 Counterweights

1. Remove all drive end cover mounting bolts. Remove the two tapered dowel pins from the drive end cover.
2. Remove top cover and spacer bars. Do not damage the sharp corners on each end of the crankcase top. They form a junction between the end covers, top cover, and base; keep them sharp and unmarred to prevent oil leaks.
3. Detach connecting rods and move them to their full outer position. See "Connecting Rod Removal".
4. Remove chain eccentric adjustment capscrews. Turn the eccentric to loosen the chain and slip it off the crankshaft sprocket. For more clearance while lifting the crankshaft, remove the force feed lubricator drive sprocket.
5. Remove main bearing caps. See "Main Bearing Removal".

6. Before removing the crankshaft, prepare wooden saddles or blocks of wood with sides high enough to prevent the webs or oil slinger from touching the floor to store the crankshaft during maintenance - even if for only a short time. Also, protect the crankshaft from above so dropped tools or equipment cannot mar pin and journal surfaces.
7. Turn the crankshaft so that sling lifting points are above the shaft center of gravity to prevent rotation when lifted. Lift straight up with the crankshaft ends parallel to the frame. The crankshaft weight requires two men and a crane to safely remove it. Use appropriate sized nylon slings to prevent marring the crankshaft running surface.

> **NOTE:**
>
> Lower half bearing shells sometimes stick to the shaft journals. After lifting the shaft about ¼ inch (6mm) clear of the saddles, verify the lower half bearing shells remain on the saddles. If not, tap them back onto the saddles before lifting the shaft further.

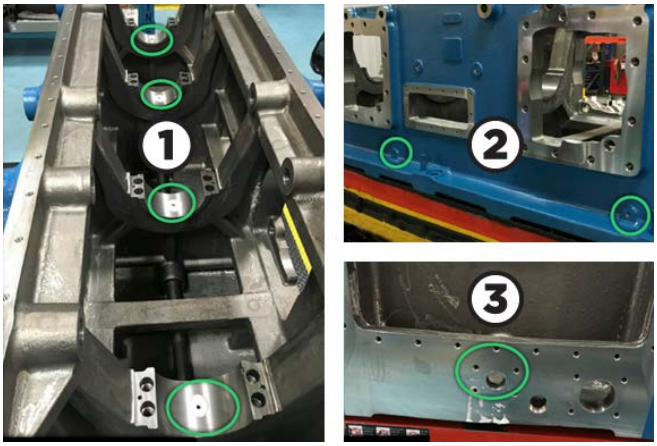
8. While one person raises the crane very slowly, the second person must grasp the crankshaft at the drive end with one hand on the counterweight or one of the throws and the other on the shaft end to keep the crankshaft level. It may be necessary to lower the crank and reposition the straps to achieve a balanced lift. Wear clean gloves for a good grip, to avoid cuts from the slinger, and to avoid marring the running surface. As the shaft slowly raises, lift the drive and auxiliary ends at the same rate. Carefully guide the crankshaft to avoid marring its surfaces.

Oil Passage and Oil Gallery Cleaning

Before crankshaft reinstallation, thoroughly clean and inspect frame oil passages and main oil gallery.

1. Remove both drive and aux end oil gallery blocks, and guide oil passage tubing and fittings.
2. With the top cover removed, use a pneumatic syphon gun to flush the entire frame with solvent from the top of frame to the bottom to clean all material left behind from the failure and/or machining. Vacuum all foreign material and solvent that settles on the frame bottom. Use clay to remove stubborn material from corners and crevices. Repeat the solvent flush and vacuum process until frame is clean and free of all debris.
3. Spray/flush cleaning solvent through each main oil hole and each oil passage hole on the side of the crankcase, forcing all debris into the oil gallery.
4. Layer two lint-free cleaning cloths cut into similarly sized squares and soak them with cleaning solvent.
5. Roll cloths and push them through the main oil, and guide oil passages with a clean rod. Repeat this step for all main and guide oil passages until swabs return clean and free of debris.

Figure: Oil Passages



1 Main Oil Passages **2** Guide Oil Passages **3** Oil Gallery

> **NOTE:**
>
> Keep exact count of all cloths used and verify none remain in the frame.

Figure: Swabbing Oil Passages



Figure: Swabbing Oil Gallery



6. Layer two larger lint-free cleaning cloths soaked in solvent, roll them, and push them through the oil gallery. Place a clean bucket below the opposite end of the oil gallery to catch cloths pushed out. Repeat until swabs return clean and free of debris.

> **NOTE:**
>
> Keep exact count of all cloths used and verify none remain in the frame.

7. Push a pen light through the oil gallery while a colleague looks down the first oil hole intersecting the oil gallery. Push the pen light until the colleague sees the light and can thoroughly inspect the hole. Verify all main and guide oil passages are clean and free of debris.
8. After cleaning and inspection, immediately install oil gallery blocks, guide oil passage tubing and fittings, and crankshaft. If the crankshaft cannot be installed immediately, tape over exposed oil passages. Verify tape removal during installation later.

Figure: Pen Light Inspection



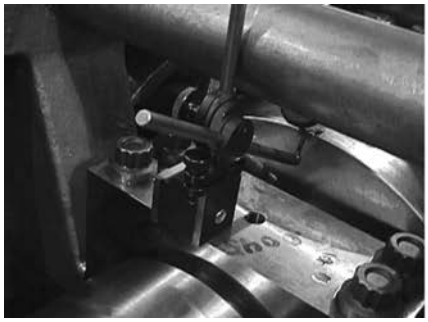
Crankshaft Installation

1. Thoroughly clean the oil gallery that runs from the drive end to the auxiliary end of the crankcase and the vertical and horizontal intersecting oil passages that lube the main bearings and crosshead guides. See "Oil Passage and Oil Gallery Cleaning". Any debris left in the oil gallery or intersecting passageways could cause bearing or crosshead failure. Deburr and clean main bearing caps to prepare for installation.
2. Thoroughly clean new main bearing half shells and install them in the frame saddles. Cover bearings with clean paper towels to prevent debris contamination.
3. Remove towels just before seating the crank in the saddles. Lubricate bearing surfaces with clean crankcase oil.
4. Move connecting rods to full outer position if not already there.
5. Install the crankshaft. While the crankshaft lowers very slowly into the crankcase (suspended by a crane with clean nylon slings), one man wearing clean gloves should grasp the drive end and slowly maneuver the drive end and auxiliary end straight down into the crankcase. Both drive end and auxiliary end journals should touch the bottom bearing shells at the same time.
6. Before connecting rod installation, measure each crankshaft journal bearing jack clearance with a dial indicator (see procedure on "Main Bearing Vertical Jack Clearance Measurement" and allowable clearances in Appendix B).

7. Reattach connecting rods (See "Connecting Rod Installation"), packing diaphragms, and unloaders/head end heads.
8. Reinstall chain drive. (See “Chain Drive System”).
9. Replace spacer bars. Locate spacer bar match mark. Install spacer bar so the match mark is up and next to the spacer bar boss with the same marking.
10. Install drive end cover with new gasket using the tapered dowel pins to properly align the cover. Verify at least 0.008 inch (0.2 mm) clearance exists between the dust seal housing and crankshaft. Then install a new dust seal.
11. Use a knife to trim excess from end cover gaskets flush to the compressor frame top cover rail.
12. Examine top cover and side cover seals. If there is doubt about their condition, replace them. Oil the o-ring and apply a light coating of corrosion inhibitor or marine grease to mating surface. Replace top cover and crosshead guide side covers. Tighten all cap screws hand wrench tight.

Main Bearing Vertical Jack Clearance Measurement

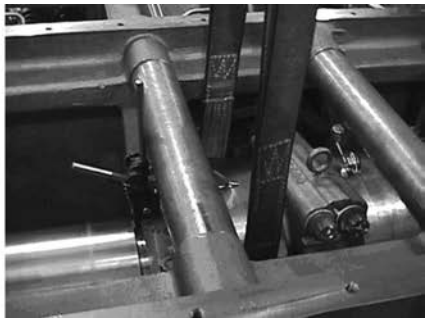
Figure: Typical Crankshaft Journal Bearing Vertical Jack Clearance Measurement



Dial indicator magnetic stand placement.



Dial indicator placement.



Place lifting strap around crankshaft web - attach strap to crane.

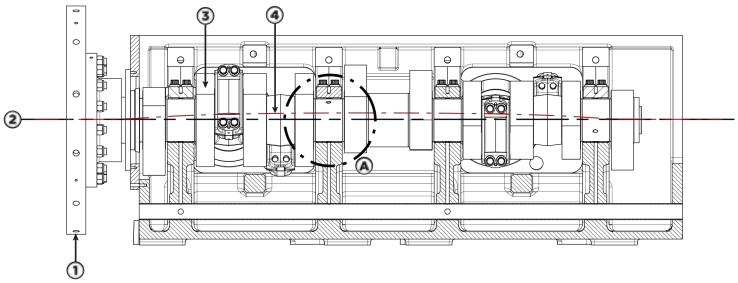
1. Turn the crankshaft until the web adjacent to the desired main bearing is vertical (see Figure: Typical Crankshaft Journal Bearing Vertical Jack Clearance Measurement).
2. Loop a clean, appropriately sized, and rated lifting strap around the crankshaft web adjacent to the main bearing and attach the strap to a crane.

> **NOTE:**
>
> If the crane has no fine adjustment, attach a chain hoist to the crane lifting hook to prevent excessive force on the crankshaft and crane.

3. Attach the magnetic base of a dial indicator to any stable frame surface. Place the point of the dial indicator against the top of the web adjacent to the measured bearing and zero the indicator.
4. GENTLY lift the crankshaft. The dial indicator needle moves steadily until the crane/hoist overcomes the crankshaft oil stiction; at this point, the needle jumps a little. Stop lifting immediately at this point, and note the reading.
5. Release crane/hoist pressure and verify indicator returns to zero. Repeat step 4 to verify the measurement.
6. In some instances, a heavy drive end flywheel can distort the crankshaft enough to lift it from the bottom main bearing, causing low main bearing jack clearance readings at certain main bearing locations.

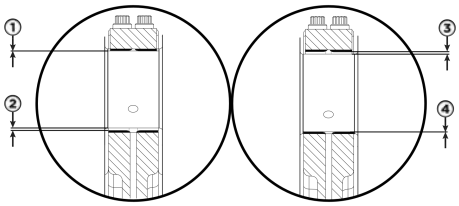
Figure: Crankshaft Distortion

Four-Throw Compressor



1 Flywheel Lifting Point
2 Crankshaft Centerline
3 First Web
4 Crankshaft Distortion

Detail A



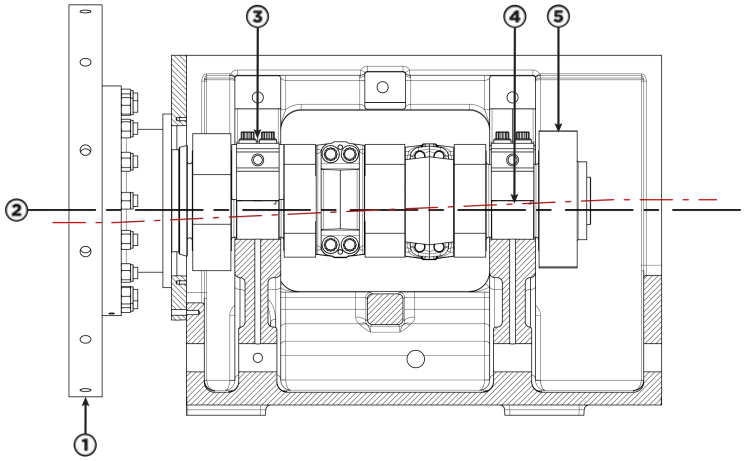
1 Main Jack Clearance before Lifting Flywheel
2 Bottom Journal Clearance before Lifting Flywheel
3 Main Jack Clearance after Lifting Flywheel
4 Bottom Journal Clearance after Lifting Flywheel

If heavy drive end flywheel affects measured values on a 4- or 6-throw compressor, follow these steps:

a. Roll the crankshaft so the first web is vertical.
b. Set up a dial indicator on the first web to measure vertical movement.
c. Slowly lift the flywheel from underneath. Initially, the indicator moves in the negative direction during flywheel lifting. When the indicator stops and begins moving in the positive direction, stop lifting and maintain the current lifting device position. This position relieves flywheel weight for accurate bearing clearance measurement.
d. Measure main bearing clearances per step 2-5 above.

For two-throw compressors with a heavy drive end flywheel:

Figure: 2-Throw Compressor with Crankshaft Distortion



1 Flywheel Lifting Point
2 Crankshaft Centerline
3 Drive End Main Bearing
4 Crankshaft Distortion
5 Non-Drive End Counterweight

a. Mount a dial indicator on the crankshaft counterweight on the non-drive end and zero it.

b. Slowly lift the flywheel from underneath until the dial indicator stops moving, which indicates the crankshaft sits in the bottom of the non-drive end main bearing shell.
c. Record reading from Step 2 as the main bearing clearance for the non-drive end.
d. Place another dial indicator on the drive end main bearing and zero it.
e. Lift the flywheel until the indicator stops moving.
f. Record reading from Step 5 as the main bearing clearance for the drive end.

7. Repeat the measurement process for each main bearing. If reading is outside tolerances in Appendix B, contact your packager or Ariel.

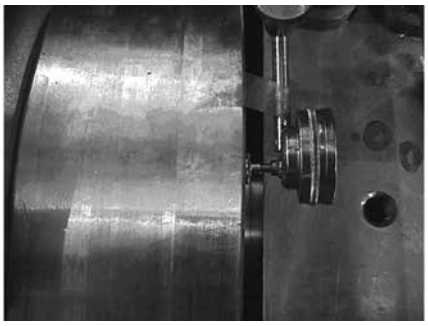
> **NOTE:**
> Remove magnetic bases, dial indicators, and lifting strap after measurement.

Crankshaft Thrust (End) Clearance Measurement

Figure: Typical Crankshaft Thrust Clearance Measurement



Dial indicator magnetic stand placement on main bearing cap.



Dial indicator placement against the side of crankshaft web.



Pry against compressor frame to thrust crankshaft back and forth.

1. Attach the magnetic base of a dial indicator to any stable frame surface. Place the button of the dial indicator against the side of the adjacent crankshaft web (see Figure: Typical Crankshaft Thrust Clearance Measurement).

> **NOTE:**
> Do NOT place magnetic base on end cover- it can flex if pried against, resulting in inaccurate readings.

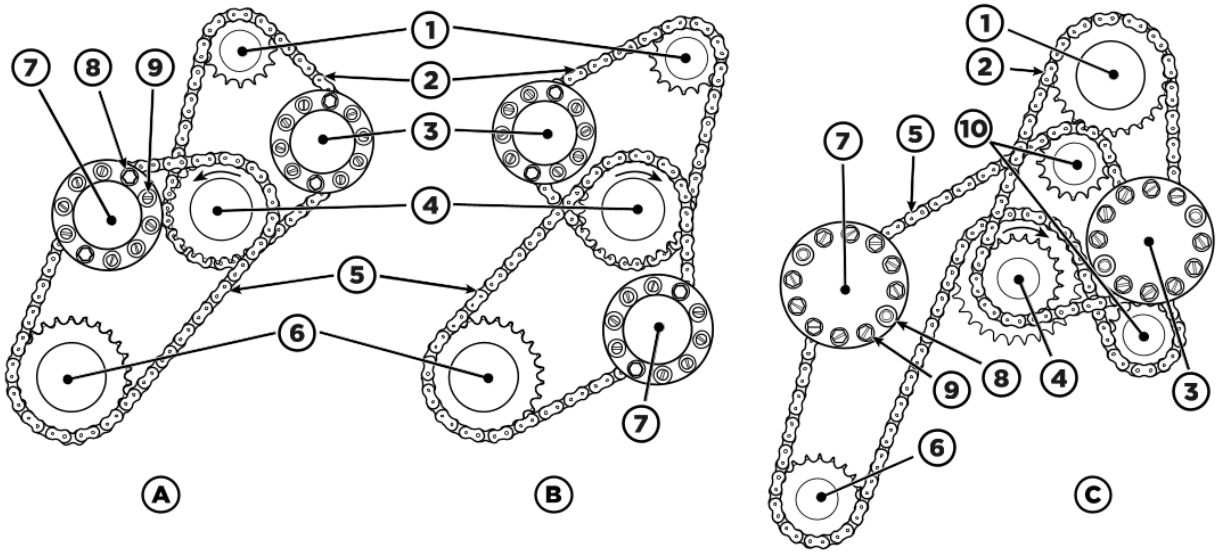
2. Use a pry bar to pry against the compressor frame and thrust the crankshaft tight toward the dial indicator. Release pressure and zero the dial indicator.
3. Thrust the crankshaft tight in the opposite direction until the dial indicator needle stops moving, then release pressure. Note reading. Repeat steps 2 and 3 to verify measurement accuracy.
4. If the measurement is outside the tolerances in Appendix B, contact your packager or Ariel.

> **NOTE:**
> Remove magnetic bases, dial indicators, and pry bar after measurement.

Chain Drive System

For KBE compressors, one chain runs the lube oil pump and another runs the force feed lubricator. A separate idler sprocket for each chain, each attached to its own eccentric adjustment cap, controls chain tightness. The lube oil pump chain dips into the crankcase oil and splash action oils the force feed lubricator chain as well. In dry sump applications, an oil sprayer lubricates the chains.

Figure: JGE:K:T/6 and KBE:K:T:C:D:F Chain Drive Systems



- **A** JGE:K:T/6 and KBE:K:T:C:D:F Standard Rotation
- **B** JGE:K:T/6 Reverse Rotation
- **C** KBK:T:C:D:F Reverse Rotation
- **1** Force Feed Lubricator Sprocket
- **2** Force Feed Lubricator Chain
- **3** Force Feed Lubricator Eccentric Adjustment
- **4** Crankshaft Sprockets
- **5** Oil Pump Chain
- **6** Oil Pump Sprocket
- **7** Oil Pump Eccentric Adjustment
- **8** Cap Screw
- **9** Dust Plug
- **10** Idler Sprocket

Chain and Sprocket Replacement

WARNING:

To decrease the risk of injury or death, only trained and approved personnel can do maintenance work. Approved personnel must read and know the applicable Ariel Technical Manual and Packager-supplied information.

Before you start maintenance:

- Always use personal protection equipment (PPE) and obey your company policies for safety and operation.
- Do not tighten or loosen fasteners or fittings while the unit is in operation or is pressurized.
- Make sure the driver and compressor cannot turn during maintenance.
- Obey your company and packager instructions to isolate and vent the system.
- Make sure there is no remaining pressure in the cylinders. If equipped with “Kiene” valves on the indicator ports, open them to release residual pressure after blowdown.

  WARNING: RELEASED GAS MAY BE POISONOUS AND/OR FLAMMABLE.

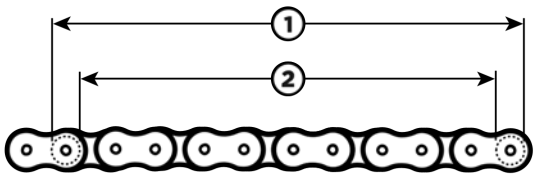
Before operation, purge the system. A pressurized hydrocarbon and air mixture can explode and cause injury or death.

Replace chain(s) if elongation exceeds 0.084 inches (2.13 mm) over a 10 pitch length. Measure the section of chain with vernier calipers while it is stretched tight in position in the compressor. Add a measurement outside the rollers at 10 pitches (X) to a measurement between the inside of the same rollers (Y), and then divide by two. Replace the chain if the result exceeds:

- 3.834 inches (97.4 mm) for 3/8-inch (No. 35) chain
- 5.084 inches (129.1 mm) for 1/2-inch (No. 40) chain

Replace sprockets showing any undercutting or other damage.

Figure: Measuring to Check Chain Elongation



1 X- Measure outside rollers at 10 pitches
2 Y - Measure inside rollers at 10 pitches

Chain Idler Sprocket Replacement (Eccentric Adjustment Caps)

WARNING:

To decrease the risk of injury or death, only trained and approved personnel can do maintenance work. Approved personnel must read and know the applicable Ariel Technical Manual and Packager-supplied information.

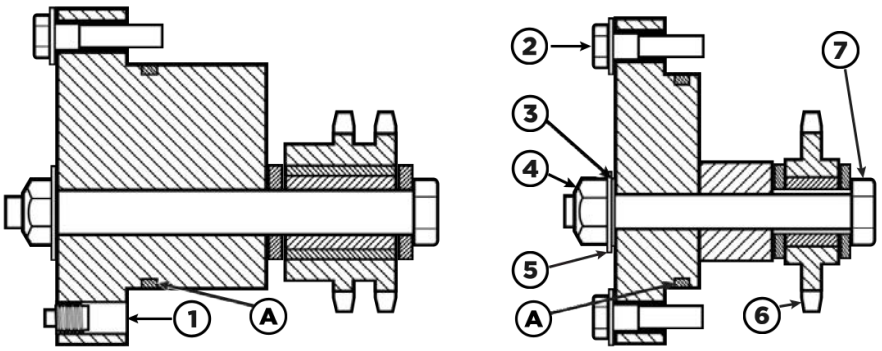
Before you start maintenance:

- Always use personal protection equipment (PPE) and obey your company policies for safety and operation.
- Do not tighten or loosen fasteners or fittings while the unit is in operation or is pressurized.
- Make sure the driver and compressor cannot turn during maintenance.
- Obey your company and packager instructions to isolate and vent the system.
- Make sure there is no remaining pressure in the cylinders. If equipped with “Kiene” valves on the indicator ports, open them to release residual pressure after blowdown.

WARNING: RELEASED GAS MAY BE POISONOUS AND/OR FLAMMABLE.

Before operation, purge the system. A pressurized hydrocarbon and air mixture can explode and cause injury or death.

Figure: Typical Chain Idler Sprockets (Eccentrics)



A Oil O-ring before assembly.
1 Dust Plug
2 Cap Screw
3 Stat-O-Seal
4 Self-Lock Nut
5 Washer
6 Sprocket
7 Thru Bolt

1. Remove frame top cover.
2. Remove all dust plugs from the eccentric cap and all cap screws holding the eccentric cap to the end cover. Rotate eccentric cap to loosen chain for removal. Drop the chain off the idler sprocket and remove entire assembly from the end cover.

NOTE:

Chain idler sprocket location varies for standard versus reverse rotation.

3. Remove and discard lock nut, hex cap screw, stat-o-seal, washer, and cap O-ring.
4. Reassemble new cap screw, stat-o-seal, washer, sprocket, and lock nut. Tighten idler lock nut to recommended torque in Appendix A.
5. Apply oil and install a new O-ring. Install assembly and chain.

6. Take care when measuring and adjusting the force feed idler sprocket; it is thinner than the force feed lubricator and drive sprockets. To center the idler sprockets in the chain, subtract the force feed idler sprocket thickness from the drive sprocket thickness and divide the difference by 2. Add this value to the measurement from the inside face of the auxiliary end cover to the outside face of the force feed idler sprocket. Adjust driven sprockets to the drive sprocket measurements, aligned within 1/32 inch (1 mm). Non-alignment may require disassembly and shimming eccentrics.
7. Adjust chain according to "Chain Adjustment".

Lube Oil Pump Sprocket Replacement

WARNING:

To decrease the risk of injury or death, only trained and approved personnel can do maintenance work. Approved personnel must read and know the applicable Ariel Technical Manual and Packager-supplied information.

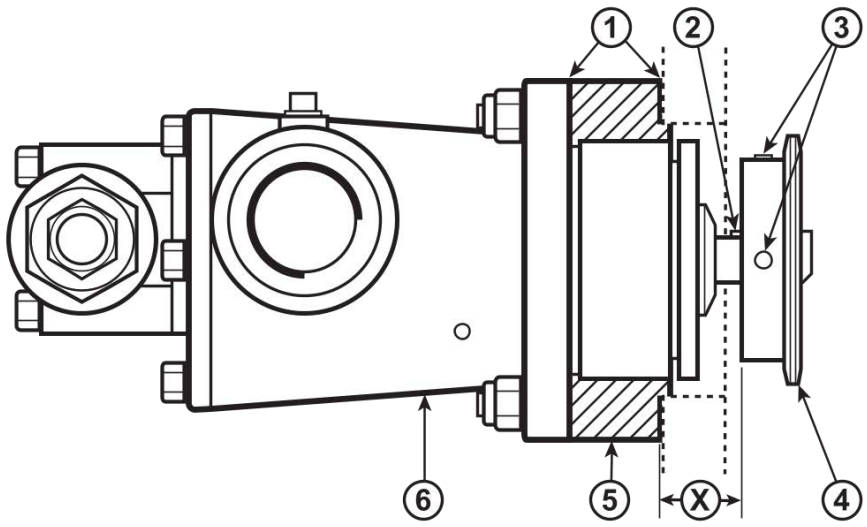
Before you start maintenance:

- Always use personal protection equipment (PPE) and obey your company policies for safety and operation.
- Do not tighten or loosen fasteners or fittings while the unit is in operation or is pressurized.
- Make sure the driver and compressor cannot turn during maintenance.
- Obey your company and packager instructions to isolate and vent the system.
- Make sure there is no remaining pressure in the cylinders. If equipped with “Kiene” valves on the indicator ports, open them to release residual pressure after blowdown.

  WARNING: RELEASED GAS MAY BE POISONOUS AND/OR FLAMMABLE.

Before operation, purge the system. A pressurized hydrocarbon and air mixture can explode and cause injury or death.

Figure: Typical Lube Oil Pump Chain Sprocket



1 Gaskets
2 Key
3 Set Screws
4 Sprocket
5 Adapter
6 Pump

1. Remove all pump piping. Remove fasteners from pump mounting flange. After chain removal, the pump with sprocket comes free through the end cover hole.
2. To position the new sprocket, use a machinist rule to measure the exact distance from sprocket drive face to pump mounting flange face (X). Note this measurement for future reference.
3. With the oil pump on a bench, use an Allen wrench to remove the sprocket set screws, then pull the sprocket from the shaft.
4. Remove the key from the shaft and file the shaft to smooth burrs raised by the set screw cup point.
5. Install a new key (see build book for type and dimensions). First verify it fits into the new sprocket. If too thick, polish the key with emery cloth on a flat surface until it easily slides into the notch. The top edge may also require a little polishing.
6. Install new sprocket to the original measurement between the sprocket drive face and the pump mounting flange face. When in position, tighten set screws.
7. Lubricate both sides of gaskets with anti-seize lubricant or multi-purpose grease to ease future removal. Install new gaskets.
8. Reinstall pump onto end cover. Using a straight edge, to within 1/32 inch (1 mm), check alignment to crankshaft drive sprocket with crankshaft centered in end play. If misaligned, adjust sprocket position as needed.
9. Adjust chain according to "Chain Adjustment".
10. Reinstall all piping to pump.

Force Feed Lubricator Dual Chain Sprocket Replacement

WARNING:

To decrease the risk of injury or death, only trained and approved personnel can do maintenance work. Approved personnel must read and know the applicable Ariel Technical Manual and Packager-supplied information.

Before you start maintenance:

- Always use personal protection equipment (PPE) and obey your company policies for safety and operation.
- Do not tighten or loosen fasteners or fittings while the unit is in operation or is pressurized.
- Make sure the driver and compressor cannot turn during maintenance.
- Obey your company and packager instructions to isolate and vent the system.
- Make sure there is no remaining pressure in the cylinders. If equipped with “Kiene” valves on the indicator ports, open them to release residual pressure after blowdown.

  WARNING: RELEASED GAS MAY BE POISONOUS AND/OR FLAMMABLE.

Before operation, purge the system. A pressurized hydrocarbon and air mixture can explode and cause injury or death.

Dual chain drive systems use a Fenner drive to attach the lubricator sprocket instead of a key and set screw. To replace a sprocket fastened with a Fenner drive:

1. With a good machinist rule, measure exact distance from inside face of auxiliary end cover to near face of lubricator sprocket. Note measurement for proper positioning of new sprocket. Remove chain.
2. Remove sprocket/Fenner drive assembly.

NOTE:

JGEKT/6 and JGBV/4/6 frames use a 5/8 ID x 1-1/2 OD Fenner while all others use a 5/8 ID X 1” OD. The 1-1/2 OD Fenner can be difficult to remove due to higher torque.

CAUTION:

Do not apply lubricants during installation. Do not use impact of any kind to move the Fenner drive assembly along the shaft.

3. Ensure shaft, new sprocket, and Fenner drive are completely free of paint, grease, oil, and debris of any kind. Return the surfaces to bare metal and clean using isopropyl alcohol.
4. Insert Fenner drive into sprocket. Center sprocket in the Fenner drive gripping area.
5. Place sprocket/Fenner drive assembly onto shaft. Position it to the desired location measured in Step 1. Verify shaft fully engages the Fenner drive gripping area.

NOTE:

At full installation torque, the 1-inch O.D. assembly travels about 0.045 inch (1.1 mm) axially along the shaft; the 1.5-inch O.D. assembly travels about 0.075 inch (1.9 mm). Position the assembly to compensate for this movement.

CAUTION:

Torque wrench required. Over-tightening the nut may damage the Fenner drive and/or sprocket.

6. Firmly hand tighten the nut (clockwise). Then use a torque wrench to tighten the nut to the specified torque in Appendix A.
7. Install chain and adjust according to "Chain Adjustment".

Crankshaft Chain Sprocket Replacement

WARNING:

To decrease the risk of injury or death, only trained and approved personnel can do maintenance work. Approved personnel must read and know the applicable Ariel Technical Manual and Packager-supplied information.

Before you start maintenance:

- Always use personal protection equipment (PPE) and obey your company policies for safety and operation.
- Do not tighten or loosen fasteners or fittings while the unit is in operation or is pressurized.
- Make sure the driver and compressor cannot turn during maintenance.
- Obey your company and packager instructions to isolate and vent the system.
- Make sure there is no remaining pressure in the cylinders. If equipped with “Kiene” valves on the indicator ports, open them to release residual pressure after blowdown.

  WARNING: RELEASED GAS MAY BE POISONOUS AND/OR FLAMMABLE.

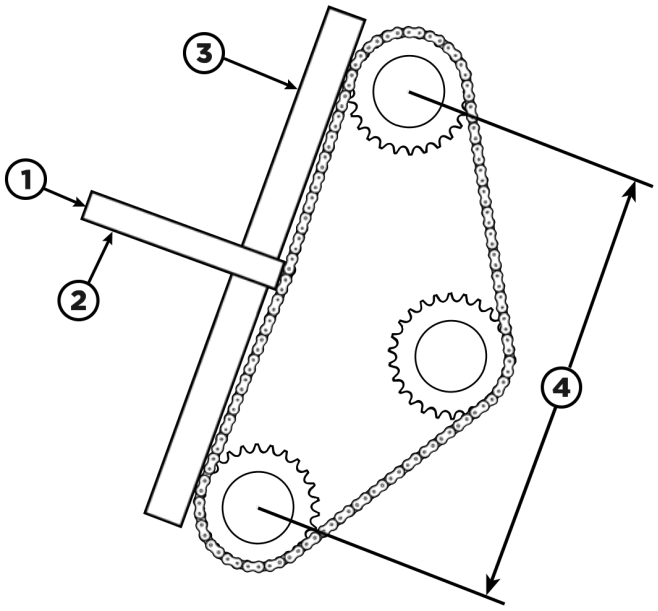
Before operation, purge the system. A pressurized hydrocarbon and air mixture can explode and cause injury or death.

For KBE: Crankshaft sprockets are on a removable adapter. Remove the adapter and sprocket and replace it with a new adapter and sprocket.

Chain Adjustment

1. Roll the crankshaft to the tightest position of the chain. This prevents snugging the chain at a slack position and breaking rollers, or ruining the pump and lubricator bearings when the chain goes through its tightest position.
2. Use a tape measure to measure the longest most easily accessible span from sprocket center to sprocket center, where the deflection will be gauged. Determine allowable deflection limits for the measured span as defined by the shaded area in Figure: Allowable Chain Deflection.
3. Measure chain deflection from a straight edge held on the chain rollers where it wraps over the two sprockets of the span (see Figure: Typical Chain Deflection Measurement). Using a machinist's rule with 0.01 inch or (1.0 mm) increments, apply finger pressure to measure the distance from the straight edge to a chain roller at the center of the span. Record the measurement and check to see if it is in tolerance.
4. If adjustment is required, remove the cap screws and plastic plugs from the eccentric cap. Rotate the cap clockwise to line up the first two new cap screw holes. If this tightens the chain too much, turn the cap counterclockwise for a different hole alignment.

Figure: Typical Chain Deflection Measurement



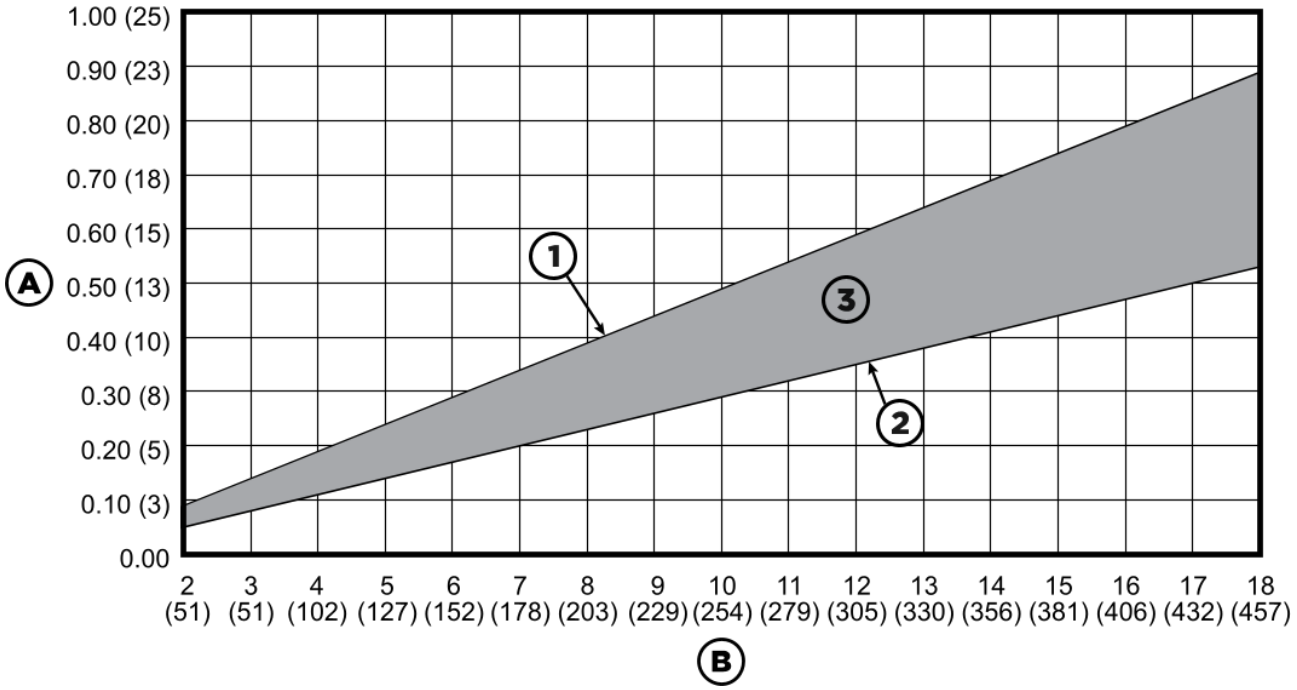
1	Finger Pressure	3	Straight Edge
2	Machinist's Scale	4	Span Length

5. Replace and tighten the two cap screws hand wrench tight.

> **CAUTION:**
>
> Over tightening may result in bent shafts.

6. Roll the crankshaft to check tightness in several positions. At its tightest position, the chain should deflect within the shaded limits in Figure: Allowable Chain Deflection. Replace plastic caps to keep the holes clean.

Figure: Allowable Chain Deflection, Inches (mm)



A	Deflection	**B**	Length of Measured Span	**1**	Maximum Deflection	**2**	Minimum Deflection	**3**	Allowable Deflection

Section 5 - Start Up

Warranty Notification - Installation List Data and Start Up Check Lists for Reciprocating Compressors

Download the Installation List Data and Start Up Check Lists forms, which is designed to ensure a successful start-up of Ariel reciprocating compressors. Ariel warranty coverage requires these completed forms be sent to:

Administrative Assistant - Sales, Ariel Corporation

35 Blackjack Road
Mount Vernon, Ohio 43050 USA
Phone: 740-397-0311 FAX: 740-397-3856
Email: warrantyregistration@arielcorp.com

Section 6 - Compressor Troubleshooting

Expect minor problems during routine operation of an Ariel compressor. These troubles most often trace to liquid, dirt, improper adjustment, or operators unfamiliar with Ariel compressors. These difficulties can usually be corrected by cleaning, proper adjustment, elimination of an adverse condition, part replacement, or proper training.

Major problems usually trace to long periods of operation with unsuitable lubrication, careless operation, lack of routine maintenance, or using the compressor for purposes not intended.

Recording inter-stage pressures and temperatures on multistage units is valuable. Any variation when operating at a given load point indicates trouble in one of the stages. Normally, a decrease in inter-stage pressure indicates trouble in the lower pressure cylinder. An increase usually indicates trouble in the higher pressure cylinder. Below is a list of common problems and possible causes.

Problem	Possible Causes
Low Oil Pressure	• Dry pump gears. Lubricate rotor and idler assembly with oil. • Oil pump pressure regulating valve set too low or sticking. • Oil pump or oil pump drive failure. • Oil foaming from counterweights striking oil surface (oil level in sight glass too high), or from vortex at strainer inlet (oil level in sight glass too low), or from leaks in pump suction line. • Cold oil. • Dirty oil filter. • Interior frame oil leaks. • Excessive leakage at bearings. • Improper low oil pressure switch setting. • Oil pump relief valve set too low. • Defective pressure gauge. • Plugged oil sump strainer. • Improper end clearance in oil pump.
High/Low Suction Pressure	• Suction control valve malfunction. • Faulty pressure gauges. • Frozen/plugged inlet line. • Plugged intake screen or filter. • Scrubber dump valve stuck open. • Recycle malfunction. • Site production equipment problems.
High/Low Inter-Stage Pressure	• Faulty valves or rings. • Gas leak. • Frozen cooler section or tubing to panel. • Faulty pressure gauges. • Scrubber dump valve stuck open. • Load change.
Noise in Cylinder	• Loose piston. • Piston hitting cylinder head end head or crank end head. • Loose crosshead balance nut. • Broken or leaking valve(s). • Worn or broken piston rings or wear bands. • Valve improperly seated or damaged seat gasket. • Liquids in cylinder.

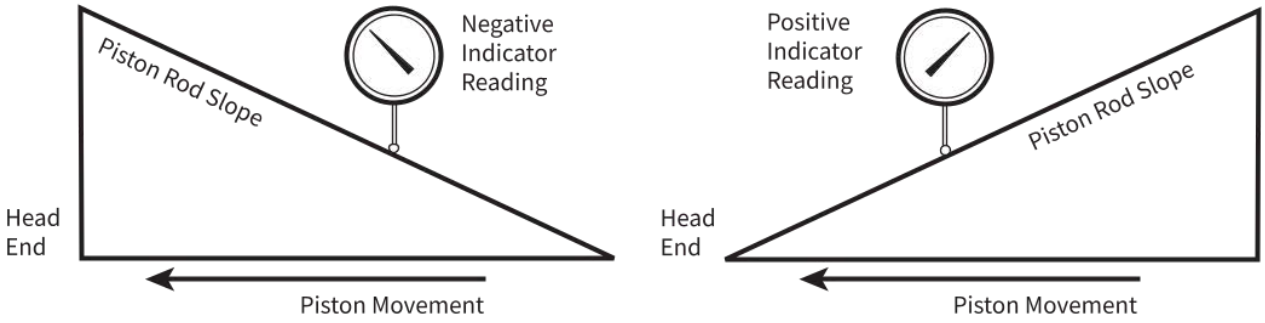
Problem	Possible Causes
Frame Knocks	• Loose crosshead pin or pin caps. • Loose or worn main, crankpin, or crosshead bearings. • Low oil pressure. • Cold oil. • Incorrect oil. • Knock is actually from cylinder end. • Low fluid level in damper.
Excessive Carbon on Valves	• Excessive lube oil. • Improper lube oil. • Oil carry-over from inlet system or previous stage. • Broken or leaking valves causing high temperature. • Excessive temperature due to high pressure ratio across cylinders.
Relief Valve Popping	• Faulty relief valve. • Leaking suction valves or rings on next higher stage. • Obstruction, closed or faulty valve in discharge line.
High Discharge Temperature	• Excessive ratio across cylinder due to leaking inlet valves or rings on next higher stage. • Fouled inter-cooler piping. • Leaking discharge valves or piston rings. • High inlet temperature. • Improper lube oil and/or lube rate.
High Frame Oil Temperature	• Faulty temperature gauge. • Dirty oil filters. • High oil level. • Faulty thermostatic element. • Faulty thermostatic control valve. • Clogged or blocked oil cooler. • Binding or tightness in the compressor.
Packing Over Heating	• Lubrication failure. • Improper lube oil and/or insufficient lube rate. • Worn packing rings. • Dirt in packing. • Improper ring side or end gap clearance. • Scored, tapered, out of round, or bent piston rod. • Excessive piston rod runout.

Problem	Possible Causes
Excessive Packing Leakage	• Worn packing rings. • Improper lube oil and or insufficient lube rate. • Dirt in packing. • Packing rings assembled incorrectly. • Improper ring side or end gap clearance. • Plugged packing vent system. • Scored, tapered, out of round, or bent piston rod. • Excessive piston rod runout. • Packing not seated or properly run in.
Drive End of Crankshaft Oil Leaks	• Clogged vent or vent piping. • Excessive cylinder packing leakage.
Excessive Piston Rod Oil Wiper Leaks	• Worn wiper rings. • Wipers incorrectly assembled. • Worn/scored rod. • Improper fit of rings to rod/side clearance.
Force Feed Lubrication Shutdown	• Force feed pump or lubricator block failure. • Loss of oil supply to force feed pump. • Lubricator drive failure. • DNFT not programmed correctly. • DNFT battery failure or power loss. • Loose or grounded control wiring. • Pin assembly not completely pushed into the DNFT housing.
Oil Leaks at Pipe Threaded Connections	• Joint not tight. • Pipe sealant was omitted. • Defective or damaged pipe threads. • NPTF Dryseal threads not being used. • Pressure too high for pipe threaded connection use. • Pipe thread sealant incompatible with the synthetic oil used. • Cracked pipe or fittings.
Vibration Shutdown	• Short in wire to panel. • Incorrectly positioned vibration switch. • Loose mounting bolts. • High scrubber level (liquid carry-over). • Broken valve, piston, or piston rod. • Main drive-line or coupling failure.

Problem	Possible Causes
VVCP adjustment handle or piston stem is locked and will not turn.	• Piston hit VVCP head. Refer to "VVCP Adjustment".

Piston Rod Runout Correction

Figure: Piston Rod Runout Measurement



Piston rod runout may be either positive or negative. A positive reading indicates the piston drops as it moves to the head end of the cylinder due to the clearance in the bore. As the piston drops, the rod rises, yielding a positive reading on the indicator. If the cylinder is not supported properly, it could droop causing the piston to drop as it moves to the head end of the cylinder. A negative reading indicates the piston climbs as it moves to the head end of the cylinder. When the piston climbs, the rod drops, yielding a negative reading. Any reading outside the Ariel rod runout tolerance requires correction for safe, efficient compressor operation (see Appendix B). Piston rod runout correction first requires verification (and correction, if needed) of the crosshead clearance. See flowcharts below.

Figure: Crosshead Clearance Verification/Correction Flowchart

> **NOTE:**
>
> Review all steps before troubleshooting. If issues persist, contact the Ariel Response Center.

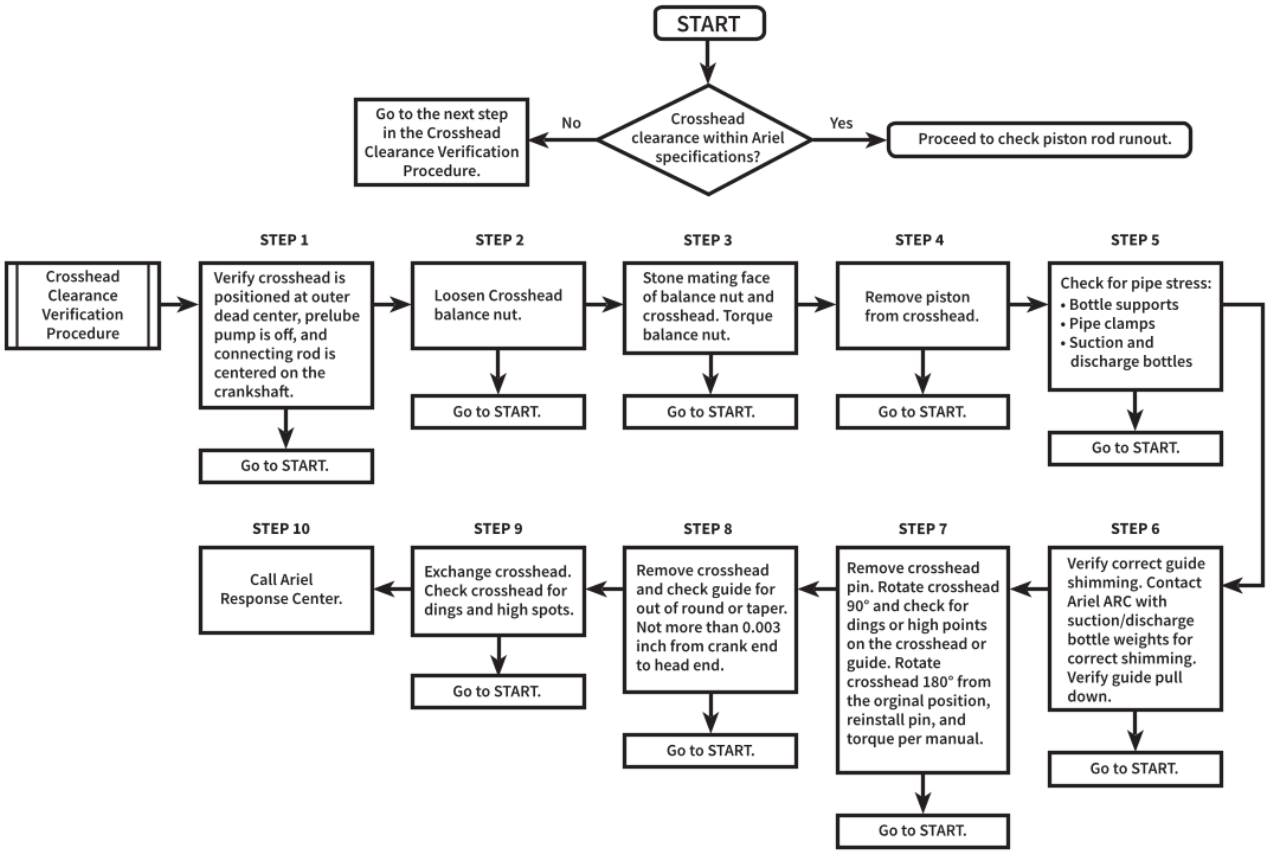
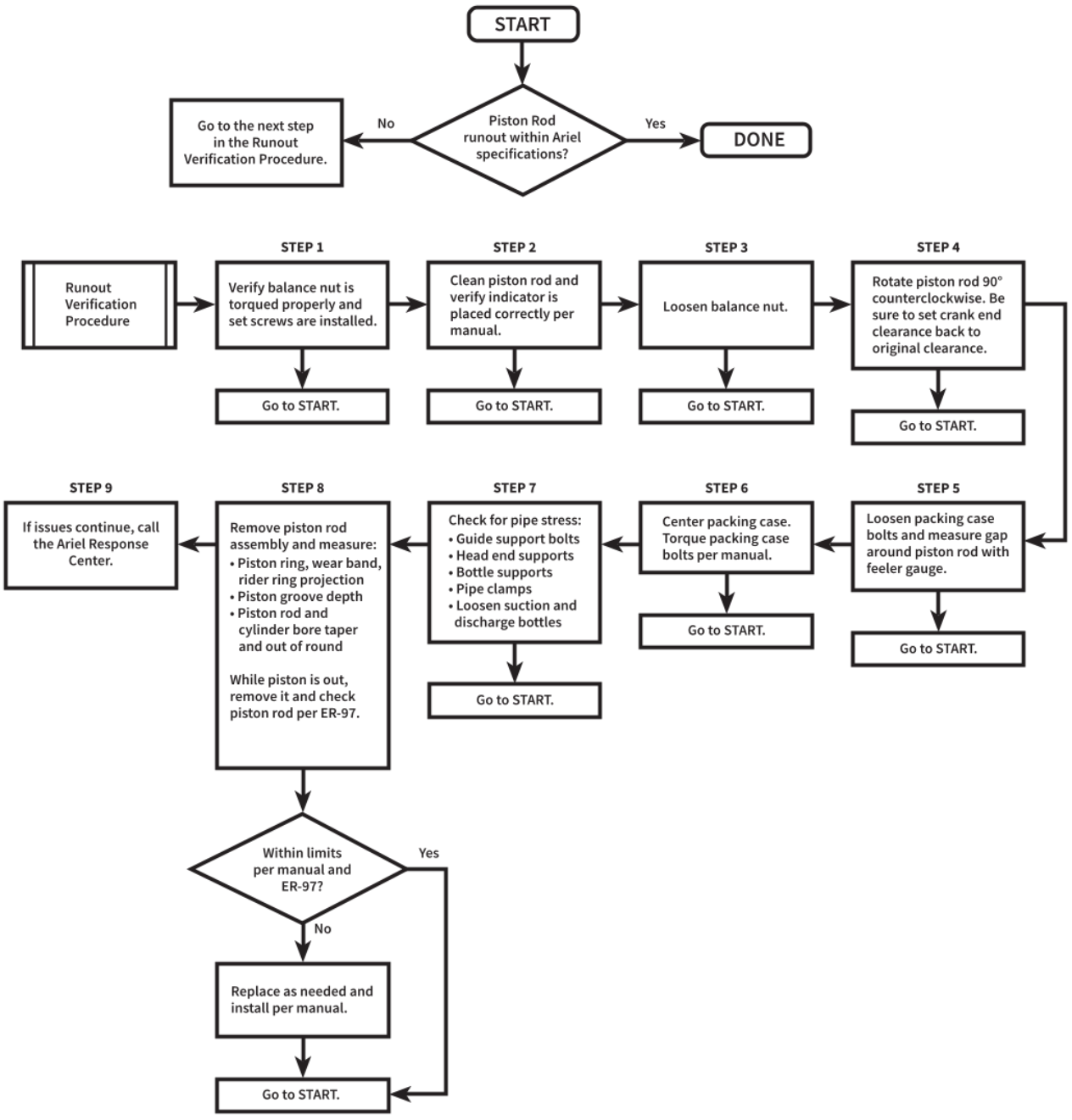


Figure: Piston Rod Runout Verification/Correction Flowchart

> **NOTE:**
> Verify crosshead clearance is within Ariel specs per the manual before checking piston rod runout.



Appendix A - Ariel Fasteners & Torques

This document lists fastener lubrication and torque requirements for proper assembly of KBE reciprocating compressors. For older units with lower specified torque values, do not assume higher values without consulting the packager and/or Ariel.

- Use only Ariel specified fasteners tightened to the correct torque.
- Connecting rod, valve cap, and suction/discharge nozzle (Ariel supplied flange) fasteners are modified to prevent fatigue; do not replace them with standard cap screws. To replace other fasteners with standard cap screws, contact your packager or Ariel.
- Clean and de-burr all threads.
- Do not use molybdenum disulfide lubricants.
- Do not use Never-Seez for fastener lubrication unless specified or excessive stresses may result with specified torques. See Table: Fastener Thread & Seating Surface Lubrication.
- Use anti-seize lubricants sparingly; excessive amounts cause oil analysis to indicate contamination and may unnecessarily increase maintenance costs.
- Re-preserve any fasteners subject to corrosion after installation.

Table: Fastener Thread & Seating Surface Lubrication

NOTE: Lubricate all fasteners both under the head and on the threads.	
Application	**Lubricant**
Cast Cylinders: valve cap, tandem cylinder mounting, packing, crank end and head end head	Mineral oil (ISO 100-150)
Frame: jack screws **Pistons:** piston nuts, set screws, piston collar, crosshead threads, tandem piston to piston rod **All stainless steel fasteners except valve fasteners listed in Table: Hoerbiger Valve Assembly Fastener Torques**	Never-Seez Regular Grade
Connecting rod cap	Lubriplate 630-AA
Crosshead balance nut, crosshead guide to frame to spacer bar, crosshead guide to frame to main bearing cap	**Under the head or mating surface:** Mineral oil (ISO 100-150). **On the threads:** Never-Seez Regular Grade.
All pipe threads	Loctite 567
Fasteners specified with Loctite on threads	Loctite only with no additional lube
All other fasteners, including crankshaft attachments, all O-rings and stat-o-seals	Mineral oil (ISO 100-150)

Figure: Bolt Head Grade and Material Identification



CAUTION:

To replace a fastener, see parts list for fastener grade and part number. Do not use a lesser or greater material grade. Use Ariel parts to replace special fasteners and fasteners with reduced body diameter for fatigue resistance. Do not torque fasteners with compressor running or pressurized. Read technical manual safety warnings.

Recommendations for Torque Accuracy

1. Critical bolted joints require accurate fastener pre-stress to ensure safe and reliable operation. On Ariel compressors, these joints have specified torque or other engineered tightening methods. Only mechanics properly trained in bolted flange joint assembly should maintain these joints.
2. When torque is specified for a fastener, use a properly calibrated torque wrench to torque the fasteners correctly.
3. Use the correct lubricant on the fastener (see Table: Fastener Thread & Seating Surface Lubrication).
4. Always turn fasteners in by hand until they are finger tight and drawn down evenly. Using a wrench, turn fasteners to draw o-rings into their bores. Using pneumatic, electric, or hydraulic drivers and runners can result in uneven tightening and over-stressed fasteners. If there is significant resistance, remove the fastener to determine and correct the cause.
5. Always apply a steady, slow force to a torque wrench, and stop immediately when the wrench clicks. Do not bounce, jerk, or double tap the wrench – it significantly increases the applied torque. Perform final tightening with a torque wrench.
6. Do not break fasteners loose with a torque wrench - it may overload the wrench and affect its accuracy.
7. Determine torque wrench accuracy range before using - most are not accurate at the extreme lower and higher ends of their range.
8. Clean and de-burr all threads before assembly. Ensure no oil or debris is in the bottom of the bolt hole when installing bolts. Apply oil only to fasteners, not in the bolt hole.
9. Inspect all joint mating surfaces for scratches, nicks, gouges, burrs, corrosion and flatness. This includes the fastener and the fastener contact area on the bolted part. Address any findings before proceeding.

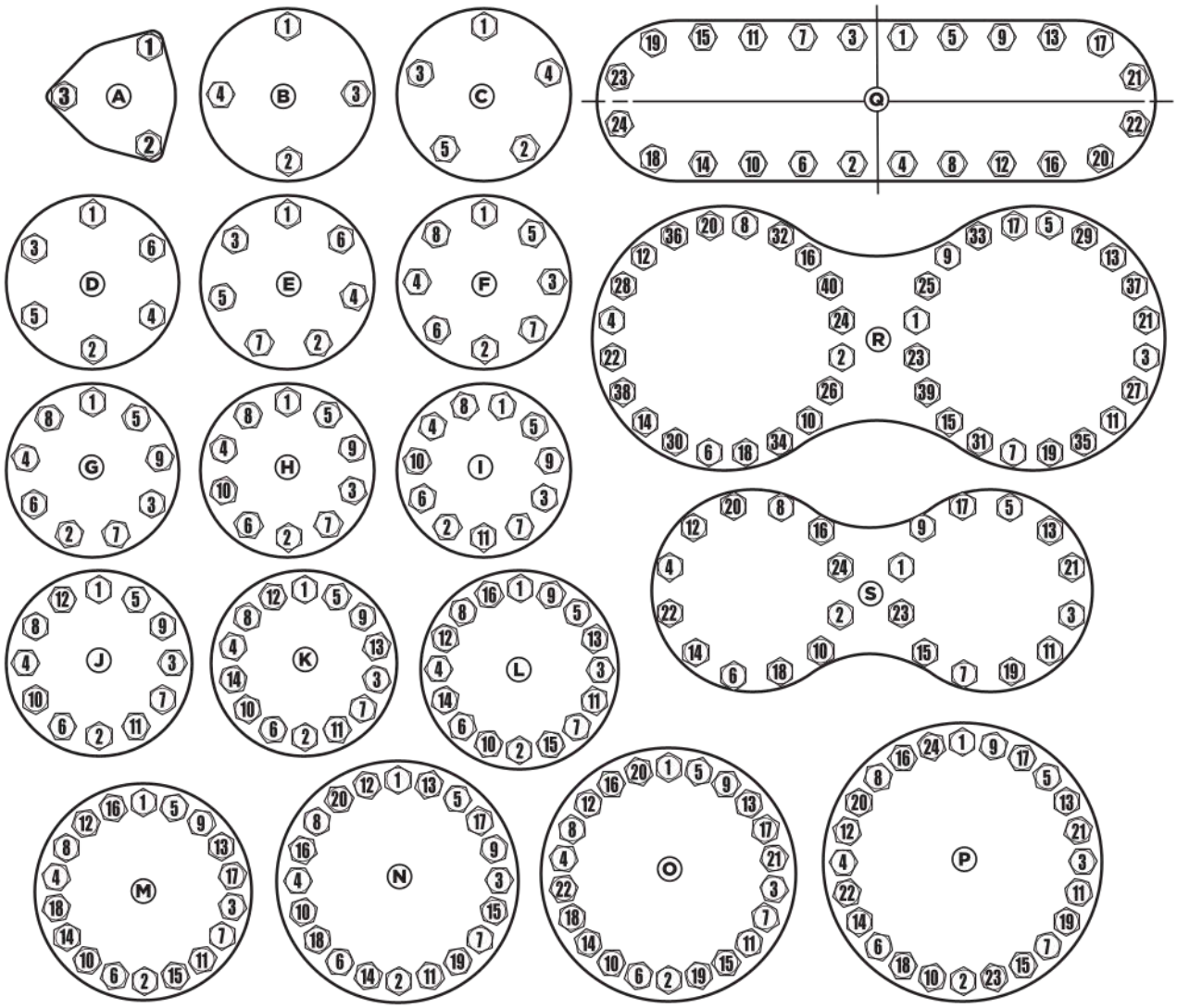
10. **For main bearing cap and connecting rod cap fasteners:** If the bearing cap is stamped with torque sequence numbers, tighten fasteners in that sequence. If not stamped, follow the Figure: Main Bearing Cap or Connecting Rod Cap Torque Sequence.

    Figure: Main Bearing Cap or Connecting Rod Cap Torque Sequence

    - **Where torque and turn IS NOT SPECIFIED:** Apply 50% of specified torque in the sequence shown to close joint. Once closed, apply 100% of specified torque in the same sequence. Repeat 100% torque in the same sequence until all fasteners stop moving.
    - **Where torque and turn IS SPECIFIED:** Apply 100% of pre-torque in the sequence shown to close joint. Repeat 100% torque in the sequence until all fasteners stop moving. Finally, apply the full turn value to each fastener one time in the same sequence.
11. **For Ariel pressure containing components, see Figure: Pressure Containing Joint Recommended Torque Sequences for recommended torque sequences (includes cylinders, valve caps, cylinder heads, unloaders, gas passage caps, flanges, and packings):**
    - Refer to ASME PCC-1 “Guidelines for Pressure Boundary Bolted Joint Flange Assembly” for commonly accepted practices.
    - Install a new gasket or O-ring.
    - For 3-bolt valve caps, tighten in this sequence: 1-2-3, 2-3-1, 3-1-2. Use the sequence to close the joint without exceeding 25% of full torque. Repeat the three-pass sequence for 25%, 50%, 75%, and 100% of full torque. Repeat 100% torque in a circular pattern until all fasteners stop moving.
    - For other pressure containing joints, tighten the fasteners in four passes using the appropriate sequence for 25%, 50%, 75% and 100% of the specified torque. After 100% torque is reached, continue passes until all fasteners stop moving.
    - Inspect the gap under the bolted flange (if visible) after each pass to ensure a consistent gap all around. If the gap is not consistent, the assembly is cocked – disassemble and inspect to determine the cause, and re-assemble after the cause is corrected.

Figure: Pressure Containing Joint Recommended Torque Sequences



A	3 Fasteners	**E**	7 Fasteners	**I**	11 Fasteners	**M**	18 Fasteners	**Q**	Obround Flange
B	4 Fasteners	**F**	8 Fasteners	**J**	12 Fasteners	**N**	20 Fasteners	**R**	40-Fastener Peanut Flange
C	5 Fasteners	**G**	9 Fasteners	**K**	14 Fasteners	**O**	22 Fasteners	**S**	24-Fastener Peanut Flange
D	6 Fasteners	**H**	10 Fasteners	**L**	16 Fasteners	**P**	24 Fasteners		

12. When finished, reset torque wrench to its lowest setting to relax the spring. A torque wrench left in a high setting stresses the spring and decreases accuracy with time.
13. To determine the torque wrench setting when using a torque multiplier, divide the desired fastener torque by the multiplier actual mechanical advantage, not the design mechanical advantage. See manufacturer documentation for multiplier.

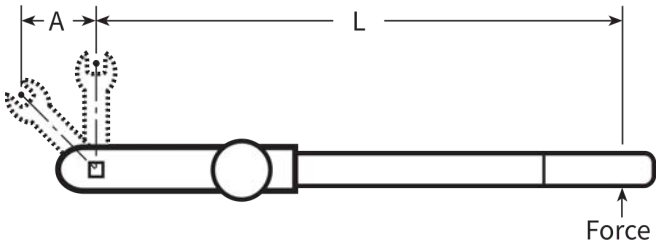
14. If using an offset adapter with a torque wrench, the torque setting is not the actual torque applied, unless the adapter is 90° to the torque wrench. To determine the correct torque setting:

    Tw = Ta x [L ÷ (L + A)]

    Tw = Torque wrench setting, Lb x Ft or N·m.

    Ta = Torque required at fastener, Lb x Ft or N·m.

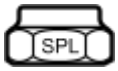
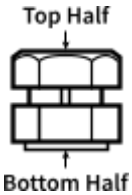
Figure: Torque Wrench with Angled Adapter



L = Length of wrench, ft or m (from square drive end to center point of force on handle).

A = Wrench length added by adapter, ft or m (measured to end of adapter on a line parallel to wrench center line).

Table: Hoerbiger Valve Assembly Fastener Torques

Fastener	Nominal Size Inch - TPI	Type Lb x Ft (N·m), unless specified	Torque Lb x Ft (N·m), unless specified
Center Cap Screw[a]	5/16 - 24	12 Point - Steel Grade 5 Material Parts: SPL3 & 4	14 (19)
	3/8 - 24		24 (33)
	7/16 - 20		35 (48)
	5/16 - 24	12 Point - Steel Grade 5 Material Parts: SPL or SPL5	22 (30)
	3/8 - 24		40 (54)
	7/16 - 20		65 (88)
	5/16 - 24	12 Point - Stainless Steel Grade B 8M Material Parts: SPL6	14 (19)
	3/8 - 24		24 (33)
	7/16 - 20		35 (48)
Peripheral Cap Screw	#10 - 32	Hex Socket Head	25 Lb x In (2.8)
	#12 - 28		43 Lb x In (4.9)
	1/4 - 20		110 Lb x In (12)
	5/16 - 18		176 Lb x In (20)
	3/8 - 16		21 (28)
Single-piece Lock Nuts (Greer, Flexloc, Selflock)  (Microlock, Spiralock[b])	**Nominal Size**	**Torque Lb x Ft (N·m), unless specified**	
	Inch - TPI	**SPL[b]**	**Other**
	#12 - 28	96 Lb x In. (11)[c]	-
	5/16 - 24	15 (20)	109 Lb x In (12)
	3/8 - 24	26 (35)	17 (23)
	7/16 - 20	48 (65)	26 (35)
	1/2 - 20	67 (91)	41 (55)
	5/8 - 18	110 (150)	82 (110)
	3/4 - 16	195 (265)	145 (195)
	7/8 - 14	295 (400)	230 (310)
Center Stud Drake 2-Piece Beam Lock Nut 	**Inch - TPI**	**Bottom Half**	**Top Half**
	1/4 - 28	103 Lb x In (12)	66 Lb x In (7.5)
	5/16 - 24	168 Lb x In (19)	96 Lb x In (11)
	3/8 - 24	16 (22)	96 Lb x In (11)
	1/2 - 20	36 (49)[d]	20 (27)
	5/8 - 18	73 (99)	40 (54)
	3/4 - 16	130 (175)	70 (95)

Fastener	Nominal Size Inch - TPI	Type Lb x Ft (N·m), unless specified	Torque Lb x Ft (N·m), unless specified
	7/8 - 14	210 (285)	115 (155)

a Center cap screw valve assemblies have Spiralock® (SPL) threads to prevent loosening. See bottom of valve assembly for SPL material parts number (3, 4, 5 or 6) and select proper torque from the table. Lubricate both threads and seating surfaces with mineral oil (ISO 100-150) ONLY. If using older valve assemblies not covered in the table, see the original torque chart provided in the compressor tool box, or contact Ariel for instructions. Spiralock® threads cannot be dressed with a standard tap. Clean center cap screws in valve assemblies not marked SPL with Loctite safety solvent and lock them with one or two drops of Loctite #272 thread locking compound. DO NOT use petroleum lubricants.

b Microlock and Spiralock nuts are marked "SPL" on one of the flats.

c Use this torque only with a 17-4PH 900H material center stud.

d Use 29 (39) for 1/2 - 20 bottom half Drake lock nut with non-metallic valve plates in liftwasher valves.

KBE Fastener Torques

Fastener	Nominal Size Inch - TPI	Type	Torque Lb x Ft (N·m), unless specified
Main Bearing Cap - Cap Screw	7/8 - 9	12 Point - Grade 8	280 (380)
Connecting Rod Cap - Cap Screw Note: Use Ariel turn indicator, B-3993	1” - 14	12 Point - Grade 8	120 (165) + 110°
Torsional Vibration Detuner - Cap Screw	1” - 14	12 Point - Grade 8	530 (715)
Crosshead Pin Thru Cap Screw - Lock Nut	1/2 - 20	Hex - Nylon Insert	61 (83)
Crosshead Guide to Frame - Cap Screw	7/8 - 9	12 Point - Grade 8	280 (380)
Crosshead Guide to Frame to Spacer Bar- Cap Screw	7/8 - 9	12 Point - Grade 8	280 (380)
Crosshead Guide to Frame to Main Bearing Cap - Cap Screw	7/8 - 9	12 Point - Grade 8	280 (380)
Crosshead Guide to Cylinder - Cap Screw	5/8 - 11	12 Point - Grade 8	97 (130)
Crosshead Guide to Support - Cap Screw	7/8 - 9	12 Point - Grade 8	255 (345)
Eccentric Vernier Cap - Cap Screw	5/16 - 18	12 Point - Grade 8	Hand Wrench Tight
Idler Sprocket Thru Cap Screw - Lock Nut	1/2 - 20	Hex - Prevailing	41 (55)
Rod Packing - Cap Screw[e]	3/4 - 10	12 Point - Grade 8	125 (170)
Packing Tie Stud Nut	#10 - 24	Hex	20 Lb x In. (2.3)
Piston Nut	1-5/8 - 12	Ariel Design	1590 (2150)[k]
Piston Nut Set Screw	5/16 - 24	Socket Head	164 Lb x In (19)[f]
Crosshead Balance Nut	1-3/4 - 12	Ariel Design	Slugging Method[g]
Force Feed Lube Box - Bearing Housing	1-3/8 - 20 - LH	Bearing Housing	70 (95)
Force Feed Lube Box - Jam Nut	1” - 14	Hex	75 (100)
	1-1/2 - 12		112 (152)
Force Feed Lube Pump Mounting	1/4 - 20	Hex - Grade 5	50 Lb x In (5.6)
Auto Relief Valve	1/2[h]	Ariel Design	75 Lb x In (8.5)
Auto Relief Valve Manifold Plug	#10 - 24	Ariel Design	50 Lb x In (5.6)
	1/4 NPT	Hex	Hand Wrench Tight
Flywheel to Hub	1” - 8	12 Point - Grade 8	460 (620)
	1” - 14	12 Point - Grade 8	530 (715)
Lifting Bracket to Frame	7/8 - 9	Dog Point - Grade 8	230 (310)
Piston Rod Oil Slinger Bolt - Lock Nut	1/4 - 28	Hex Jam - Prevailing	95 Lb x In (11)
Frame Foot Foot Hold Down	1-1/4 - 7	Hex Nut	850 (1150)[i]
Flywheel/Damper/Sprocket Adapter to Crankshaft (auxiliary end) - Cap Screw	5/8 - 18	12 Point - Grade 8	125 (170)
Fenner Drive at FF Lube Sprocket, A-11560	7/8	Hex	200 Lb x In (23)

Fastener	Nominal Size Inch - TPI	Type	Torque Lb x Ft (N·m), unless specified
Cap Screw[j] • Valve Cap • Cylinder Head[l] • VVCP or eVCP Adapter to Head[l] • VVCP, FVCP, or eVCP to Cylinder • Ariel supplied companion flanges, except “Peanut” Dual Nozzle • VVCP Seal Carrier	5/16 - 18	Hex - Grade 8 or 9, 12 Point - Grade 8 or 17-4PH, or Socket Head	108 Lb x In. (12)
	3/8 - 16		193 Lb x In (22)
	1/2 - 13		40 (54)
	5/8 -11		79 (105)
	5/8 -18		92 (125)
	3/4 - 10		140 (190)
	3/4 - 16		160 (215)
	7/8 - 9		230 (310)
	7/8 -14		260 (350)
	1” - 8		345 (465)
	1” - 14		395 (535)
	1-1/8 - 12		560 (760)
	1-1/4 - 12		780 (1060)
Unloader Actuator to Valve Cap - Cap Screw	1/2 - 13	Hex - Grade 8	48 (65)
ANSI Flange to Cylinder Nozzle	See Cylinder Outline	Provided by Packager	See ER-127.
Taper-Lok Flange[m]	1/2 - 13	Hex Nut on Grade 8 Stud or 17-4 stud	40 (54)
	1/2 - 20		46 (62)
	5/8 -11		79 (105)
	5/8 -18		92 (125)
	3/4 - 16		160 (215)
	7/8 - 9		230 (310)
	7/8 -14		260 (350)
	1” - 14		395 (535)
	1-1/8 - 12		560 (760)
Tandem Cylinder to Cylinder[j]	5/8 - 11	Hex - Grade 8 or 9	88 (120)
Stud Seating	3/8 - 16	Dog Point Grade 8 or 17-4PH	107 Lb x In (12)
	1/2 - 13		22 (30)
	5/8 - 11		44 (60)
	3/4 - 10		79 (105)
	3/4 - 16		90 (120)
	7/8 - 9		130 (175)

Fastener	Nominal Size Inch - TPI	Type	Torque Lb x Ft (N·m), unless specified
	7/8 - 14		145 (195)
	1” and larger		200 (270)
Distribution Block Tie Rod - Nut	1/4 - 28	Hex	68 Lb x In (7.7)
Distribution Block Divider Valve - Cap Screw	1/4 - 28	Socket Head	75 Lb x In (8.5)[n]
Grade 5 - Hex Cap Screw	ALL	Hex - Grade 5	Hand Wrench Tight

e Repeat final torque for rod packing bolts until they no longer turn before the torque wrench clicks.
f Or tighten set screw 15° past the Allen wrench yield point. Discard deformed wrench, and use a new Allen wrench to tighten the other set screw. Stake set screws per manual after tightening.
g Or use 3500 psig (241 barg) hydraulic pressure on separately purchased crosshead (balance) nut torquing tool.
h "1/2" indicates the wrench size. Auto relief valve can swivel in the manifold once torqued. See assembly drawing.
i Minimum torque to stress recommended 1-1/4 - 7 hold-down stud to 55,000 psi (380 MPa). Use studs with an ultimate strength of 100,000 psi (690 MPa) or greater. If greater, increase torque stress to about 55% of ultimate strength, as specified by packager.
j For studs specified for cylinder applications, tighten nuts to the same torque as cap screws in similar applications.
k Or use 3500 psig (241 barg) hydraulic pressure and 200 (270) torque on spanner with piston rod tensioning tool A-28362 or A-24512. Tighten, loosen, then re-tighten piston nut to ensure proper torque.
l Excludes socket head cap screws, which are installed hand wrench tight.
m See ER-136 for Taper-Lok installation instructions.
n Step-torque first to 40 Lb x In. (4.5 N•m), and then to 75 Lb x In. (8.5 N•m).

Appendix B - Clearances

Table: KBE Frame and Running Gear Clearances, in. (mm)

Description	Clearance
Crankshaft Journal Bearing (Jack)	0.0015 to 0.0050 (0.038 to 0.127)
Crankshaft Thrust (End)	0.0085 to 0.0200 (0.216 to 0.508)
Connecting Rod Bearing (Jack)	0.0035 to 0.0070 (0.089 to 0.178)
Connecting Rod Thrust (Side)	0.007 to 0.018 (0.178 to 0.457)
Crosshead to Guide	0.008 to 0.013 (0.20 to 0.33) (Feeler Gauge)
Crosshead Pin to Connecting Rod Bushing	0.002 to 0.004 (0.05 to 0.10)
Crosshead Pin to Crosshead Bushings	0.0020 to 0.0042 (0.05 to 0.11)
Shaft Seal Housing, Centered on Crankshaft	0.008 (0.20) (Feeler Gauge All Around)

Table: KBE Piston End Clearances, in. (mm)[a, b]

Cylinder Class	Crank End	Head End	Total
All KBE	0.040 (1.0)	0.050 to 0.110 (1.3 to 2.8)	0.090 to 0.150 (2.3 to 3.8)

a Measured clearances may not agree because of oil films, assembly tolerances, wear, etc. Do not use plastigages, solder, etc.

b If total piston end clearance is not within table tolerance, contact your Packager or Ariel.

Table: KBE Max. Acceptable Piston Rod Runout, in. (mm)

Description	Runout
Vertical Runout	0.0020 (0.0508)
Horizontal Runout	0.0010 (0.0254)

Table: Packing Seal Ring Side Clearances, in. (mm)

Ring Set	Actual Groove Width	Side Clearance
RS3	0.625 to 0.627 (15.88 to 15.93)	0.030 to 0.036 (0.76 to 0.91)
VR	0.375 to 0.377 (9.53 to 9.58)	0.023 to 0.027 (0.58 to 0.69)

Table: Pressure Breaker Side Clearance, in. (mm), (PB1)

Actual Groove Width	Side Clearance
0.375 to 0.377 (9.53 to 9.58)	0.016 to 0.020 (0.41 to 0.51)

Table: Wiper Ring Side Clearance, in. (mm)

Wiper Ring Type	Side Clearance
OW3	0.006 to 0.009 (0.15 to 0.23)

For piston ring, rider ring, and wear band clearances, refer to the Ariel Torque Guide:

Table: KBE Cylinder Clearances, in. (mm)

Bore Diameter	Piston-to-Bore Diametral Clearance	
	Conventional Piston Rings & Wear Band	Piston/Rider Rings[a]
5.5 (140)	0.013 to 0.018 (0.33 to 0.46)	-
6.0 (152)	-	0.090 to 0.096 (2.29 to 2.44)
6.375 (162)	-	
9.25 (235)	-	
9.75 (248)	-	
11 (279)	-	
12.25 (311)	-	
13.5 (343)	-	
15.75 (400)	-	0.090 to 0.097 (2.29 to 2.46)

a Piston/rider rings are standard on all E & KBE cylinders except 5.5 (140).

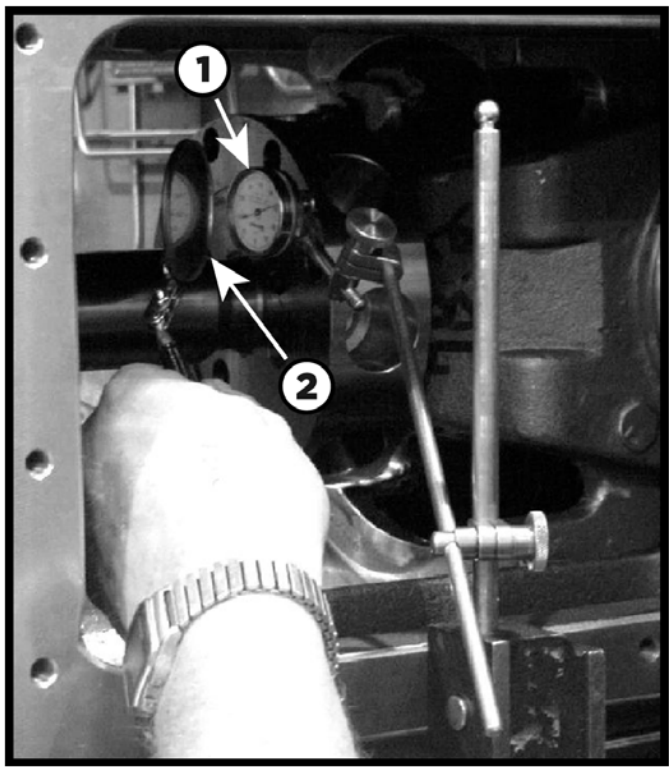
Measuring Head End Clearance for Forged Steel Tandem Cylinders with Concentric Valves

CAUTION:

Completely vent compressor to relieve ALL gas pressure and remove coupling spacer prior to removing access covers for compressor internal maintenance. For electric drive motors, lock out the breaker. Failure to follow these steps can result in serious personal injury or death.

1. Set crank end piston clearance to the proper feeler gauge specification stamped on the cylinder data plate or printed in the Ariel Technical manual.
2. With head end components removed, position the piston for the desired cylinder at outer dead center. Position a dial indicator to index off of the crosshead or crosshead nut outer face, with the cylinder near to outer dead center. Manually turn crankshaft to find the 0.010 inch (0.25 mm) position, before and after outer dead center. Then turn crankshaft to position piston at outer dead center (see Figure: Position Piston At Outer Dead Center).
3. **Dimension A:** Measure from cylinder head end to piston head end to the nearest thousandth inch and record. See illustration.
4. **Dimension B:** With gasket removed, measure from head end cylinder face to the gasket seat and record.
5. **Dimension C:**

Figure: Position Piston At Outer Dead Center



1 Button Dial Indicator **2** Mirror

Norm. Thickness	Use
.030	.028
.060	.055
.060	.070*

* Use when the cylinder has a removable liner with a metallic tapered seat gasket between it and the cylinder body (see ER-118).

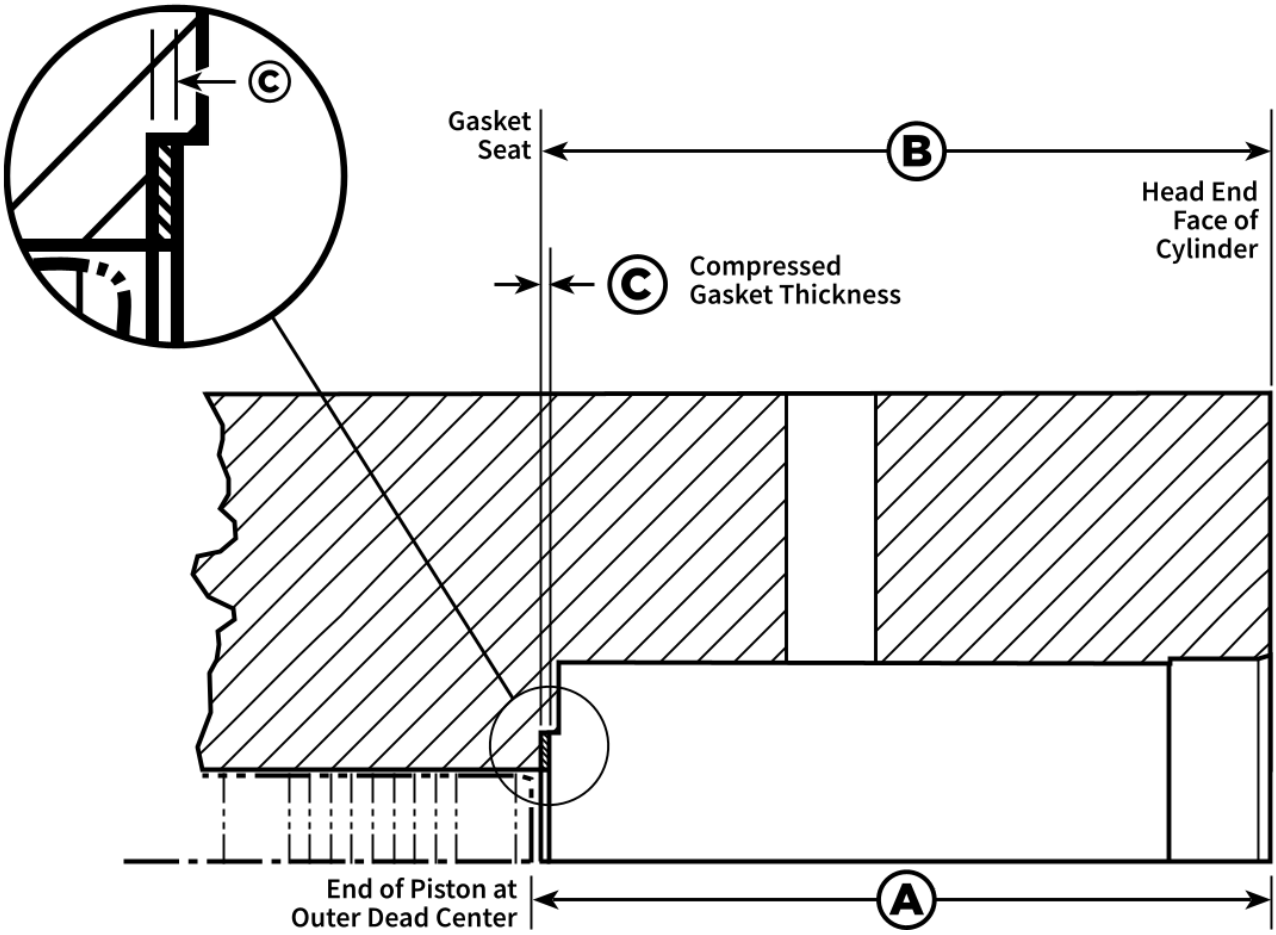
6. Calculate Head End Clearance: **A - B + C = Head End Clearance**

A _____

- B _____

+ C _____

= _____



7. Compare calculated head end piston clearance to clearance tolerances stamped on the cylinder data plate or printed in the Ariel Technical manual. **If tolerances are unavailable or if calculated clearance is outside tolerance, contact your packager or Ariel before proceeding.**
8. Measure head end piston clearance prior to new start up, as required by recommended maintenance intervals in Section 3, or after installation, removal, or replacement of any one of the following:
   - Crankshaft
   - Connecting Rod
   - Crosshead Guide
   - Piston and/or Piston Rod
   - Either Tandem Cylinder Body
   - Crosshead and/or Crosshead Nut
9. Re-assemble all components according to the Ariel Maintenance and Repair Manual supplied with your compressor. Verify installation of all gaskets and manually bar over the compressor to confirm the crankshaft rotates freely.

Appendix C - Frame Specifications

Table: KBE Frame Specifications

Specification	KBE/2	KBE/4
Stroke, in. (mm)	5.0 (127)	
Maximum Allowable Speed [a], RPM	1500	
Minimum Speed [b], RPM	700	
Piston Speed [c], FPM (m/s)	To 1250 (6.35)	
Horsepower, hp (kW)	1340 (999)	2680 (1998)
Maximum Length, in. (m)	54.0 (1.37)	98.0 (2.49)
Maximum Overall Height, in. (m)	29.0 (0.74) to top of aluminum cover	
Maximum Width with Cylinders	See appropriate frame, guide, and cylinder outline drawings.	
Height - Bottom to Crankshaft Centerline, in. (mm)	17.0 (432)	
Approximate Weight with Cylinders	See Ariel Performance Program.	
Connecting Rod Centerline to Centerline, in. (mm)	13.75 (349)	
Sump Capacity, US gal. (L)	18.0 (68)	39.0 (148)
Oil Pump Flow Rate [d], GPM (L/s)	28.0 (1.77)	35.0 (2.21)
Oil Heat Rejection BTU/hr (kW)	37000 (10.8)	74000 (21.7)
Piston Rod Diameter, in. (mm)	2.0 (51)	
Internal Rod Load - Double Acting		
Compression + Tension, lbf. (kN)	74000 (329)	
Tension, lbf. (kN)	37000 (165)	
Compression, lbf. (kN)	40000 (178)	
Internal Rod Load - Single Acting		
Tension, lbf. (kN)	37000 (165)	

a Maximum Allowable Speed is the highest (potential) speed at which the frame design permits continuous operation. Compressor frame data plate "Frame Rated Speed (RPM)" is application specific, and may be lower than Maximum Allowable Speed. Do not exceed the lower of frame rated speed, lowest cylinder rated (RPM), or driver rated speed.

b Minimum Speed is the lowest frame speed needed to provide adequate oil flow to the compressor bearings.

c Average Piston Speed is based on Maximum Allowable Speed (RPM). The cylinder data-plate rated speed (RPM) or frame rated speed may be less, resulting in a lower piston speed rating.

d Flow rate at maximum rated speed and 180°F (82°C) oil.

For more information, see the Ariel Performance Program.

Opposed Throw - Reciprocating Weight Balancing

Ariel recommends a reciprocating weight differential between opposing throws of 2.5 pounds (1.1 kg) or less for KBE compressors.

To replace a connecting rod assembly, piston, piston and rod assembly, balance nuts, or crosshead, weigh component parts on a scale calibrated to 0.1 pound (0.05 kg) and compare to the Balancing Record in the compressor Parts Book. If the weight changes, recalculate opposing throw reciprocating weight differential. If not within recommended limits, the compressor may require new balance nuts and/or crossheads.

To exchange opposing throw cylinder locations, exchange all reciprocating components to the opposite throw, except the connecting rod assemblies. Check the Balancing Record and recalculate reciprocating weight differential, including the weight of the connecting rods. If not within recommended limits, the compressor may require new crosshead balance nuts to reduce differential.

If unable to balance opposing throws within recommended limits, contact the packager or Ariel.

Table: KBE Approximate Component Weights, Lbs (Kg)

Component	Weight	Component	Weight
Main/Connecting Rod Bearing	2 (1)	Crosshead Guide (long single)	750 (340)
Top Cover 2-Throw	65 (29)	Crosshead Guide (long two)	780 (354)
Top Cover 4-Throw	126 (57)	Crosshead	See Note a
Spacer Bar	32 (15)	Crosshead Pin	25 (12)
End Cover Drive End	83 (38)	Lube Oil Pump 2-Throw	44 (20)
End Cover Auxiliary End	130 (60)	Lube Oil Pump 4-Throw	58 (26)
Crankshaft 2-Throw [b]	550 (249)	Frame Assembly w/o Cylinders	See Ariel Performance Program.
Crankshaft 4-Throw [b]	1065 (483)	VVCP	
Main Journal Caps	30 (14)	Cylinder Assembly	
Connecting Rod	91 (41)	Piston & Rod Assembly [a]	

a For exact weights, see Balancing Record sheet provided by Ariel in the Parts Book for each compressor.
b Crankshaft weight is without flywheel.

Appendix D - Compressor Clearance, Oil, & Temperature Record

SERIAL NO. F-	**MODEL**	**DATE**

CRANKSHAFT THRUST (END) CLEARANCE, In. (mm)	
Crankshaft Serial Number	Thrust Clearance, In. (mm)

CONNECTING ROD THRUST (SIDE) CLEARANCE, In. (mm)					
Throw 1	Throw 2	Throw 3	Throw 4	Throw 5	Throw 6

JACK CLEARANCES, In. (mm)						
Throw #	1	2	3	4	5	6
Main Bearing						
Conn. Rod Bearing						

After new bearing installation, if measured clearances exceed tolerances in Appendix B, contact your packager or Ariel before proceeding.

OIL PRESSURE AND TEMPERATURE						
Date	Time	RPM	Filter Inlet Oil Pressure psig (barg)	Filter Outlet Oil Pressure psig (barg)	Oil Temp. into Frame °F (°C)	Remarks

Throw	BEARING CAP TEMPERATURE, °F (°C), AFTER RUN TIME OF [a]:					
	3 Minutes Idle Speed (engine, VFD) 1 Minute (single speed motor) No Gas Load		Additional 3-5 Minutes Full Speed, No Gas Load		Additional 10-15 Minutes Full Speed, Gas Load	
	Main	Rod	Main	Rod	Main	Rod
1						
2						
3						
4						
5						
6						

a These minimum recommended test intervals may change based on driver restrictions. Regularly inspect running gear for any debris or signs of abnormality. Bearing cap temperature should not be more than 10°F from each other (compare mains and rods separately).

Appendix E - Balance Valve Log

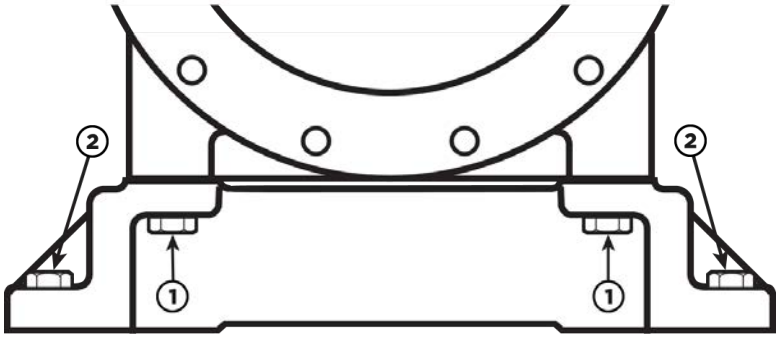
Table: Balance Valve Maintenance Log

Date	Time	Pressure Gauge Description	Div. Block Pressure		Balance Valve Set Pressure	Div. Block Cycle Time (Seconds)	Notes
			Min.	Max.			

Appendix F - Hold-down Bolting to Resist Shaking Forces and Couples

Requirements

Figure: Hold-down Fasteners



1 Crosshead Guide-to-Support Fastener (see Table: Crosshead Guide-to-Support Fastener Torques)
2 Crosshead Guide Support Foot and Frame Foot Hold Down Fastener (see Table: Crosshead Guide Support Foot & Frame Foot Hold-down Fasteners)

1. The torque values provided will achieve the minimum clamp load necessary for operation.
2. The torque values in Table: Crosshead Guide-to-Support Fastener Torques achieve 50,000 psi (345 MPa) prestress in the fasteners. Use grade 8 or 9 bolting to hold crosshead guides to supports.

Table: Crosshead Guide-to-Support Fastener Torques [a]

Compressor Model	Size inch - TPI	Torque lb-ft (N•m)
KBE	7/8 - 9	255 (345)

a Guide supports purchased from Ariel ship with guide-to-support fasteners.

3. To use the torques in Table: Crosshead Guide Support Foot & Frame Foot Hold-down Fasteners, the ultimate strength (Su) of the stud material must equal or exceed 100,000 psi (690 MPa); higher strength stud material is preferred. In all cases, torque the nuts to stress the stud to about 55% of the ultimate strength of the stud material. Torques shown in Table: Crosshead Guide Support Foot & Frame Foot Hold-down Fasteners are based on 100,000 psi ultimate strength material for imperial sizes and 830 MPa ultimate strength material for metric sizes to meet the minimum clamp load necessary for operation. Higher strength materials require higher torques than shown in the tables for added fastener stretch and preload. Material strength and torque are directly proportional. Example: for 120,000 psi (830 MPa) ultimate strength Imperial bolting: 120,000 ÷ 100,000 = 1.2 x min. torque for Imperial size from table = required torque for given nut size. To use a different stud diameter, consult Ariel for feasibility and torque.

Table: Crosshead Guide Support Foot & Frame Foot Hold-down Fasteners - Min. Torques [a]

Compressor Model	Foot Hold-down Fastener Application [b]	Hole Diameter inch (mm)	Imperial (basis 100,000 psi)		Metric (Class 8.8) [c]	
			Recom. Stud Size inch - TPI	Min. Nut Torque [e] lb-ft (N•m)	Recom. Stud Size [d] mm x pitch	Min. Nut Torque [d] lb-ft (N•m)
KBE	Guide Support	1-1/4 (32)	1-1/8 - 7	600 (810)	M27x3	630 (850)
					M30x3.5	850 (1150)
KBE	Frame	1-3/8 (35)	1-1/4 - 7	850 (1150)	M30x3.5	850 (1150)
					M33x3.5	1170 (1590)

a See #1, "Requirements".
b Crosshead guide support feet and frame feet with counterbored holes are designed to accept 12-point fasteners. To use hex head fasteners or studs and nuts instead, a washer thick enough to clear the counterbore is required to provide adequate clearance for the fastener and any tooling used for installation. A washer is also recommended for feet with holes that are milled flat.
c Ultimate Strength: 830 MPa (120,000 psi); Yield Strength: 660 MPa (95,700 psi).
d Where two sizes appear, the first is the preferred size, and the second is an alternate size.
e For a stud pitch other than those listed above in the same nominal diameter size, use the values in Table: Crosshead Guide Support Foot & Frame Foot Hold-down Fasteners and this formula to calculate torque: Torque = Table Fastener Torque value x (Actual Fastener Root Area ÷ Table Fastener Root Area value).

4. Compressor frame foot hold-down studs and crosshead guide support foot hold-down studs listed in Table: Crosshead Guide Support Foot & Frame Foot Hold-down Fasteners should be as long as possible. Ariel recommends **a minimum 12 to 1 length-to-diameter ratio.**

> **NOTE:**
>
> Fasteners long enough to reach only through the compressor foot and a deck plate or flange of an I-beam can loosen if not provided with engineered devices to maintain fastener pre-load.

5. Tighten nuts as recommended in step 3 to maintain fastener pre-load during operation. Ariel also recommends rolled threads and hardened spherical washers for hold down fasteners to help compensate for fastener-to-frame surface angular misalignment.
6. Re-preserve any fasteners subject to corrosion after installation.
7. Follow lubrication guidelines and recommended procedures for torque accuracy in Appendix A.

Appendix G - Soft Foot & Top Plane Flatness Checks for Main Bearing Bore Alignment in Compressors

It is very important to main bearing and crankshaft life that the main bearing bores are aligned. The two methods to make sure the main bearing bores are aligned are Soft Foot measurement and Top Plane Flatness measurement. Soft Foot measurements are necessary on all compressors. Top Plane Flatness measurement does not apply to KBE frames.

Make sure the main bearing bores are aligned at these times:

- When a new compressor is installed or when a compressor is moved to a new location. Measure before skid grouting and again before you start the compressor.
- 6 months or 4000 hours after you start the compressor. If no change in measurement is found, measure again at intervals of 12 months or 8000 hours.

> **NOTE:**
>
> If you find loose hold-down bolts, do the necessary measurements for the frame model. If a crankcase is damaged, correct Soft Foot before repairs.

Ariel recommends that the packager installs reference pads when the skid is made. Use the pads to make sure the skid is level and flat during assembly. With correct documentation, this practice decreases the work necessary to align primary components during installation.

During compressor installation, make sure the compressor is level before you adjust Soft Foot. For level limits, refer to Appendix I in the Packager Standards Manual.

Soft Foot Measurement

With the skid and compressor level:

1. Torque compressor hold-down bolts. Refer to Appendix A for bolt size and torque.
2. Set a calibrated dial indicator on one foot, then loosen the hold-down bolt. Measure how much the foot moves away from the skid. For measurements on the middle hold-down feet (feet located between two guides) on KBE:K:T:C:D:F frames, it is necessary to loosen the two feet when you measure each foot. Feet near to each other can act as one foot. Refer to Figure: Flatness Measurement Locations for KBE:K:T:C:D:F Frames to see the two feet to loosen. **Soft Foot measurements are necessary on each foot.**
3. Add a shim to all feet that move more than 0.002 in. (0.05 mm) when loosened.
4. If the foot moves less than or equal to 0.002 in. (0.05 mm), record the Soft Foot measurement.
5. Torque the hold-down bolt or bolts.

Complete this procedure again for each frame foot. Main bearing bores are aligned when frame feet do not move more than 0.002 in. (0.05 mm).

Appendix H - Attachment of Wiring, Tubing, or Pipe Clamps to Cylinders

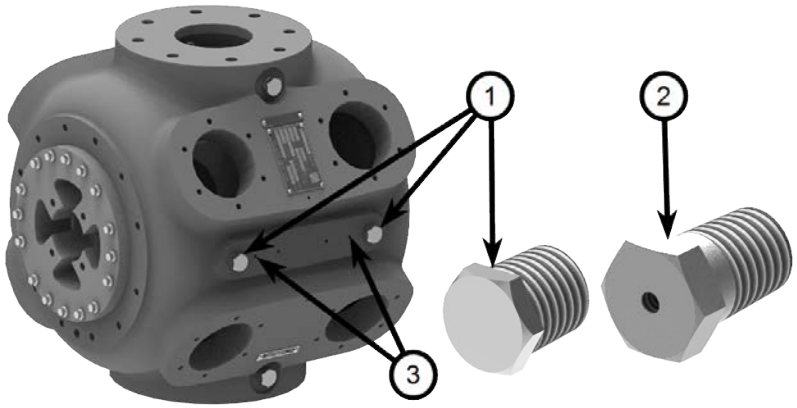
DO NOT drill holes to mount wiring, tubing, instrumentation, or pipe clamps to any pressure containing component. Pressure containing components include cylinders, valve caps, cylinder heads, unloaders, gas passage caps, and flanges. DO NOT use pressure containment fasteners to mount brackets or other structural attachments. These practices can compromise the integrity of pressure containing components.

CAUTION:

Do not drill holes in any pressure containing components for any reason.

Ariel offers a 1/2-inch NPT pipe plug with a 1/4-20 UNC tapped hole to accommodate a tubing clamp or bracket (see Figure: Typical Cylinder Plugs and Tapped Holes). Installing this plug in cylinder indicator ports provides locations to secure unloader tubing or instrumentation lines. The plug is designed for high pressure and will not affect existing Ariel cylinder MAWP in any application. Also, standard cylinder plugs may be replaced with this plug without a hydrotest. Packagers may purchase this plug (A-28373) through Ariel Parts Sales.

Figure: Typical Cylinder Plugs and Tapped Holes



1 Standard Plug **2** Plug A-28373 **3** New 1/4-20 UNC Tapped Holes

Newer Ariel cylinder designs include 1/4-20 UNC tapped holes to the milled pad that contains the cylinder indicator port holes (see Figure: Typical Cylinder Plugs and Tapped Holes). These extra tapped holes will provide additional accommodation to clamp unloader tubing or instrumentation lines.

Alternate Attachment Method

An alternate method to attach clamps to compressor cylinders is to use an epoxy resin. Ariel recommends Loctite 331 or Loctite 332 applied to clean, unpainted metal. Follow all manufacturer instructions and recommendations to prepare the area for the adhesive. The clamp and unpainted surface can be painted after the adhesive cures. This clamp attachment method does not compromise cylinder integrity.

Appendix I - Recommended Maintenance Intervals for Intermittent Duty Operation

Ariel compressors, like all industrial equipment, require maintenance. The severity of compressor service directly influences the frequency and amount of maintenance needed. What follows are the recommended intervals for inspections and replacements for **intermittent duty** operation. This will help you determine the appropriate intervals for a given compressor application.

> **NOTE:**
>
> For continuous duty service, see "Recommended Maintenance Intervals for Continuous Duty Operation".

> **NOTE:**
>
> Careful documentation of inspection results is critical to establish whether recommended intervals are adequate or require more or less frequency.

400-Hour Maintenance

1. Perform Daily maintenance.
2. Sample frame oil and send to a reputable lubricant lab for analysis. See "Oil Sampling" for a list of what an oil analysis should provide.
   - If analysis results indicate increasing levels of lead, tin, or copper particles in the oil, shut down unit. Remove frame top cover and crosshead guide side covers. Visually inspect for debris. Do not disassemble further without good reason.
   - If debris indicates, replace affected parts, then change oil, oil filter, and clean the strainer with a suitable solvent.

   > **NOTE:**
   >
   > Operate the unit to bring oil temperature to a minimum operating temperature of 150°F (66°C) to drive off any entrained water

2,000-Hour Maintenance

1. Perform Daily and 400-Hour maintenance.
2. Drain and replace force feed lubricator box oil.
3. Where applicable, clean sintered element in the small oil filter on the force feed lubrication system now or at every main oil filter change. Use a suitable solvent. Where applicable, replace force feed lubricator inlet filter in independent supply system.
4. Change compressor oil filter. See top cover filter information plate or “Oil Filter” in Frame Oil System Components for procedure. Do not exceed 12 month installed period for filter element.

   > **NOTE:**
   >
   > For replaceable element style filters, drain the canister completely before removing the element. Inspect elements for cleanliness and damage. Do not install dirty or damaged elements.

5. Change compressor oil. Extremely dirty environments, oil supplier recommendations, or oil analysis may dictate a different oil change interval. Follow these steps:
   a. Drain oil from frame, associated piping, and oil cooler.
   b. Clean oil strainer with a suitable solvent. In dry sump applications, the strainer is located at the oil reservoir.
   c. Remove frame top cover and crosshead guide side covers. Visually inspect for debris. Do not disassemble further without good reason. If debris indicates, replace affected parts, then change the oil filter.
   d. Refill frame with fresh, clean oil.
   e. Visually inspect the crankcase for debris or other foreign material. Disassembly is not recommended unless a reason is found to do so.
   f. Check for adequate cylinder lubrication. See “Cylinder Lubrication Paper Test” for procedure.
6. Grease VVCP stem threads at grease fitting, with 2 to 3 pumps of cylinder lube compatible grease using a standard manual grease pump. Turn VVCP adjustment handle all the way in and all the way out to lubricate the threads. Measure or count turns to return the handle to its original position.

4,000-Hour Maintenance

1. Perform Daily, 400-Hour, and 2,000-Hour maintenance.
2. Verify safety shutdown functionality.
3. Inspect crankcase breather filter and clean with suitable solvent.
4. Open force feed lubricator box and visually inspect pump followers, cams, and gears for wear.
5. Pressure test distribution blocks. See “Distribution Block Assembly” for procedure.
6. Measure, log, and trend the following:
   a. Main bearing, connecting rod bearing, and crankshaft jack and thrust clearances. See "Main Bearing Vertical Jack Clearance Measurement", "Connecting Rod Bearing Vertical Jack Clearance Measurement", and "Crankshaft Thrust (End) Clearance Measurement".

   > **NOTE:**
   > Clearance trends along with oil analysis and crankcase visual inspection can indicate the need for bearing replacement. Do not disassemble to check clearances. Disassemble only if clearance checks indicate excessive clearance. Contact Ariel for original assembly clearances.

   b. Crosshead to guide clearances. See “Crosshead Installation” in Section 4.
   c. Piston rod runout. See “Piston Rod Runout” in Section 4.

   If any of the above items are outside limits , find the cause and correct it. It may require part replacement.
7. Remove valves and valve gaskets:

   > **NOTE:**
   > If using CP/CPs/XP valves, valve maintenance may be extended to the 8,000 hour or 22,000 starts interval.

   a. Visually inspect valve pockets for damage. Verify all old valve seat gaskets are removed.
   b. Have a qualified valve repair shop disassemble, visually inspect, and refurbish the valves where needed. Provide the valve repair shop an Ariel torque chart and valve service guide.
   c. Visually inspect cylinder gas passages and clean them of all debris.
8. If applicable, visually inspect suction valve unloader actuator stems for damage or wear. Visually inspect stem seals for damage or wear and confirm that the actuator moves freely.

9. If applicable, visually inspect pneumatic clearance pockets for damage or wear (seating surface and stems/stem seals). Confirm that actuator moves freely.
10. Inspect cylinder bores for damage or wear. Replace the cylinder body or restore the bore if any of the following conditions exist:
    - Bore surface blemishes or gouges.
    - Bore out of round more than 0.001 inch per inch of bore diameter (0.001 mm/mm) or tapered.
    - Main bore is worn to counterbore diameter or larger.
11. Inspect piston rings and wear bands:
    - Measure and log piston ring condition, end gap, side clearance, and piston ring groove width.
    - Replace rings that are damaged or outside limits listed in Appendix B.
    - When replacing rings, re-measure and log ring side clearance to check for groove wear.
    - Measure and log radial projection of wear band.
12. Inspect piston rods for damage and excessive wear. Replace rod if any of these conditions exist (see "Piston Rod Inspection"):
    - Gouges or scratches on the rod.
    - Under size more than 0.005 inch (0.13 mm).
    - In the packing travel length, out of round more than 0.001 inch (0.03 mm).
    - Anywhere in the packing travel area, tapered more than 0.002 inch (0.05 mm).
13. Re-tighten hold down fasteners to proper torque. Visually inspect cylinder gas passages and clean them of all debris. Inspect for frame twist or bending to verify main bearing bore alignment. See Appendix E for flatness and soft foot requirements.
14. Perform a coupling alignment (see "Initial Maintenance").
15. Rebuild piston rod pressure packing cases and oil wiper cases. See “Piston Rod Packing” in Section 4 for procedure.
16. Re-install valves, retainers, and valve caps using new valve gaskets and valve cap o-rings/seals. Use proper installation techniques and torque procedures for valve caps. See “Valve Installation” in Section 4.
17. Check and re-calibrate all required instrumentation.
18. If the compressor is equipped with crankcase over-pressure relief valves, visually inspect and exercise valves to manufacturer recommendations.
19. Check auxiliary end chain drive for undercut sprocket teeth and chain for excessive stretching. Replace as required. See “Chain Adjustment” in Section 4 for procedure.
20. Check fastener torques of gas nozzle flange, head end head, valve cap, piston rod packing, crosshead pin through bolt, frame hold down, crosshead guide to cylinder, distance piece to cylinder, and tandem cylinder to cylinder.

8,000-Hour or 22,000 Starts Maintenance (whichever occurs first)

1. Perform Daily, 400-Hour, 2,000-Hour, and 4,000-Hour maintenance.
2. Replace crankshaft main and connecting rod bearing shells if no replacement has occurred in the past 8,000 hours or 22,000 starts.
3. If using CP/CPs/XP valves, remove valves and valve gaskets:
   a. Visually inspect valve pockets for damage. Verify all old valve seat gaskets are removed.
   b. Have a qualified valve repair shop disassemble, visually inspect, and refurbish the valves where needed. Provide the valve repair shop an Ariel torque chart and valve service guide.

c. Visually inspect cylinder gas passages and clean them of all debris.

16,000-Hour Maintenance

1. Perform Daily, 400-Hour, 2,000-Hour, 4,000-Hour, and 8,000-Hour maintenance.
2. Remove crosshead pins.
   a. Measure and log crosshead pin to crosshead pin bore and connecting rod bushing bore clearances.
   b. Check the crosshead pin end caps and through bolt for wear.
   c. Replace if needed.

32,000-Hour Maintenance

1. Perform Daily, 400-Hour, 2,000-Hour, 4,000-Hour, 8,000-Hour, and 16,000-Hour maintenance.
2. Replace lubricator distribution blocks.
3. Replace crosshead and connecting rod bushings. See “Crossheads” in Section 4 and “Connecting Rods” in Section 4.

Appendix J - Replacement Frame Installation

> **NOTE:**
>
> When removing the compressor frame for replacement, do not move the driver. Aligning the compressor to the driver helps position the new frame in the correct location. In addition, note the amount and location of shims used under each foot and crosshead guide for reference. This can be a good starting point but will likely require adjustment to account for manufacturing tolerances.

1. Frame footprints, customer connection, and crankshafts may have changed. Call ARC and confirm changes between the replacement and existing equipment.
2. Determine frame weight and gather appropriate lifting devices. Contact ARC with the frame serial number for weight and center of gravity information.
3. Loosen or remove fasteners on shipping timbers or shipping skid.
4. Lift frame with appropriate lifting device attached to lifting eyes on frame or crosshead guides.
5. Lift the frame high enough to clean the feet easily without getting under the equipment. If jack stands are available, place the frame above the jack stands with lifting devices still attached for additional safety.
6. Remove any additional frame feet preservation materials.
7. Clean frame feet with solvent. Stone or file feet to ensure they are flat and free of nicks, scratches, gouges, rust, or paint.

> **NOTE:**
>
> Remove only high spots or corrosion; do not grind on feet and remove material.

8. If applicable, clean, stone, or file, and install sole plates. Installing a 0.020-inch shim at installation between the sole plate and foot is a best practice.
9. Set jack bolt lengths for desired compressor elevation.
10. Prepare frame hold-down fasteners and jack bolts for grouting, if applicable.
11. Clean and prepare skid for frame installation:
    a. Inspect and clean frame foot mating surfaces on the pedestal. This may include stoning or filing pedestal and welded chocks, if applicable.
    b. Prepare grout boxes, if applicable.
12. Install compressor frame hold-down fasteners into compressor frame feet or skid
13. Carefully lift the frame and move it into position. Lower onto the pedestal once hold-down fastener holes and hardware align.
14. Align the compressor to the driver per ER-8.2 and confirm thrust clearances for both pieces of equipment.
15. On 4 and 6-throw compressors, loosen jack bolts on middle feet between throws and install a dial indicator near the hold-down fastener. Pre-load foot 0.001 inch and then back jack-bolt off until the indicator returns to zero.
16. Confirm frame meets applicable ER-82 requirements. Adjust as needed.
17. If applicable, install crosshead guides and cylinders per Ariel Packager Standards Section 3 and Section 10.
18. Verify pulsation bottle and piping connections are aligned. Confirm flange faces are in planer alignment, have proper gaps, and fasteners can be installed without forcing them in. See Ariel Packager Standards Section 4.
19. If applicable, prepare and pour grout. Allow grout to be fully set.
20. Torque frame feet fasteners per ER-26.

21. Shim guides per Ariel Packager Standards Section 3.
22. Torque crosshead guide hold-down fasteners per ER-26.
23. Check frame and confirm it meets applicable ER-82 soft foot and flatness requirements.
24. Install bottles and process piping per Ariel Packager Standards Section 4.
25. After all re-assembly work is finished, complete Ariel Start-up Checklist ER-10.6 before restart.
26. Perform bearing temperature check at start-up per ER-96.5.6.12.

Appendix K - Purchasing Assembly Lubricants, Sealants, & Preservatives

Ariel maintenance procedures refer to specific lubricants, thread sealants, and preservative coatings for compressor maintenance. Ariel thoroughly tested these products, and they provide the correct properties to perform as needed. Other products may claim to be equal, but Ariel has not tested them and cannot guarantee their performance.

The following products specified are not for sale by Ariel, but are available commercially, and may be available locally via your hardware distributor.

Figure: Ariel Lubricants, Thread Sealants, and Preservatives



1. Cortec VpCI-368 or Marine Grade Grease (used on top covers and guide covers sealed with o-rings)
2. Cortec VpCI-329 Corrosion Inhibitor (used for long-term preservation)
3. Loctite 567 (used on all pipe thread fittings)
4. Lubriplate 630-AA
5. Hylomar Universal Blue (used on force-feed lubricator bearing housings)
6. Loctite 242 (used on valve center studs)
7. Never-Seez Regular Grade

Latest Manual Revisions

08/10/2026

Topic	Change Description
Compressor Troubleshooting	Added problem 'VVCP adjustment handle or piston stem is locked and will not turn.' Applies to all manuals except KBH, KB100, and KBBV.
VVCP Adjustment	Added procedure to correct locked adjustment handle or piston stem (applies to all manuals except KBH, KB100, and KBBV).